

Microsoft ADC Cybersecurity Skilling Program

Week 9 Assignment 18

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SERVICE ENDPOINTS AND SECURING STORAGE

Introduction

In this lab, we will focus on implementing network security and access control for Azure Storage using **Virtual Networks**, **Service Endpoints**, and **Network Security Groups (NSGs)**. The goal is to understand how to securely isolate resources and manage traffic between Azure services and virtual machines. We will begin by **creating a virtual network** and **adding subnets**, followed by configuring **service endpoints** to securely connect Azure resources to a storage account. To enforce security policies, we will apply **NSGs** that restrict access to specific subnets and allow **Remote Desktop Protocol (RDP)** only on designated public subnets. Additionally, we will **create a storage account with a file share** and **deploy virtual machines** into the configured subnets. Finally, we will **test the connectivity** to the storage account from both the private and public subnets to verify that access control rules are functioning as intended. This lab provides hands-on experience in securing storage access in Azure environments using service endpoints and network-level restrictions—critical components in designing secure and efficient cloud architectures.

Tasks Completed

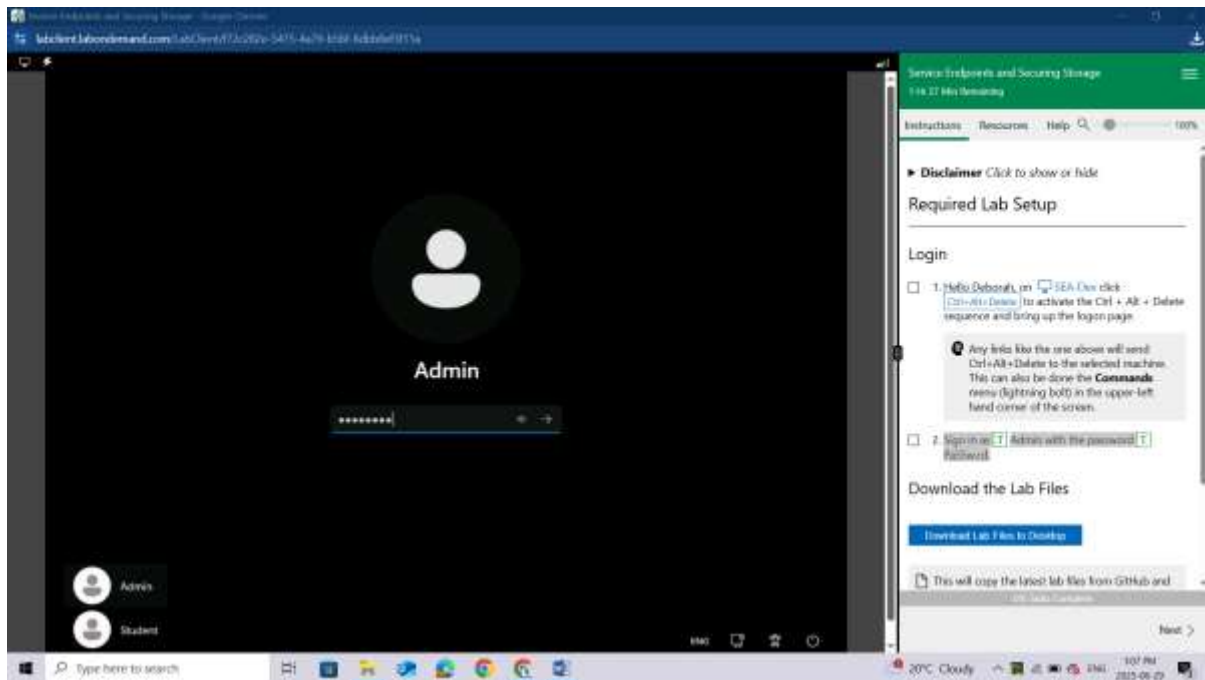
Required Lab Setup

Login

Hello Deborah, on [SEA-Dev](#) click [Ctrl+Alt+Delete](#) to activate the Ctrl + Alt + Delete sequence and bring up the logon page.

*Any links like the one above will send Ctrl+Alt+Delete to the selected machine. This can also be done the **Commands** menu (lightning bolt) in the upper-left hand corner of the screen.*

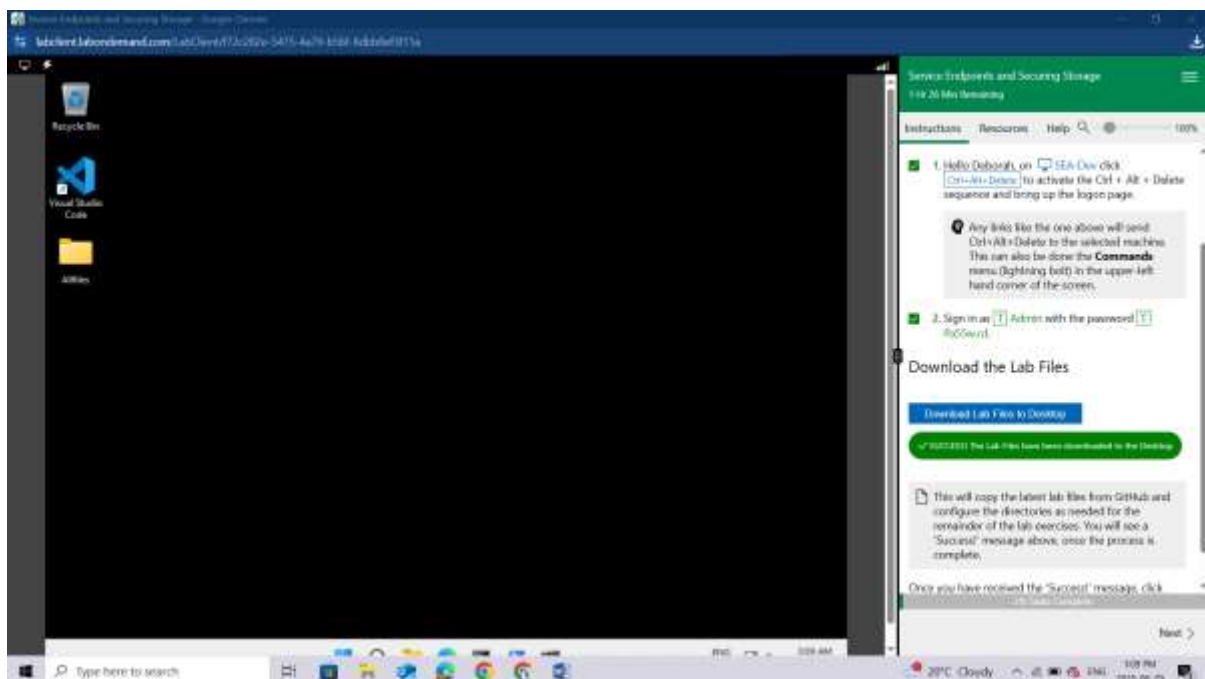
Sign in as **Admin** with the password **Pa55w.rd**.



Download the Lab Files

This will copy the latest lab files from GitHub and configure the directories as needed for the remainder of the lab exercises. You will see a 'Success!' message above, once the process is complete.

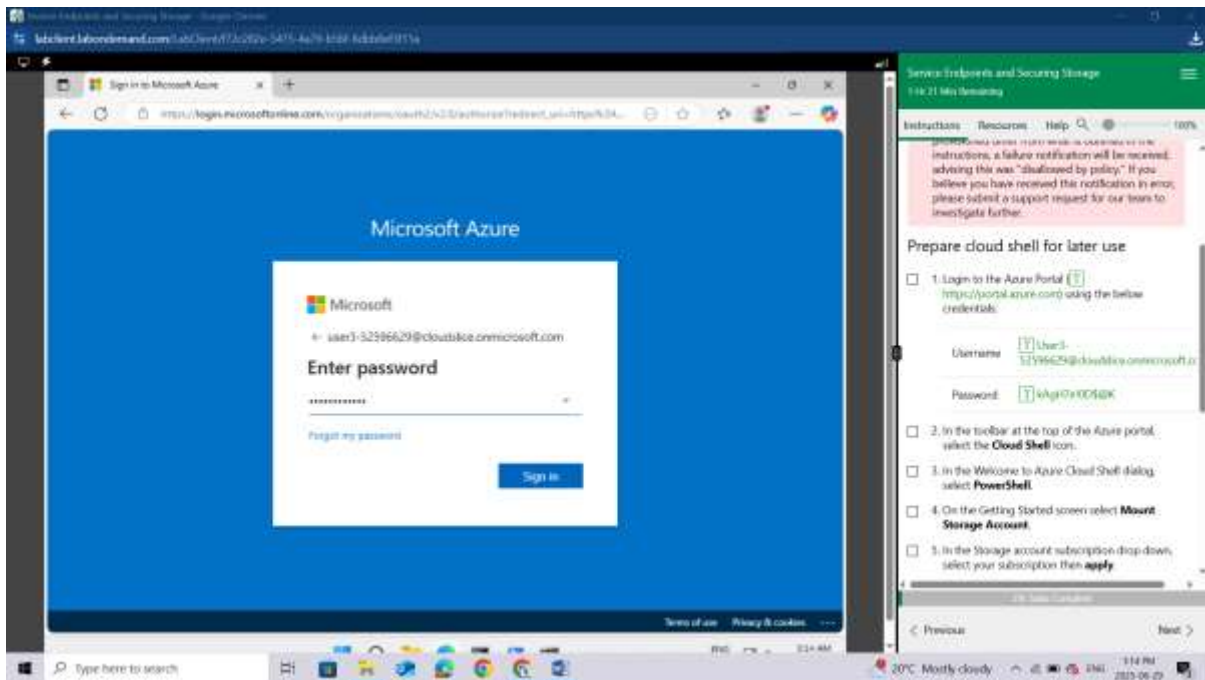
Once you have received the 'Success!' message, click **Next** to proceed to the lab.



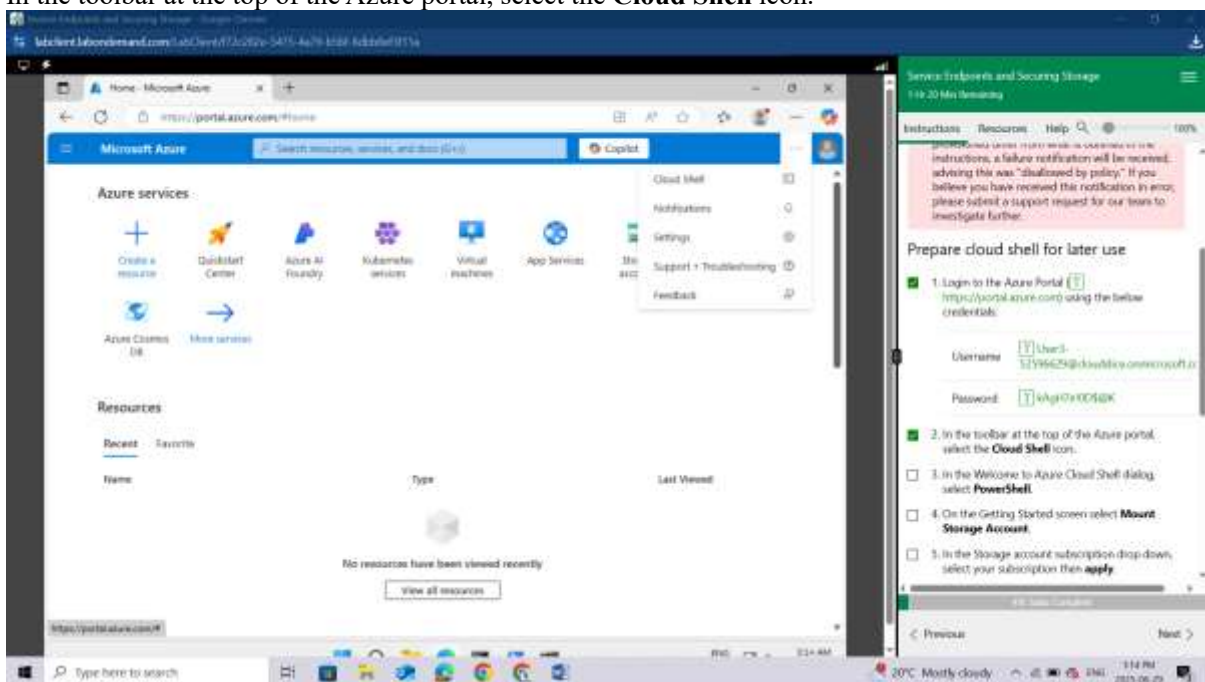
Prepare cloud shell for later use

Login to the Azure Portal (<https://portal.azure.com>) using the below credentials:

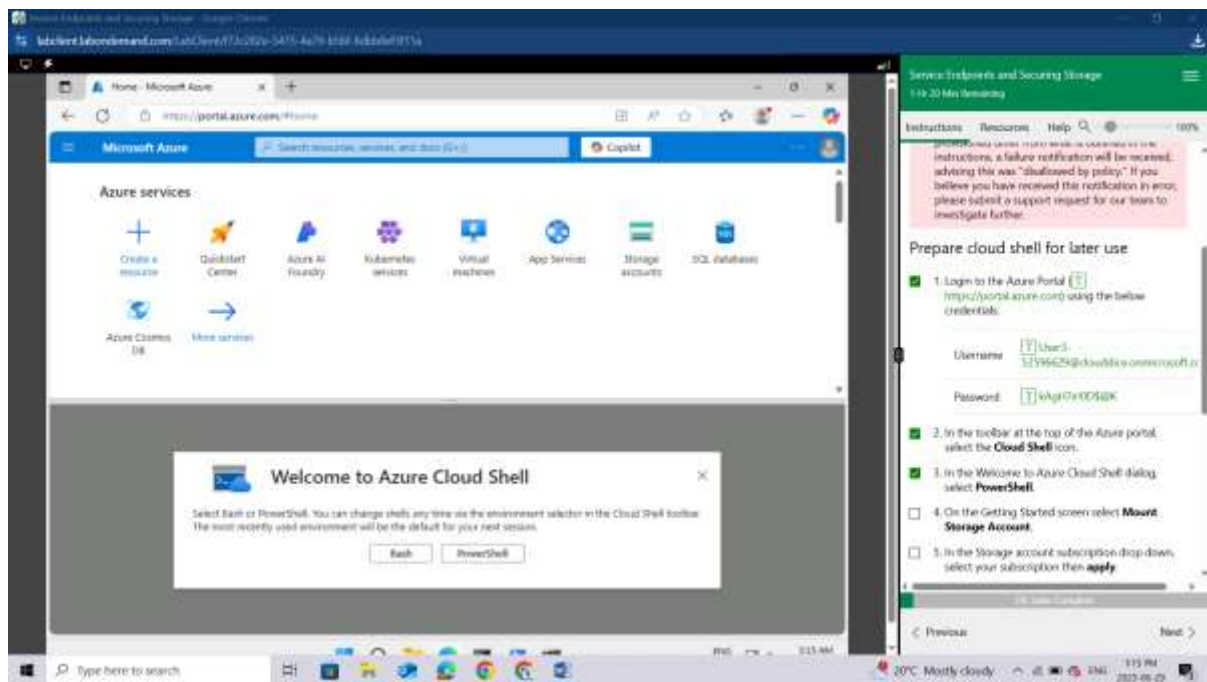
Username	User3-52596629@cloudslice.onmicrosoft.com
Password	kAgH7e!0D\$@K



In the toolbar at the top of the Azure portal, select the **Cloud Shell** icon.

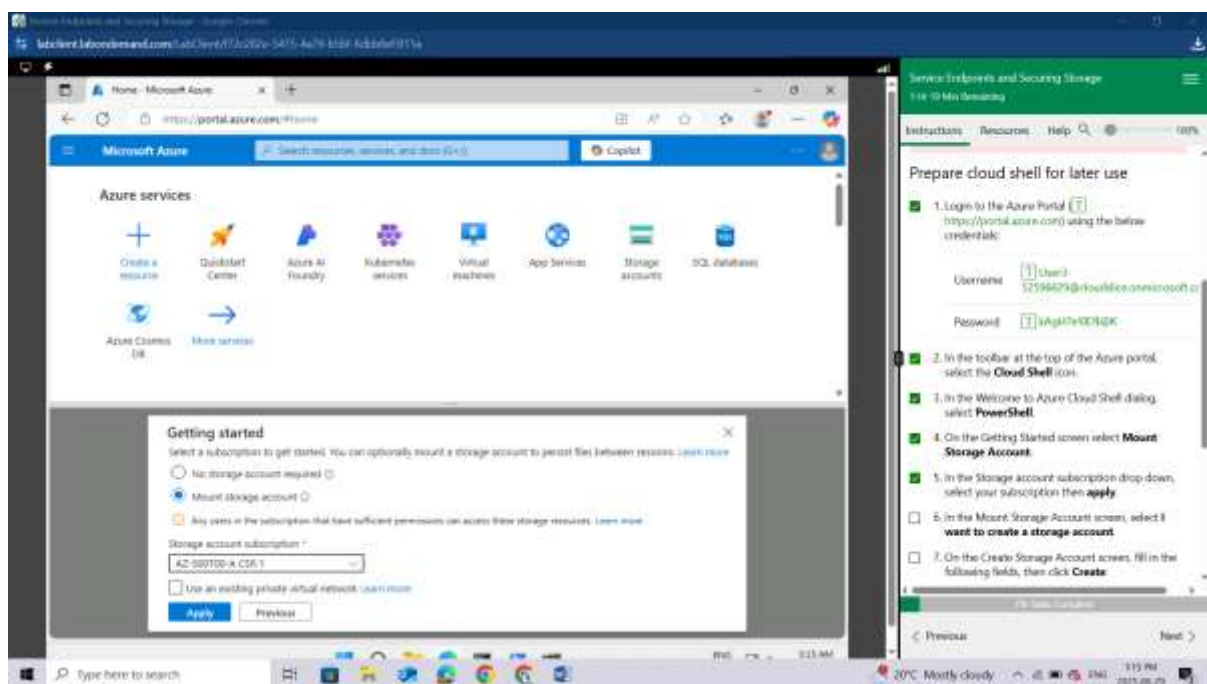


In the Welcome to Azure Cloud Shell dialog, select **PowerShell**.

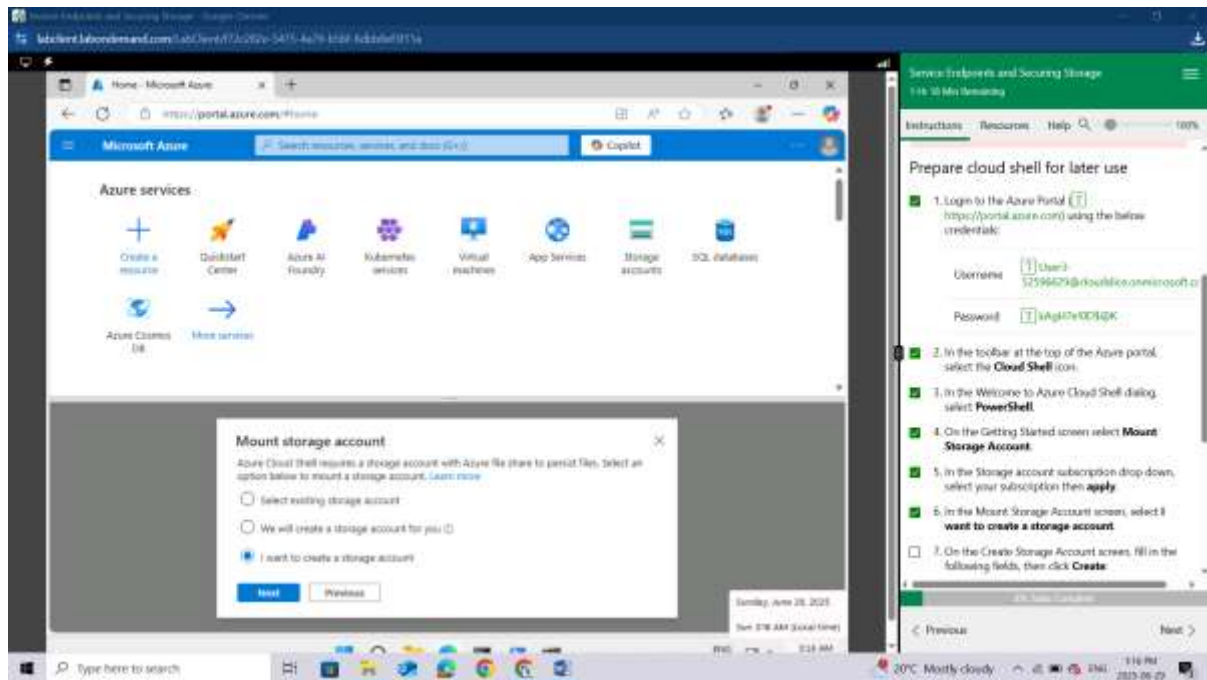


On the Getting Started screen select **Mount Storage Account**.

In the Storage account subscription drop down, select your subscription then **apply**.



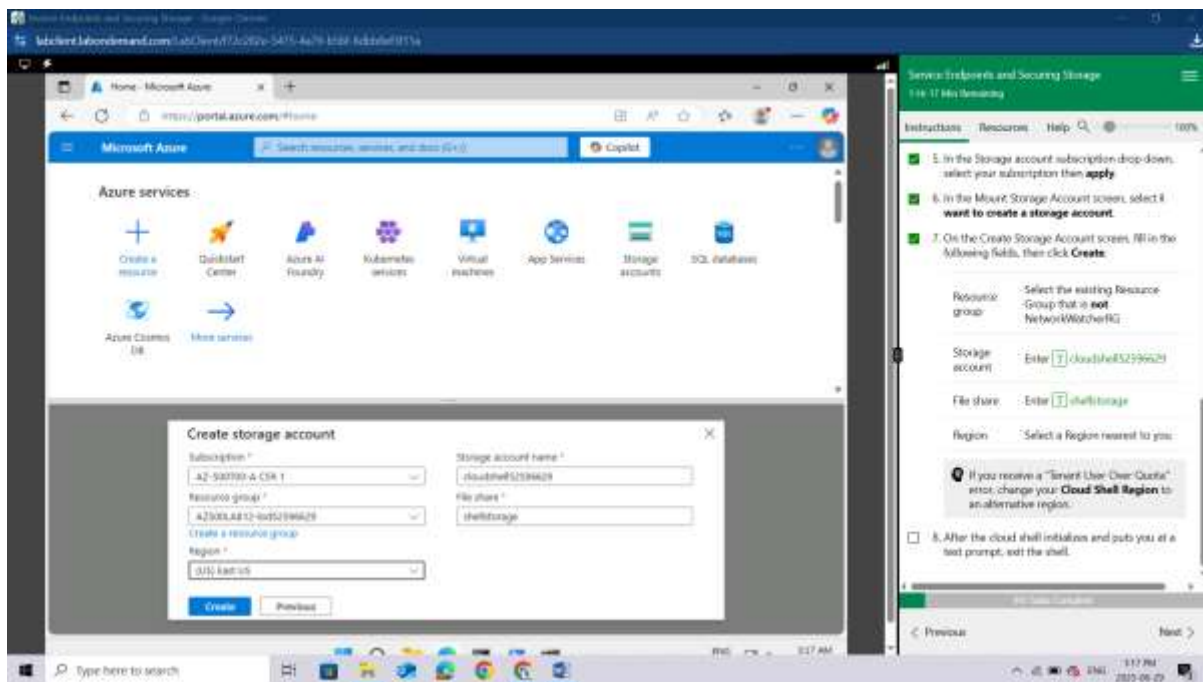
In the Mount Storage Account screen, select **I want to create a storage account**.



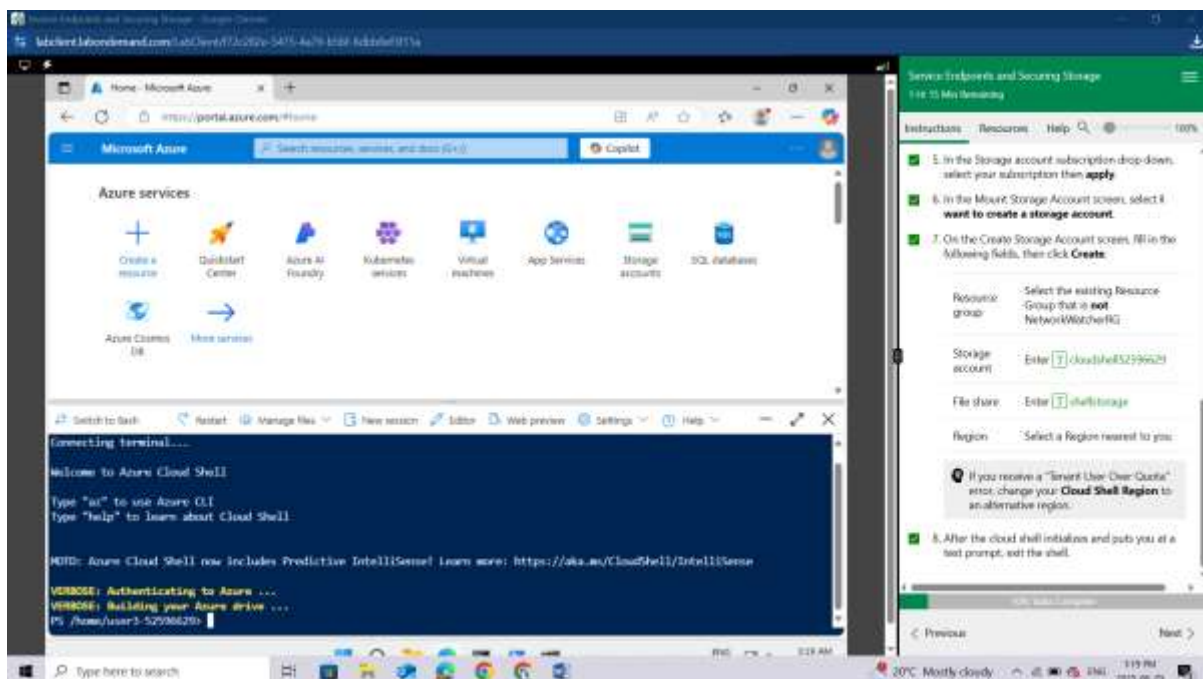
On the Create Storage Account screen, fill in the following fields, then click **Create**:

Resource group	Select the existing Resource Group that is not NetworkWatcherRG
Storage account	Enter cloudshell52596629
File share	Enter shellstorage
Region	Select a Region nearest to you

If you receive a "Tenant User Over Quota" error, change your **Cloud Shell Region** to an alternative region.



After the cloud shell initializes and puts you at a text prompt, exit the shell.



Lab 06: Service Endpoints and Securing Storage

Student lab manual

Lab scenario

You have been asked to create a proof of concept to demonstrate securing Azure file shares. Specifically, you want to:

- Create a storage endpoint so traffic destined to Azure Storage always stays within the Azure backbone network.
- Configure the storage endpoint so only resources from a specific subnet can access the storage.
- Confirm that resources outside of the specific subnet cannot access the storage.

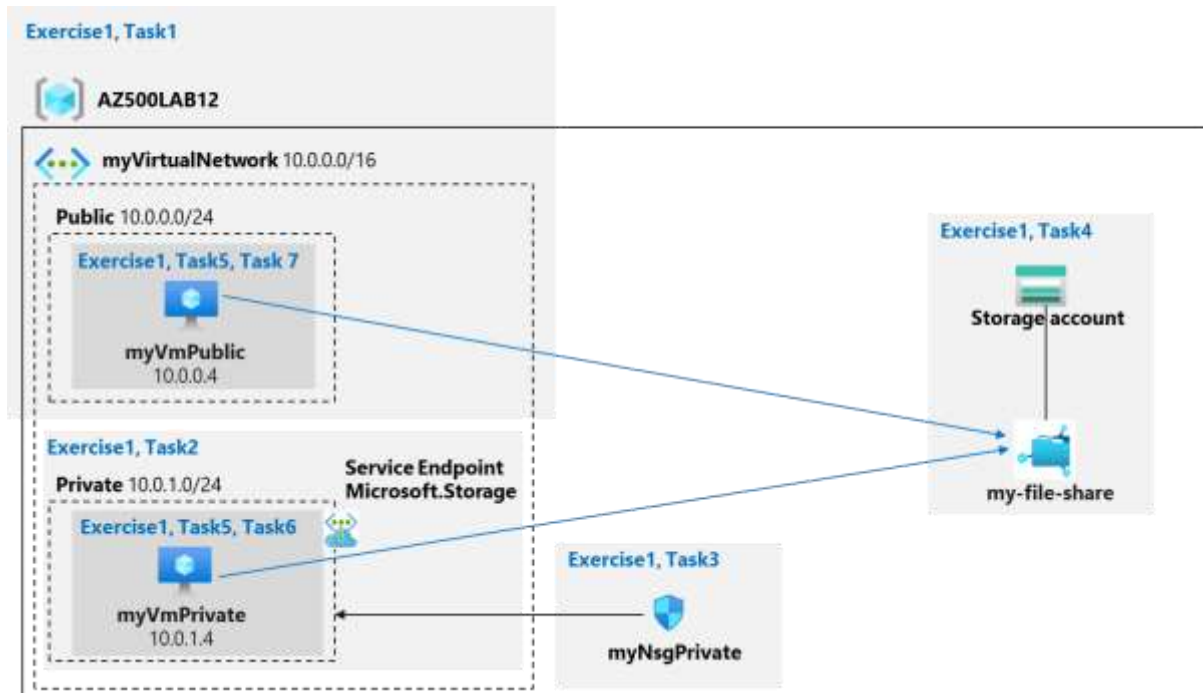
For all the resources in this lab, we are using the **East US** region. Verify with your instructor this is the region to use for class.

Lab objectives

In this lab, you will complete the following exercise:

- Exercise 1: Service endpoints and security storage

Service Endpoints and Securing Storage diagram



Instructions

Exercise 1: Service endpoints and security storage

Estimated timing: 45 minutes

In this exercise, you will complete the following tasks:

- Task 1: Create a virtual network
- Task 2: Add a subnet to the virtual network and configure a storage endpoint
- Task 3: Configure a network security group to restrict access to the subnet
- Task 4: Configure a network security group to allow rdp on the public subnet
- Task 5: Create a storage account with a file share
- Task 6: Deploy virtual machines into the designated subnets
- Task 7: Test the storage connection from the private subnet to confirm that access is allowed
- Task 8: Test the storage connection from the public subnet to confirm that access is denied

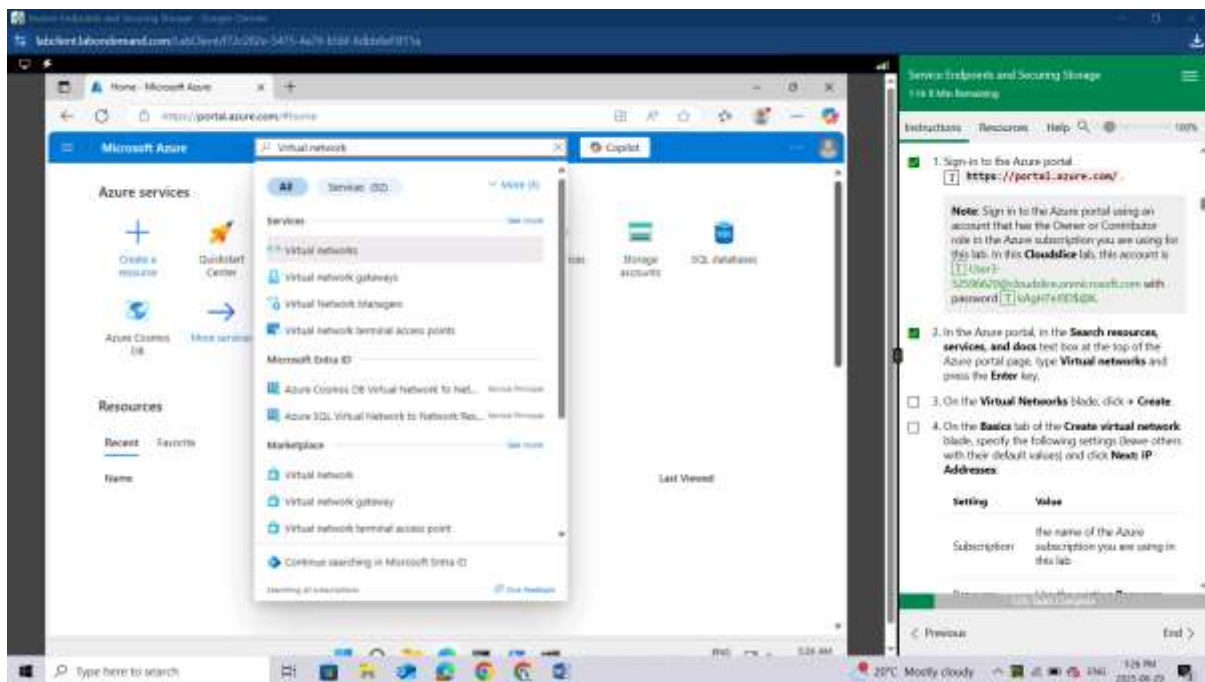
Task 1: Create a virtual network

In this task, you will create a virtual network.

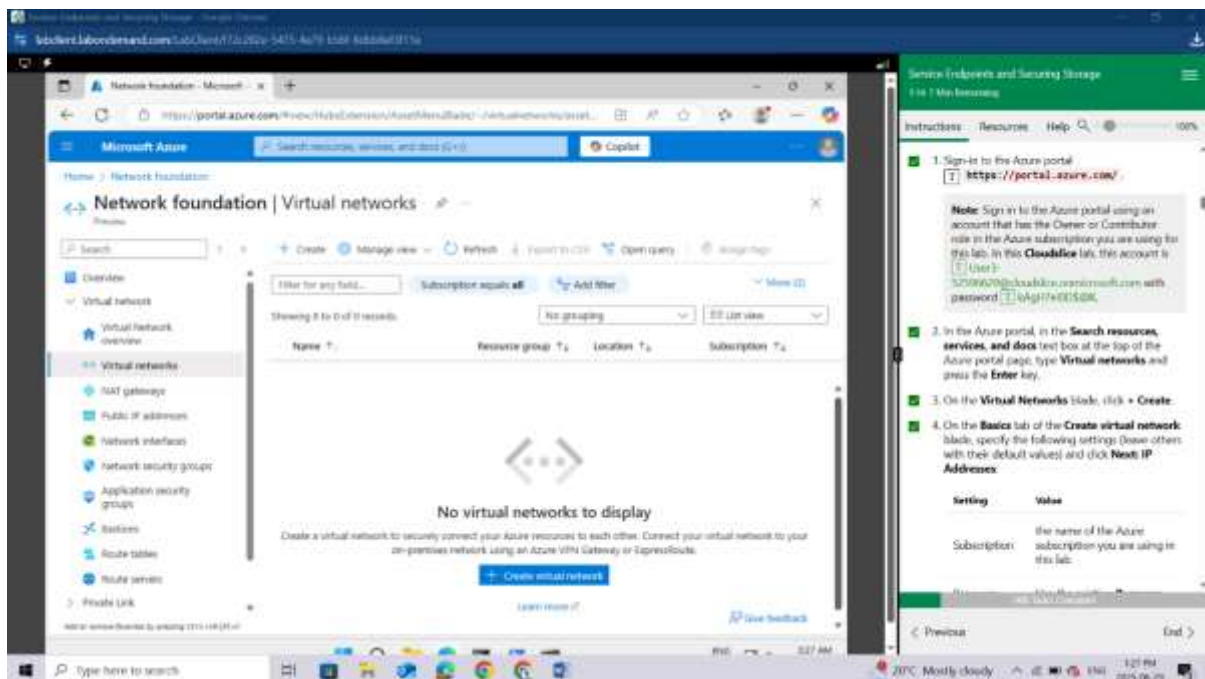
Sign-in to the Azure portal <https://portal.azure.com/>.

Note: Sign in to the Azure portal using an account that has the Owner or Contributor role in the Azure subscription you are using for this lab. In this **Cloudslice** lab, this account is **User3-52596629@cloudslice.onmicrosoft.com** with password **kAgH7e!0D\$@K**.

In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Virtual networks** and press the **Enter** key.



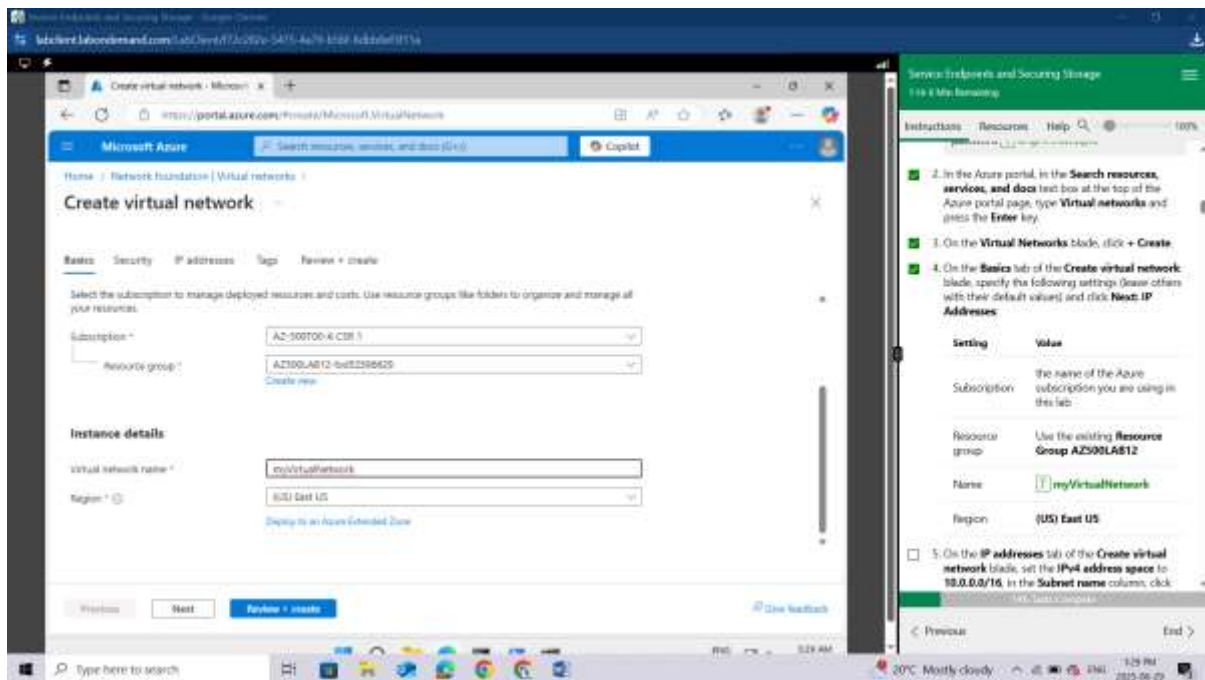
On the **Virtual Networks** blade, click **+ Create**.



On the **Basics** tab of the **Create virtual network** blade, specify the following settings (leave others with their default values) and click **Next: IP Addresses**:

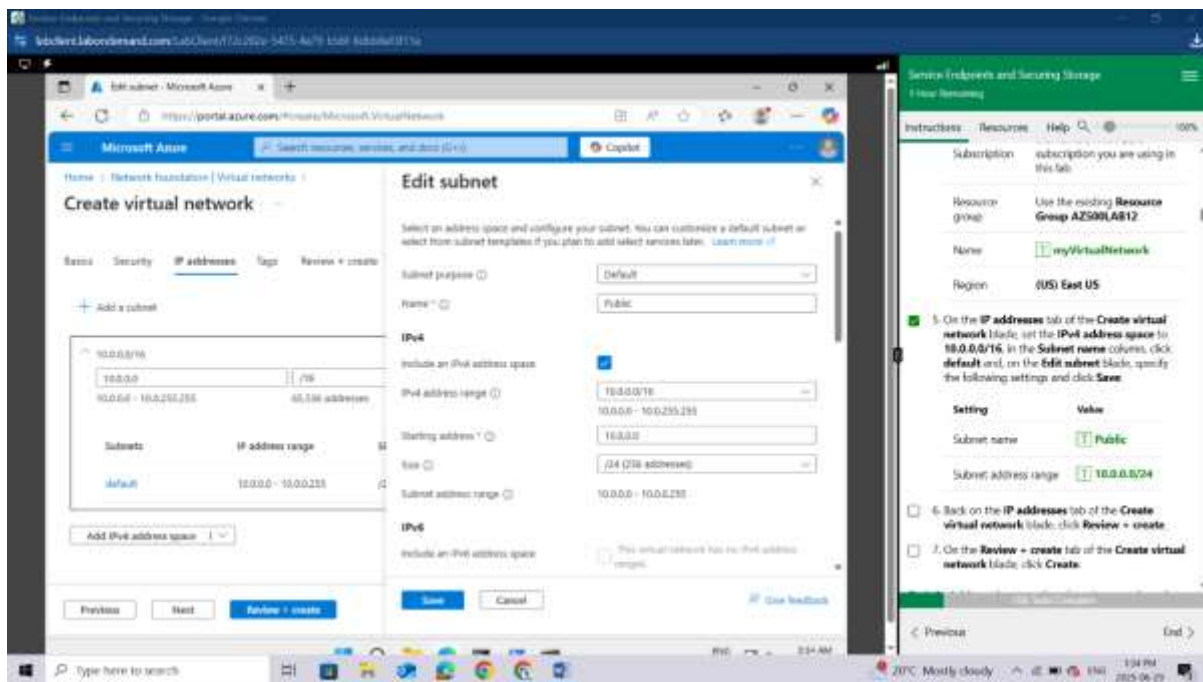
Setting	Value
Subscription	the name of the Azure subscription you are using in this lab

Setting	Value
Resource group	Use the existing Resource Group AZ500LAB12
Name	myVirtualNetwork
Region	(US) East US

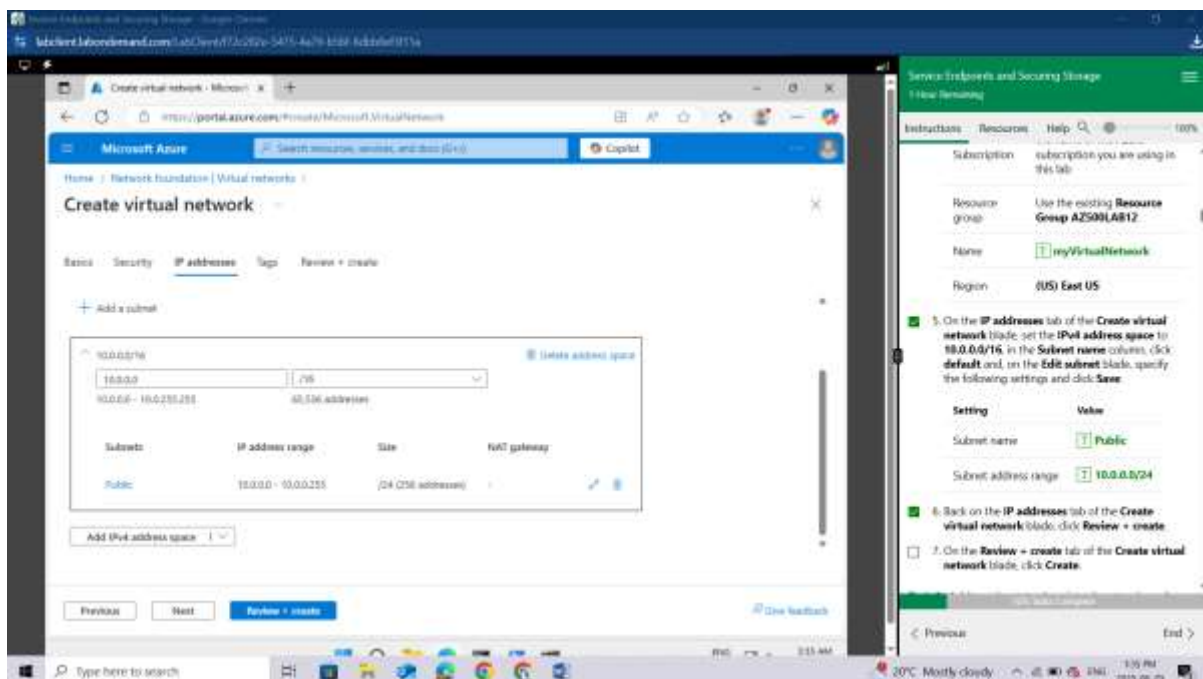


On the **IP addresses** tab of the **Create virtual network** blade, set the **IPv4 address space** to **10.0.0.0/16**, in the **Subnet name** column, click **default** and, on the **Edit subnet** blade, specify the following settings and click **Save**:

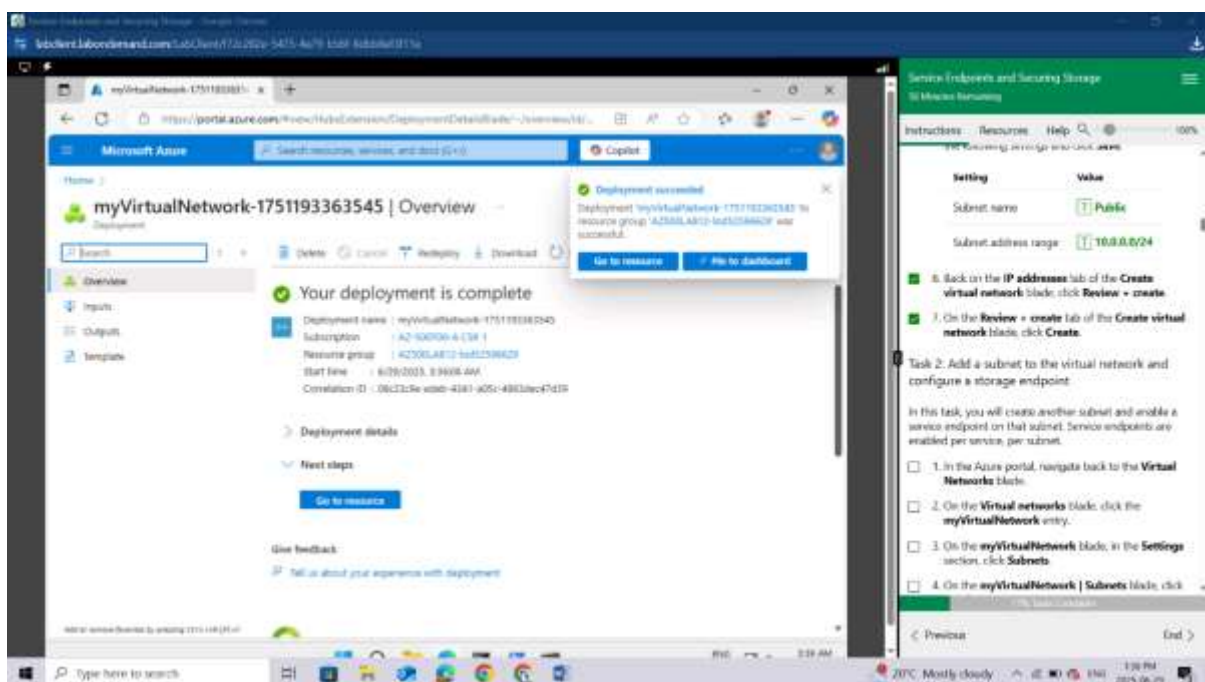
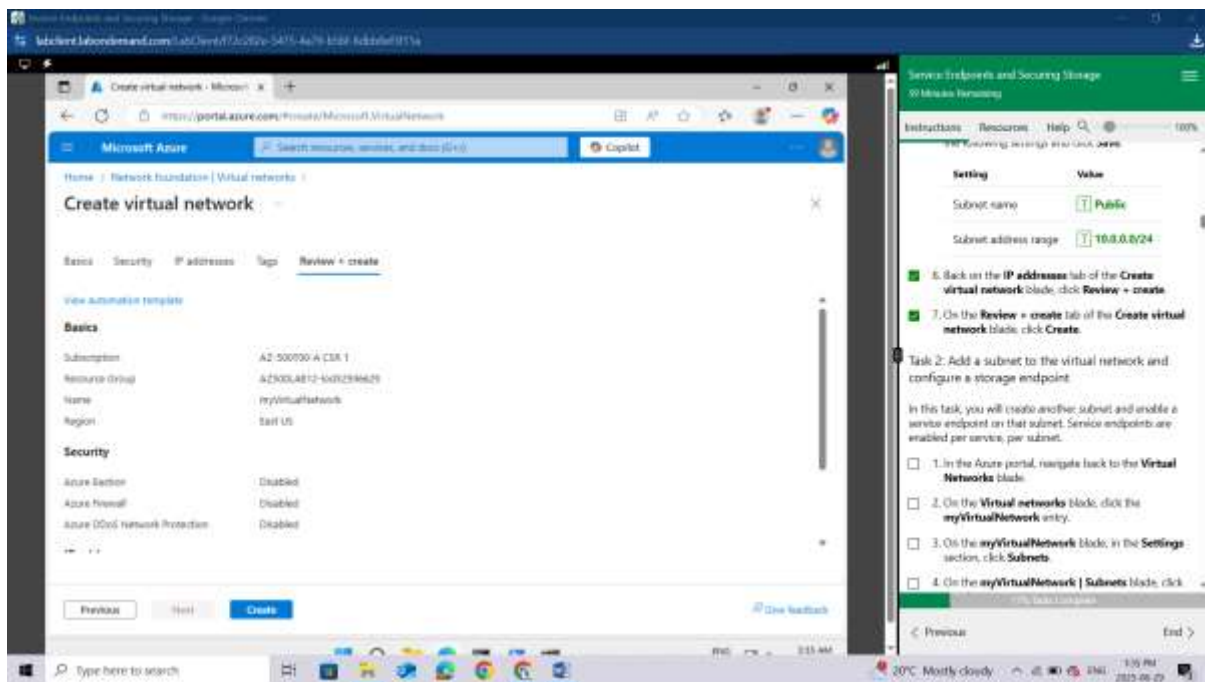
Setting	Value
Subnet name	Public
Subnet address range	10.0.0.0/24



Back on the **IP addresses** tab of the **Create virtual network** blade, click **Review + create**.



On the **Review + create** tab of the **Create virtual network** blade, click **Create**.

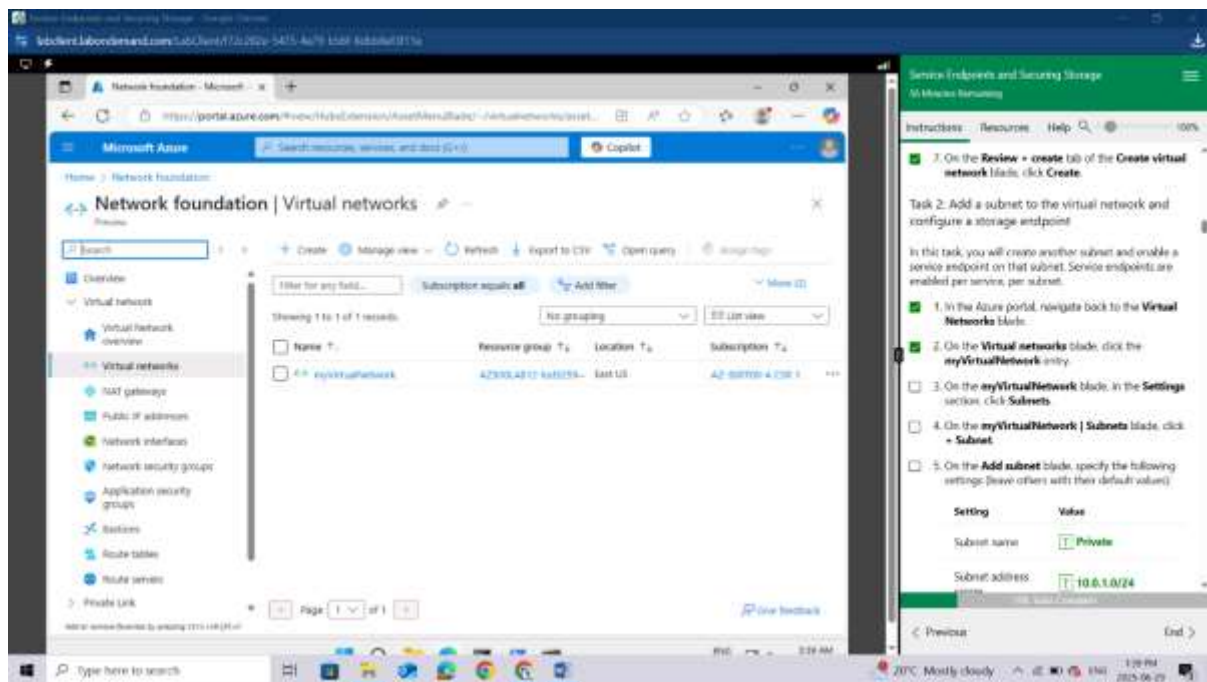


Task 2: Add a subnet to the virtual network and configure a storage endpoint

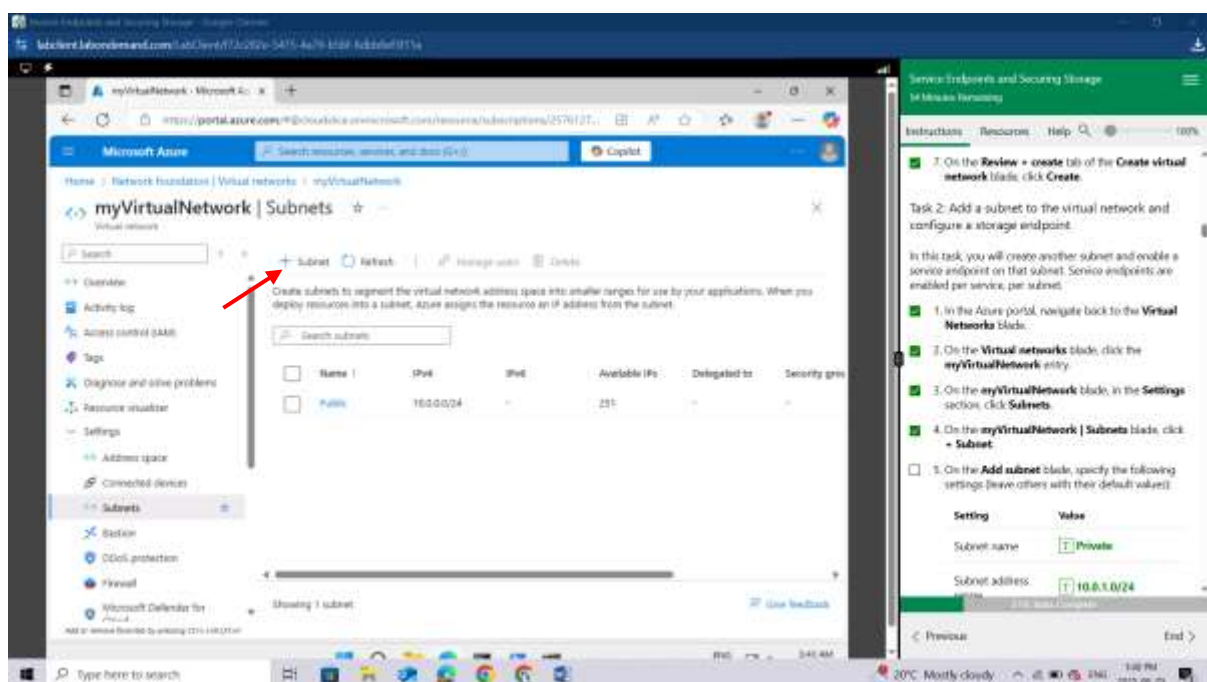
In this task, you will create another subnet and enable a service endpoint on that subnet. Service endpoints are enabled per service, per subnet.

In the Azure portal, navigate back to the **Virtual Networks** blade.

On the **Virtual networks** blade, click the **myVirtualNetwork** entry.



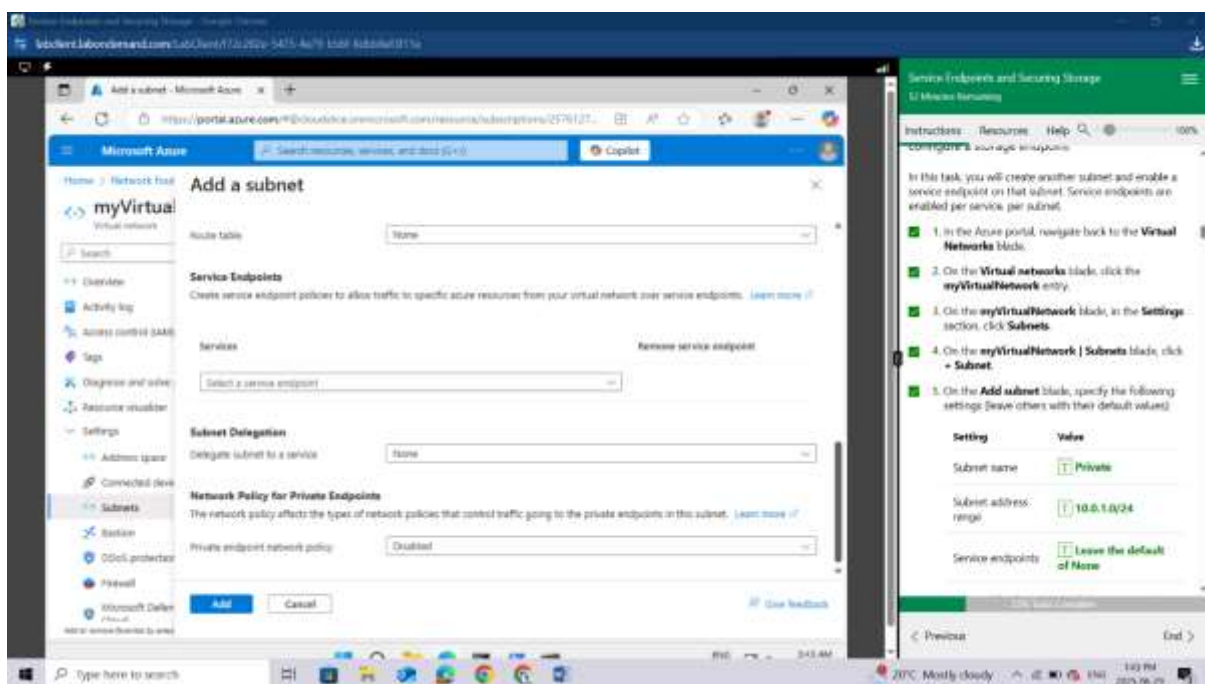
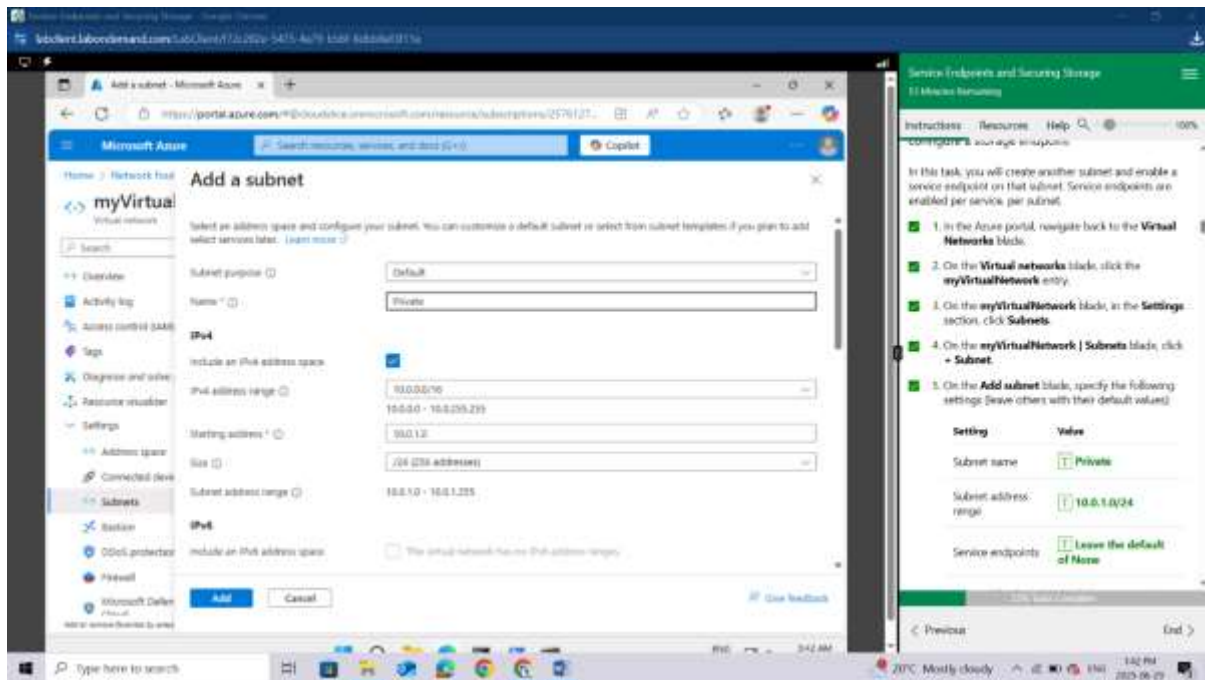
On the **myVirtualNetwork** blade, in the **Settings** section, click **Subnets**. On the **myVirtualNetwork | Subnets** blade, click + **Subnet**.



On the **Add subnet** blade, specify the following settings (leave others with their default values):

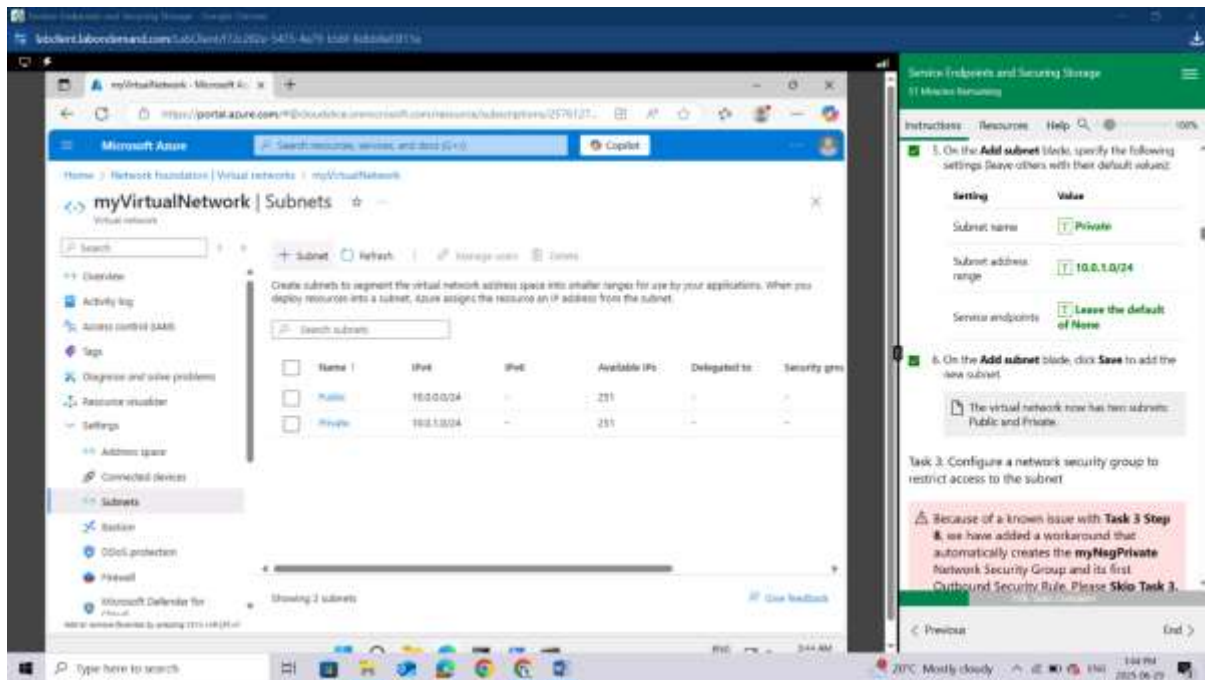
Setting	Value
Subnet name	Private
Subnet address range	10.0.1.0/24

Setting	Value
Service endpoints	Leave the default of None



On the **Add subnet** blade, click **Save** to add the new subnet.

The virtual network now has two subnets: Public and Private.



Task 3: Configure a network security group to restrict access to the subnet

Because of a known issue with Task 3 Step 8, we have added a workaround that automatically creates the myNsgPrivate Network Security Group and its first Outbound Security Rule. Please Skip Task 3, Steps 1-8 and proceed with the rest of the task starting at Step 9.

In this task, you will create a network security group with two outbound security rules (Storage and internet) and one inbound security rule (RDP). You will also associate the network security group with the Private subnet. This will restrict outbound traffic from Azure VMs connected to that subnet.

In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Network security groups** and press the **Enter** key.

On the **Network security groups** blade, click **+ Create**.

On the **Basics** tab of the **Create network security group** blade, specify the following settings:

Setting	Value
Subscription	the name of the Azure subscription you are using in this lab
Resource group	AZ500LAB12
Name	myNsgPrivate
Region	East US

Click **Review + create** and then click **Create**.

In the next steps, you will create an outbound security rule that allows communication to the Azure Storage service.

In the Azure portal, navigate back to the **Network security groups** blade and click the **myNsgPrivate** entry.

On the **myNsgPrivate** blade, in the **Settings** section, click **Outbound security rules**.

On the **myNsgPrivate | Outbound security rules** blade, click + **Add**.

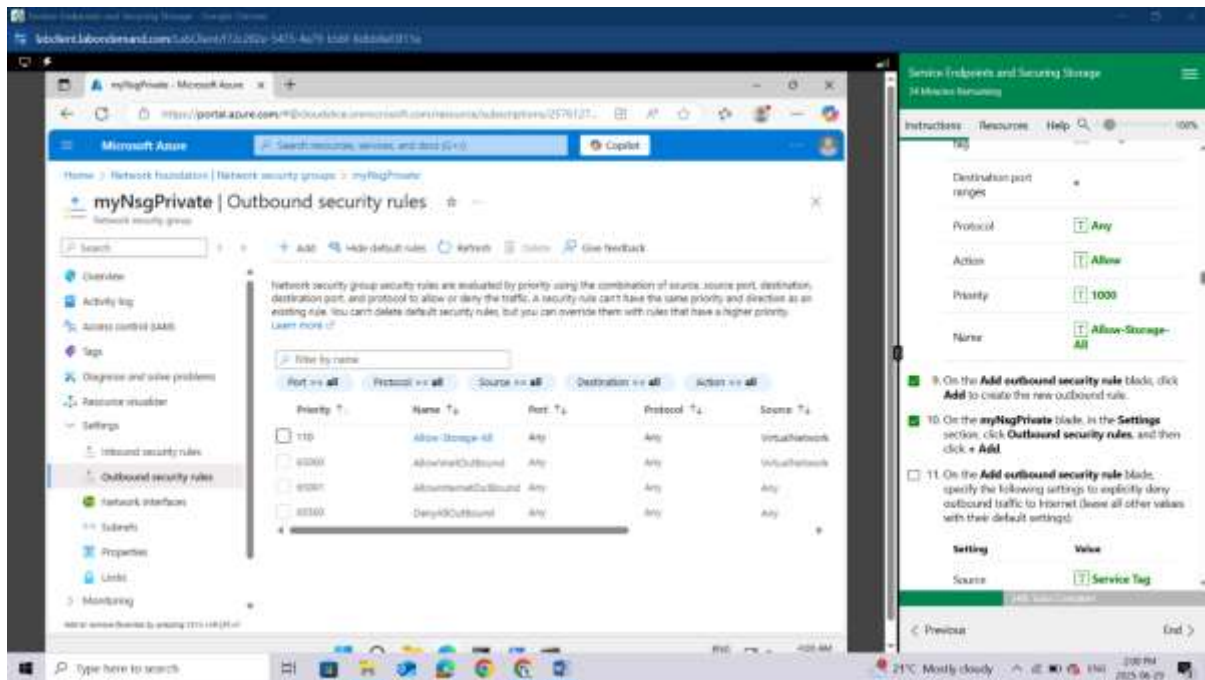
On the **Add outbound security rule** blade, specify the following settings to explicitly allow outbound traffic to Azure Storage (leave all other values with their default settings):

Setting	Value
Source	Service Tag
Source service tag	VirtualNetwork
Source port ranges	*
Destination	Service Tag
Destination service tag	Storage
Destination port ranges	*
Protocol	Any
Action	Allow
Priority	1000
Name	Allow-Storage-All

REMINDER: Please Skip Task 3, Steps 1-8 and proceed with the rest of the task starting at Step 9.

TASK 9: On the **Add outbound security rule** blade, click **Add** to create the new outbound rule.

On the **myNsgPrivate** blade, in the **Settings** section, click **Outbound security rules**, and then click + **Add**.



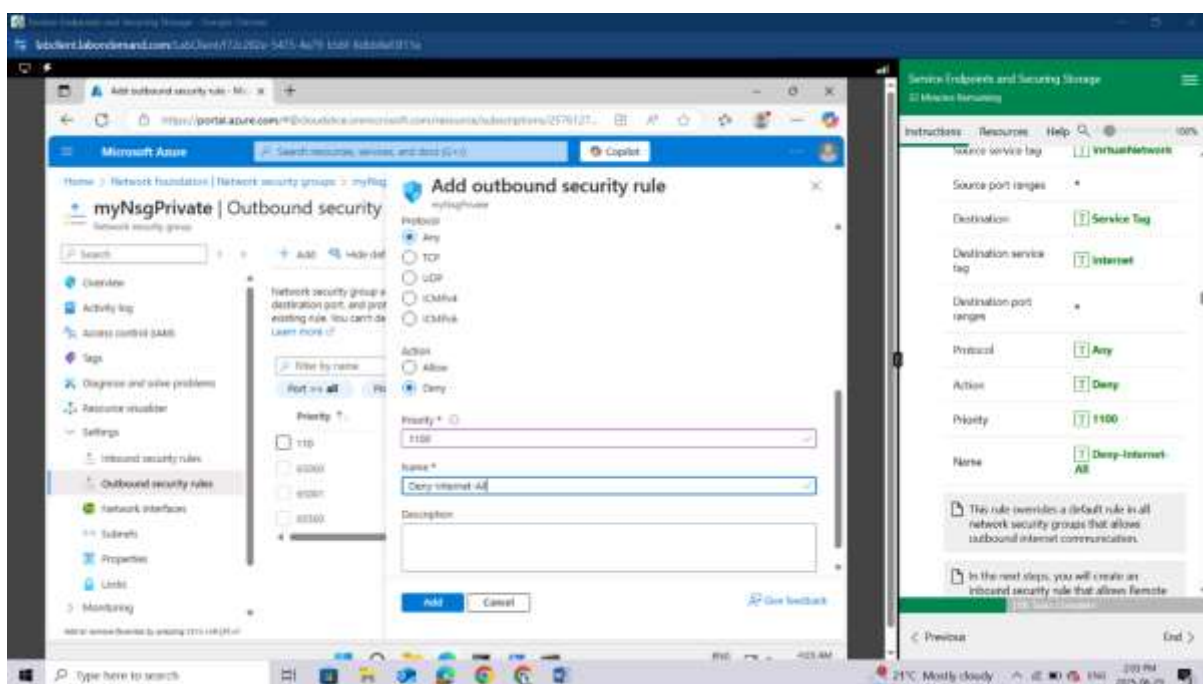
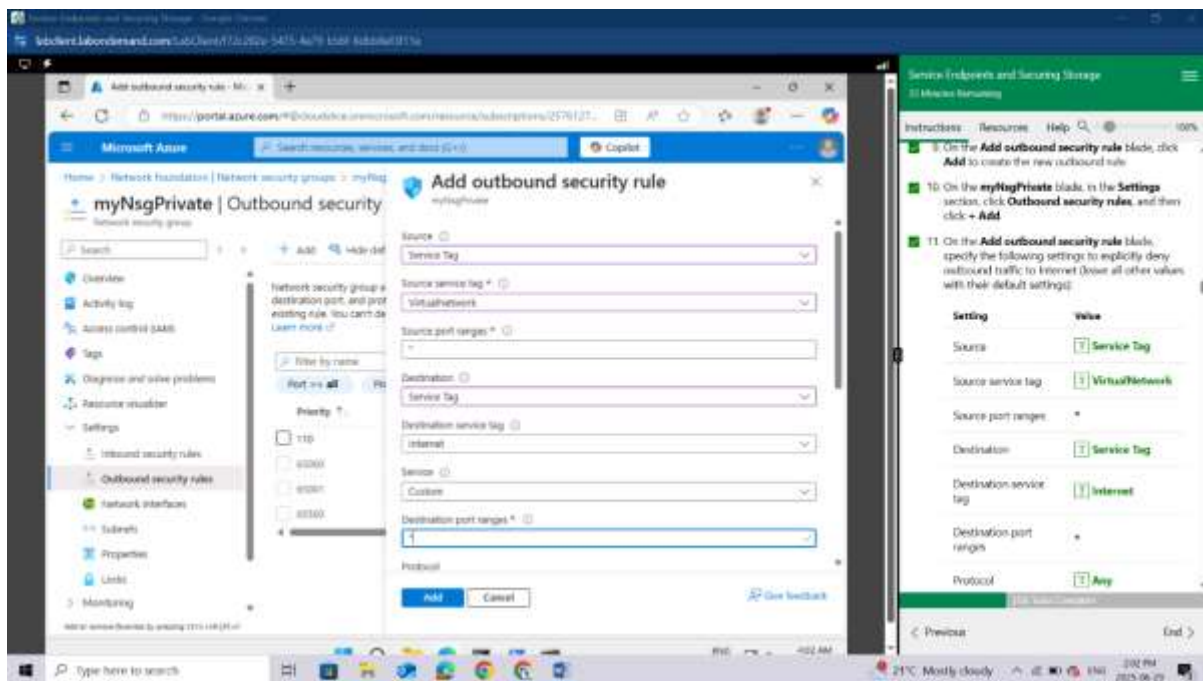
On the **Add outbound security rule** blade, specify the following settings to explicitly deny outbound traffic to Internet (leave all other values with their default settings):

Setting	Value
Source	Service Tag
Source service tag	VirtualNetwork
Source port ranges	*
Destination	Service Tag
Destination service tag	Internet
Destination port ranges	*
Protocol	Any
Action	Deny
Priority	1100

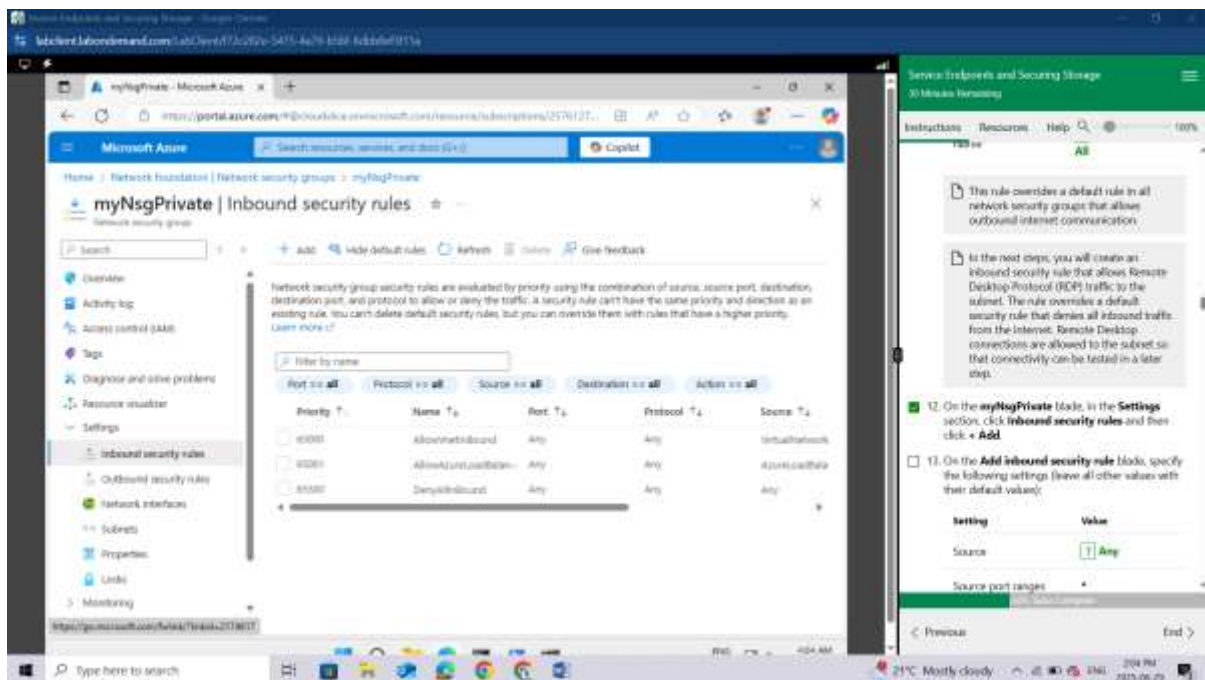
Setting	Value
Name	Deny-Internet-All

This rule overrides a default rule in all network security groups that allows outbound internet communication.

In the next steps, you will create an inbound security rule that allows Remote Desktop Protocol (RDP) traffic to the subnet. The rule overrides a default security rule that denies all inbound traffic from the internet. Remote Desktop connections are allowed to the subnet so that connectivity can be tested in a later step.



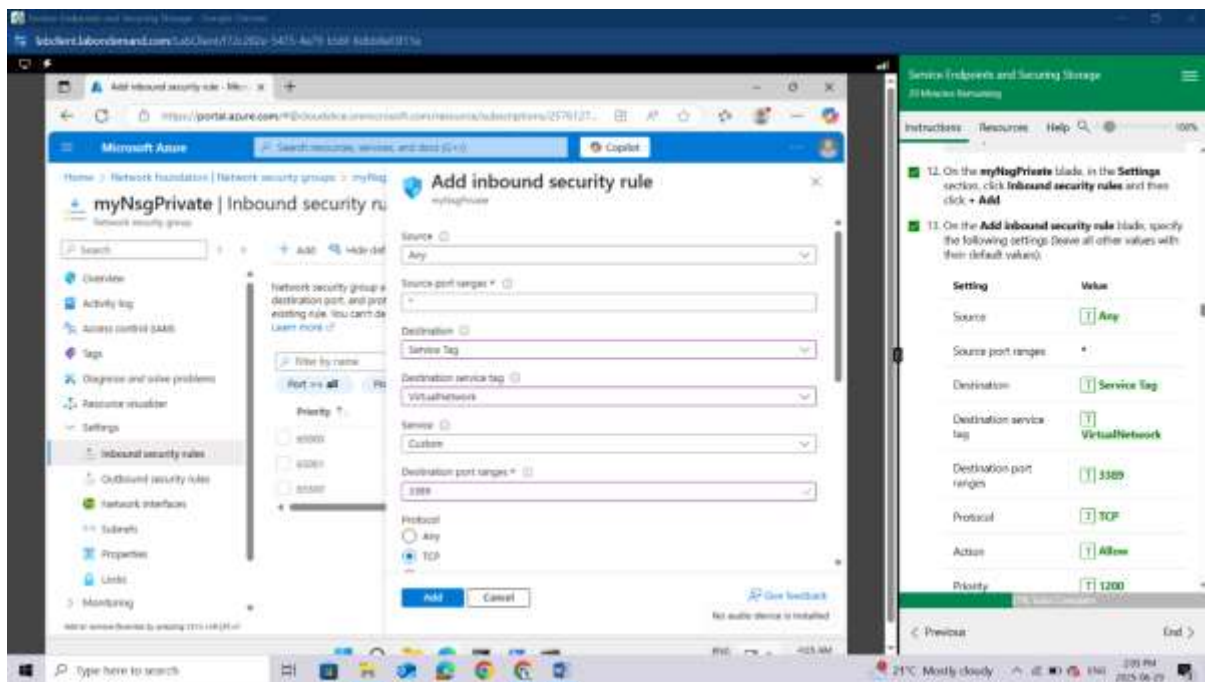
On the **myNsgPrivate** blade, in the **Settings** section, click **Inbound security rules** and then click **+ Add**.



On the **Add inbound security rule** blade, specify the following settings (leave all other values with their default values):

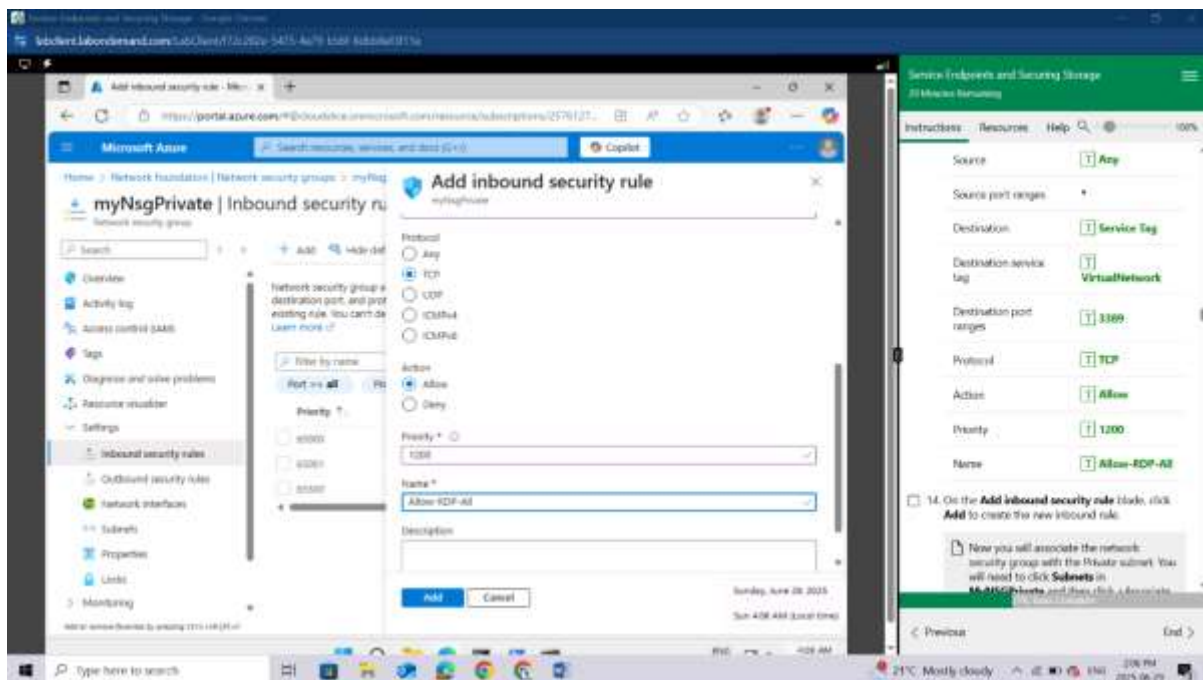
Setting	Value
Source	Any
Source port ranges	*
Destination	Service Tag
Destination service tag	VirtualNetwork
Destination port ranges	3389
Protocol	TCP
Action	Allow

Setting	Value
Priority	1200
Name	Allow-RDP-All



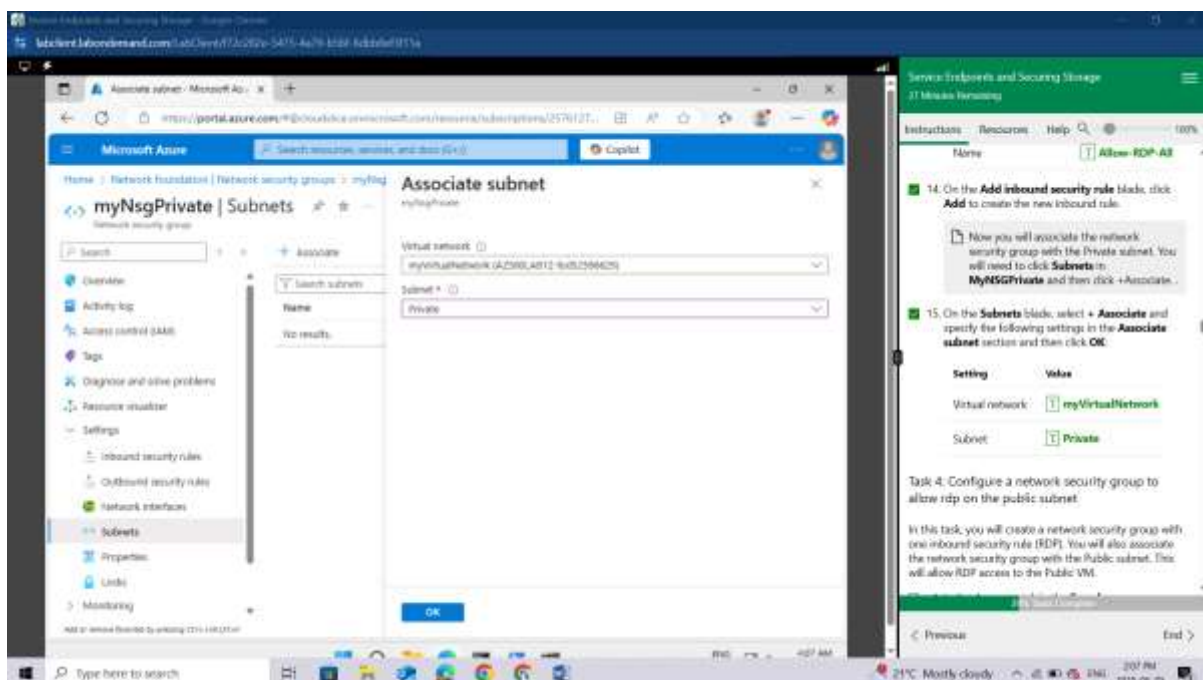
On the **Add inbound security rule** blade, click **Add** to create the new inbound rule.

Now you will associate the network security group with the Private subnet. You will need to click **Subnets** in **MyNSGPrivate** and then click **+Associate**.



On the **Subnets** blade, select **+ Associate** and specify the following settings in the **Associate subnet** section and then click **OK**:

Setting	Value
Virtual network	myVirtualNetwork
Subnet	Private

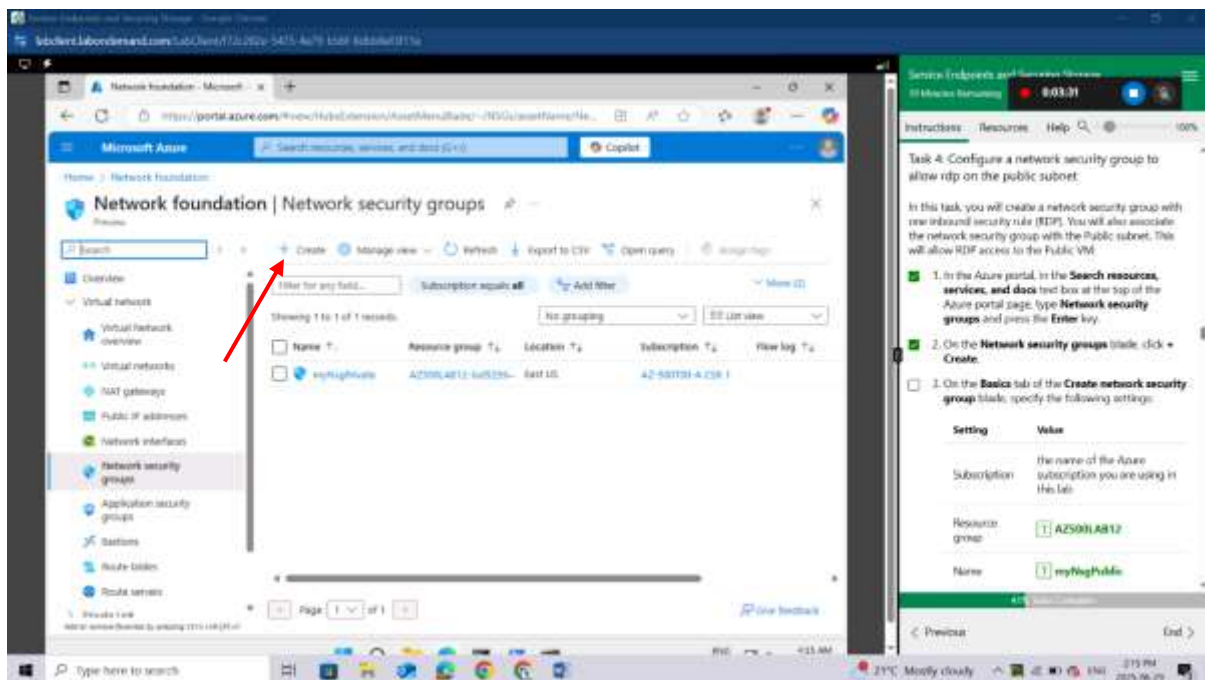


Task 4: Configure a network security group to allow rdp on the public subnet

In this task, you will create a network security group with one inbound security rule (RDP). You will also associate the network security group with the Public subnet. This will allow RDP access to the Public VM.

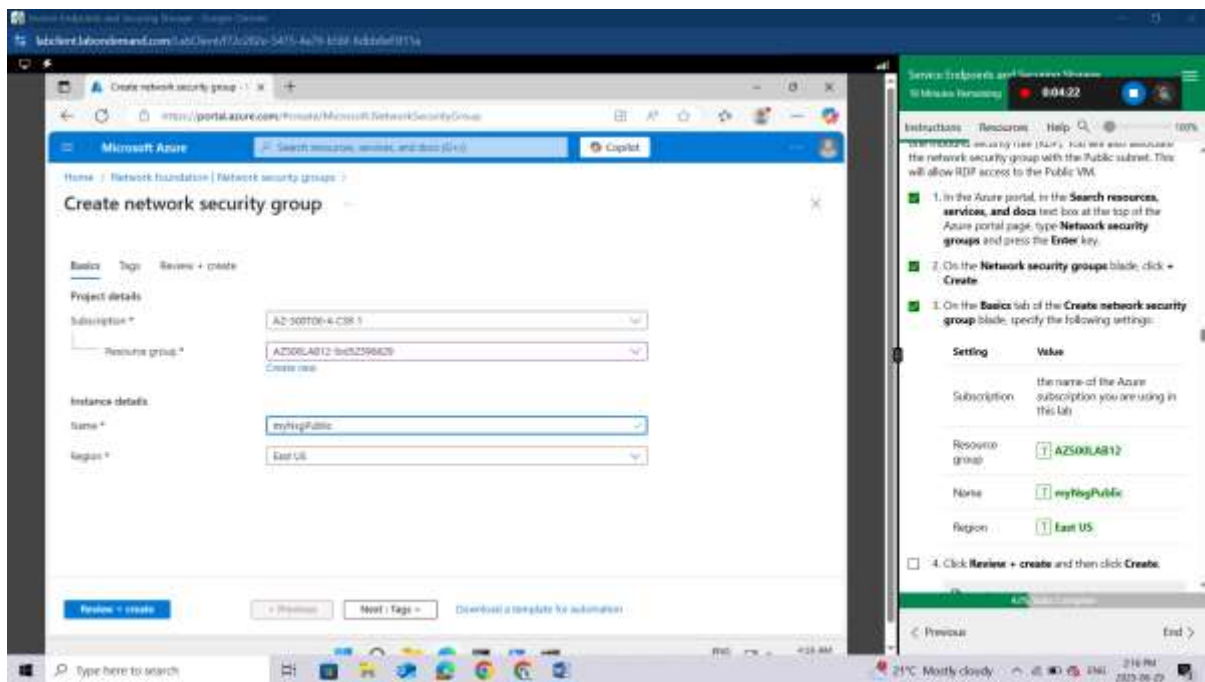
In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Network security groups** and press the **Enter** key.

On the **Network security groups** blade, click **+ Create**.



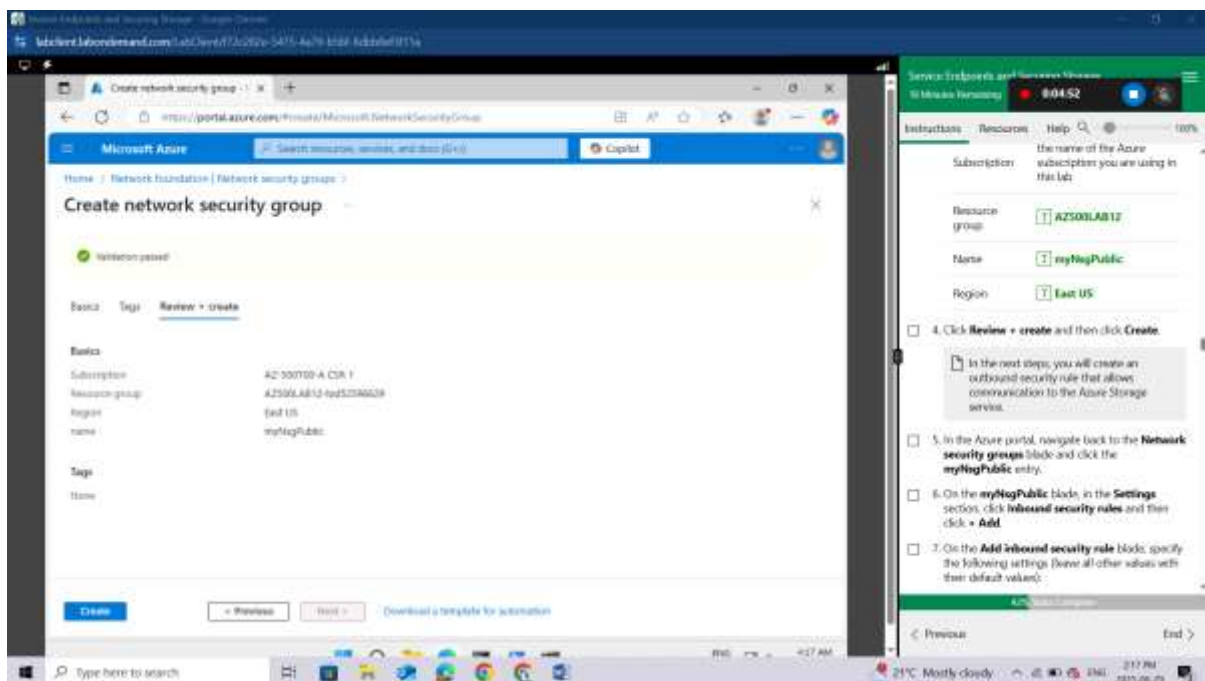
On the **Basics** tab of the **Create network security group** blade, specify the following settings:

Setting	Value
Subscription	the name of the Azure subscription you are using in this lab
Resource group	AZ500LAB12
Name	myNsgPublic
Region	East US

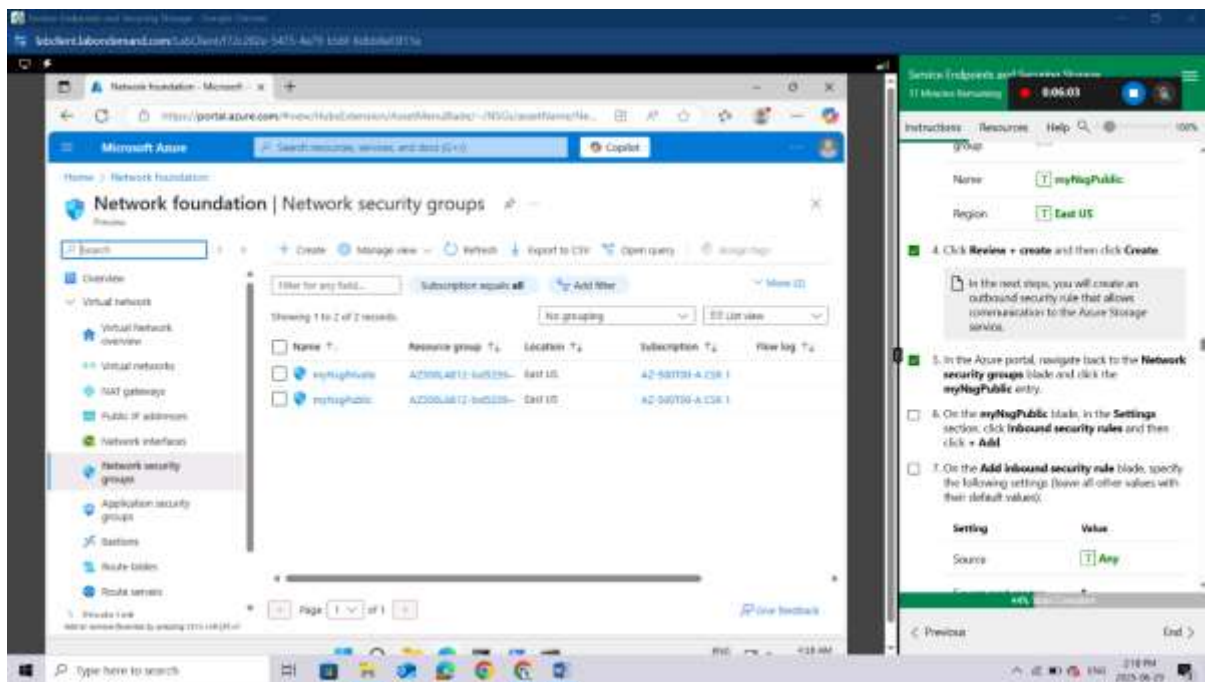


Click **Review + create** and then click **Create**.

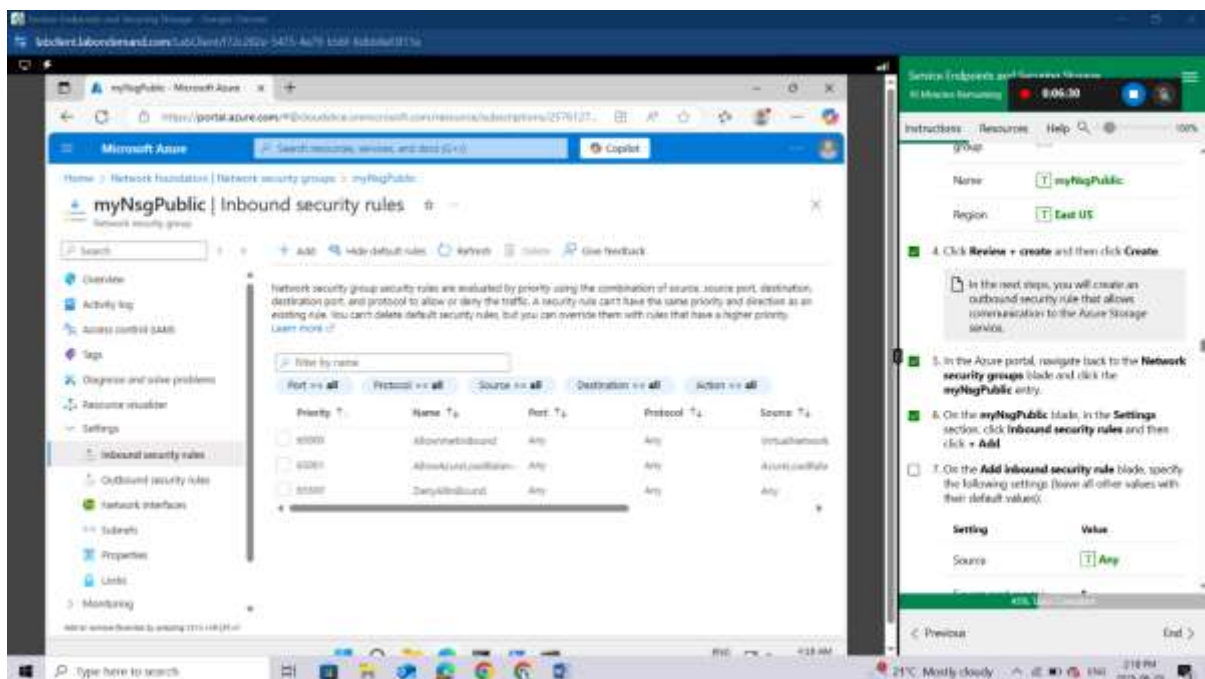
In the next steps, you will create an outbound security rule that allows communication to the Azure Storage service.



In the Azure portal, navigate back to the **Network security groups** blade and click the **myNsgPublic** entry.



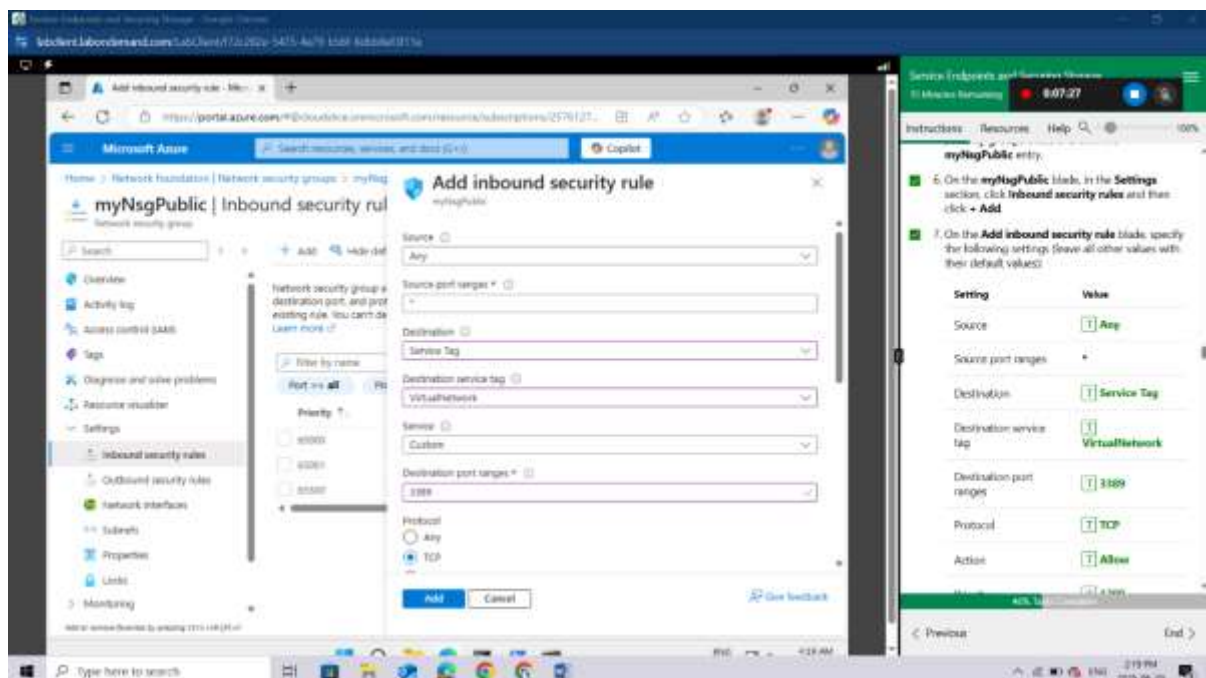
On the **myNsGPublic** blade, in the **Settings** section, click **Inbound security rules** and then click **+ Add**.



On the **Add inbound security rule** blade, specify the following settings (leave all other values with their default values):

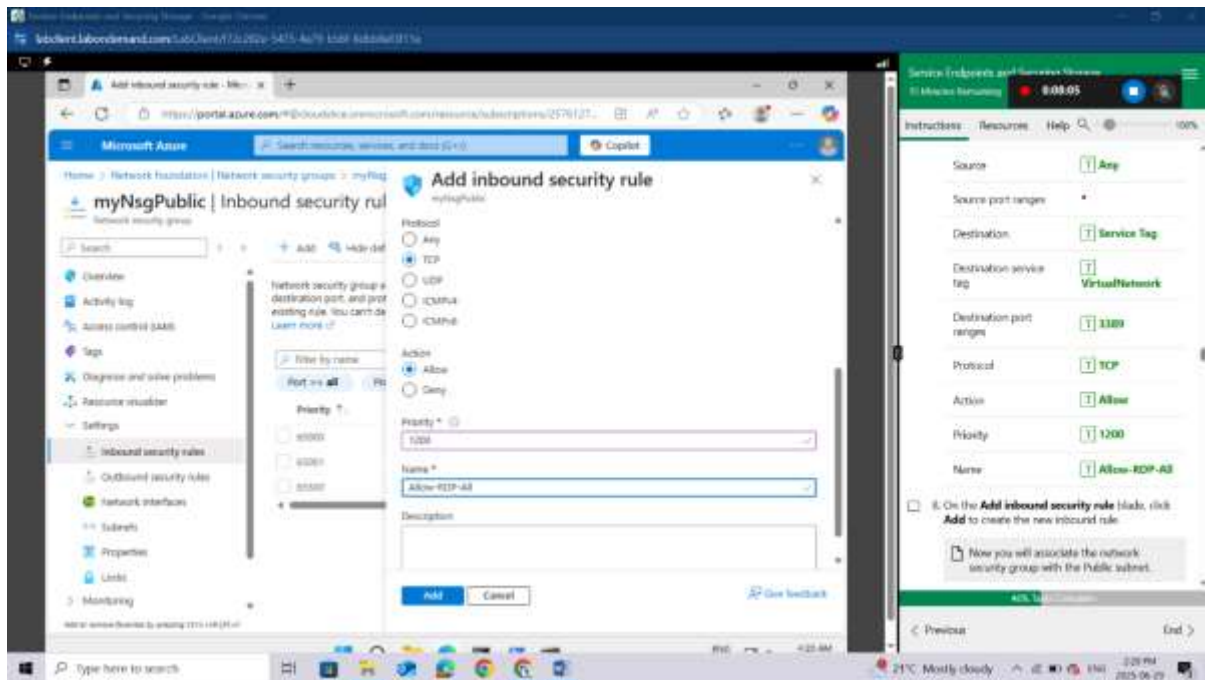
Setting	Value
Source	Any

Setting	Value
Source port ranges	*
Destination	Service Tag
Destination service tag	VirtualNetwork
Destination port ranges	3389
Protocol	TCP
Action	Allow
Priority	1200
Name	Allow-RDP-All



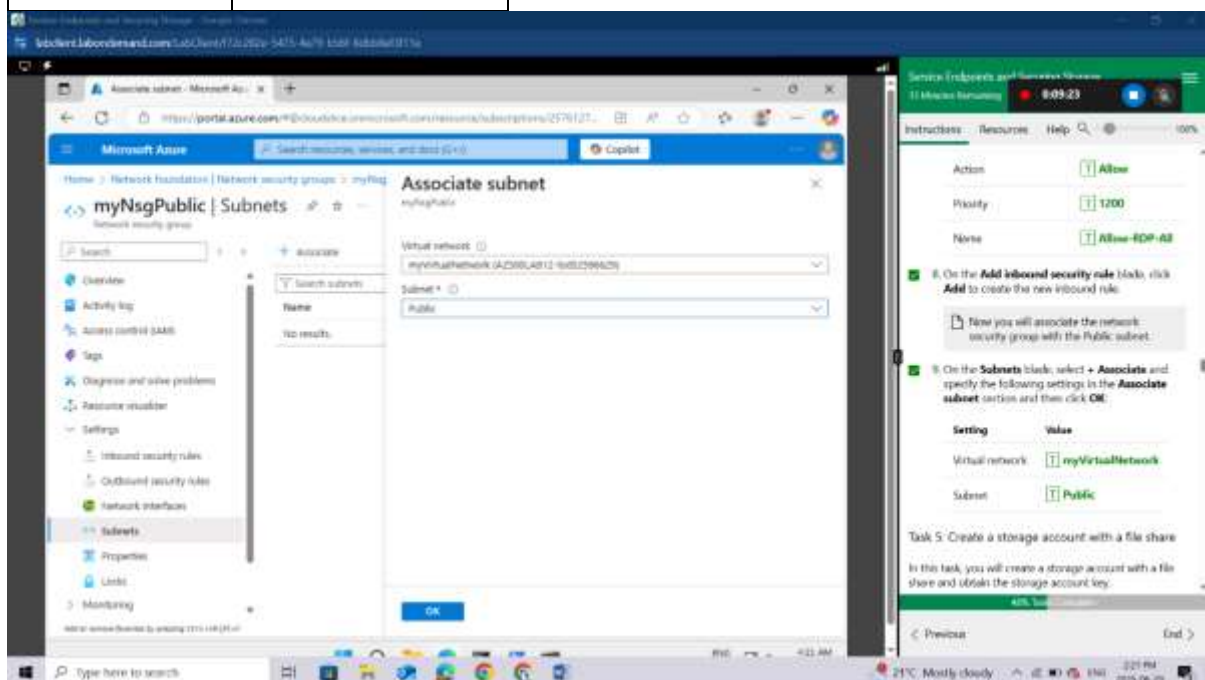
On the **Add inbound security rule** blade, click **Add** to create the new inbound rule.

Now you will associate the network security group with the Public subnet.



On the **Subnets** blade, select + **Associate** and specify the following settings in the **Associate subnet** section and then click **OK**:

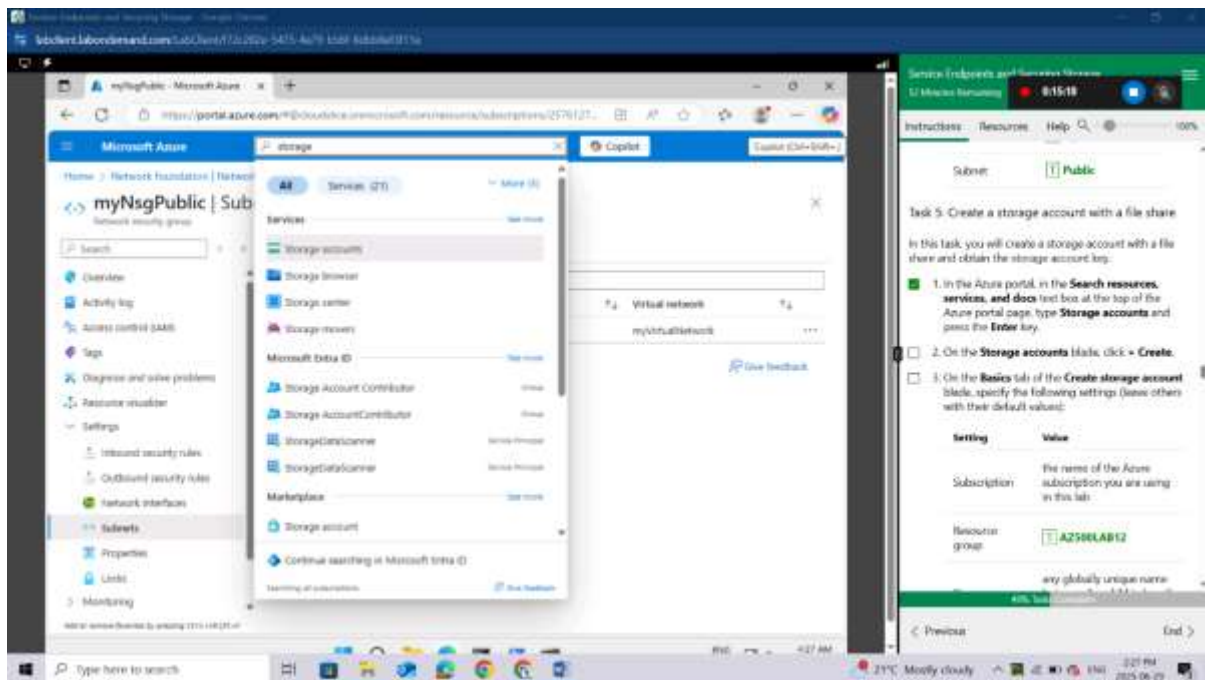
Setting	Value
Virtual network	myVirtualNetwork
Subnet	Public



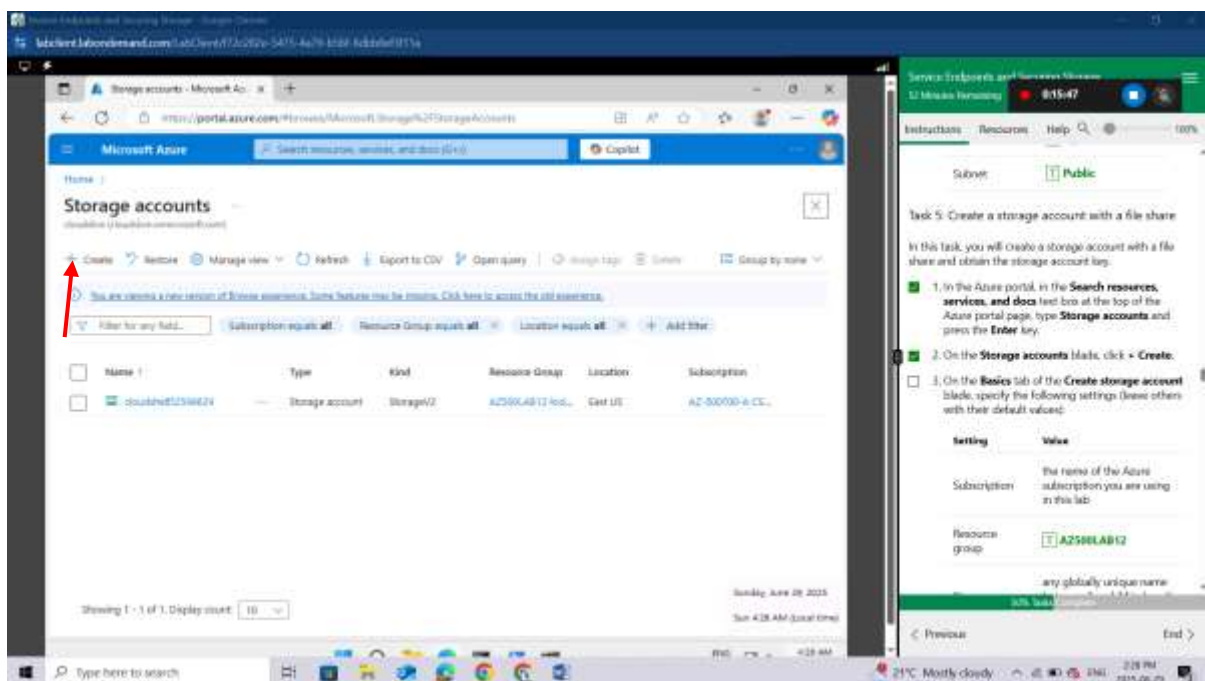
Task 5: Create a storage account with a file share

In this task, you will create a storage account with a file share and obtain the storage account key.

In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Storage accounts** and press the **Enter** key.

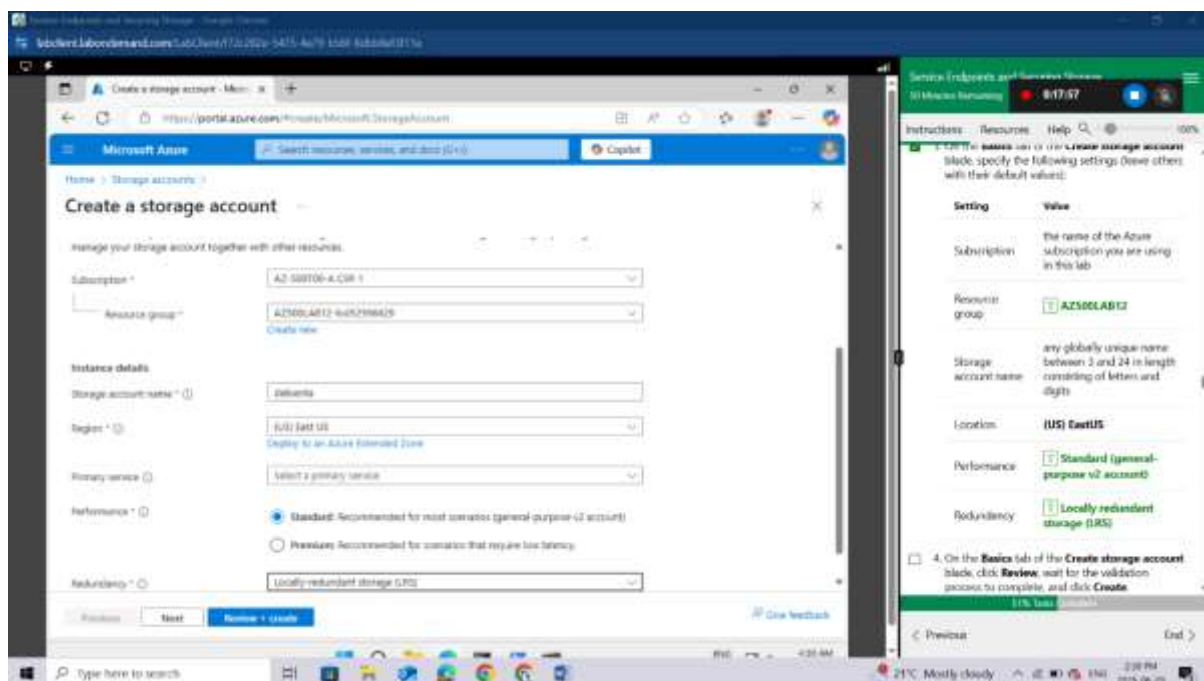


On the **Storage accounts** blade, click **+ Create**.



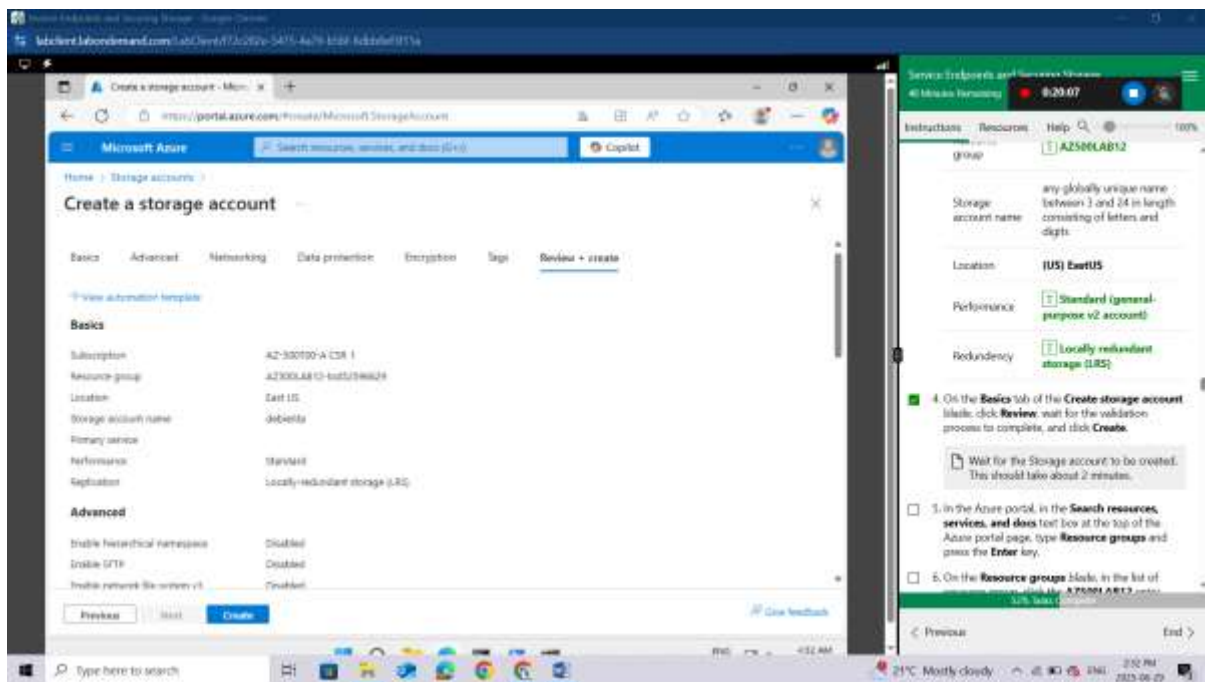
On the **Basics** tab of the **Create storage account** blade, specify the following settings (leave others with their default values):

Setting	Value
Subscription	the name of the Azure subscription you are using in this lab
Resource group	AZ500LAB12
Storage account name	any globally unique name between 3 and 24 in length consisting of letters and digits
Location	(US) EastUS
Performance	Standard (general-purpose v2 account)
Redundency	Locally redundant storage (LRS)

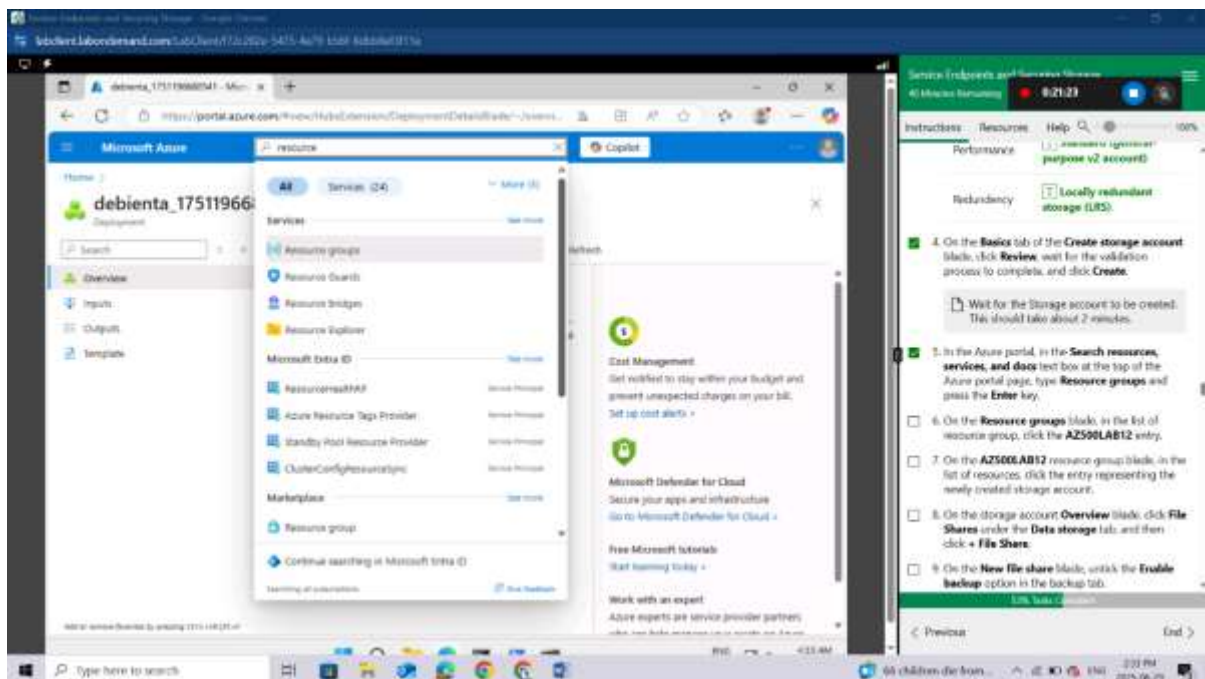


On the **Basics** tab of the **Create storage account** blade, click **Review**, wait for the validation process to complete, and click **Create**.

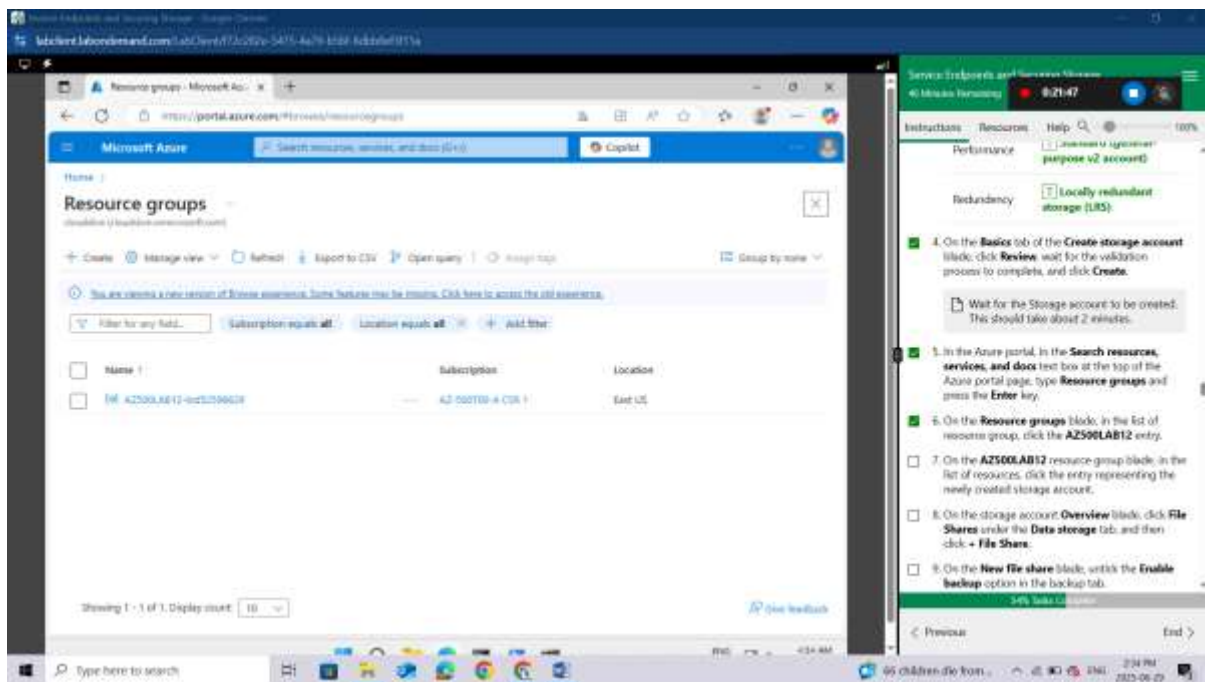
Wait for the Storage account to be created. This should take about 2 minutes.



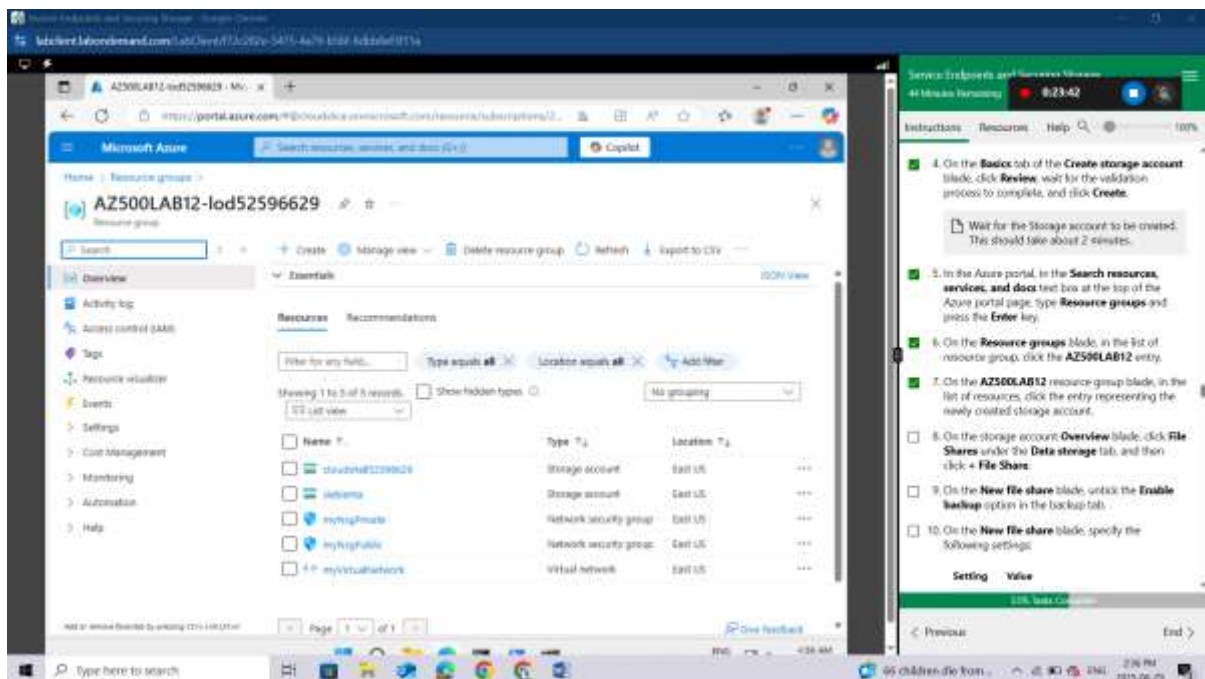
In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Resource groups** and press the **Enter** key.



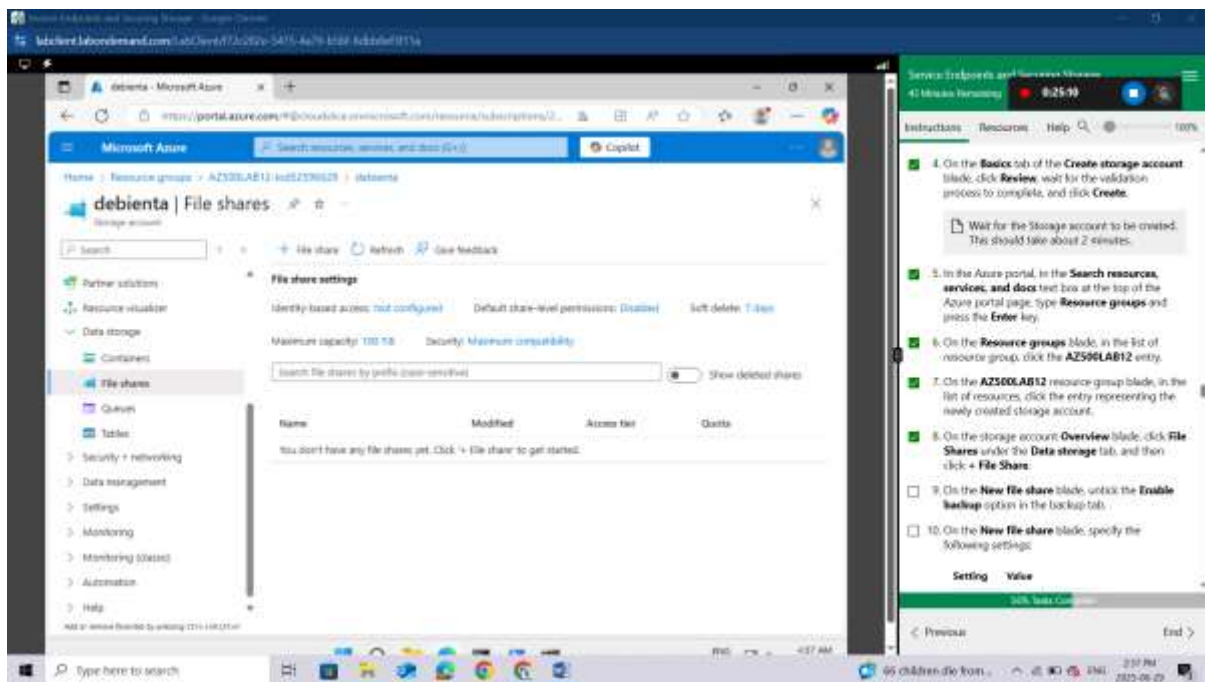
On the **Resource groups** blade, in the list of resource group, click the **AZ500LAB12** entry.



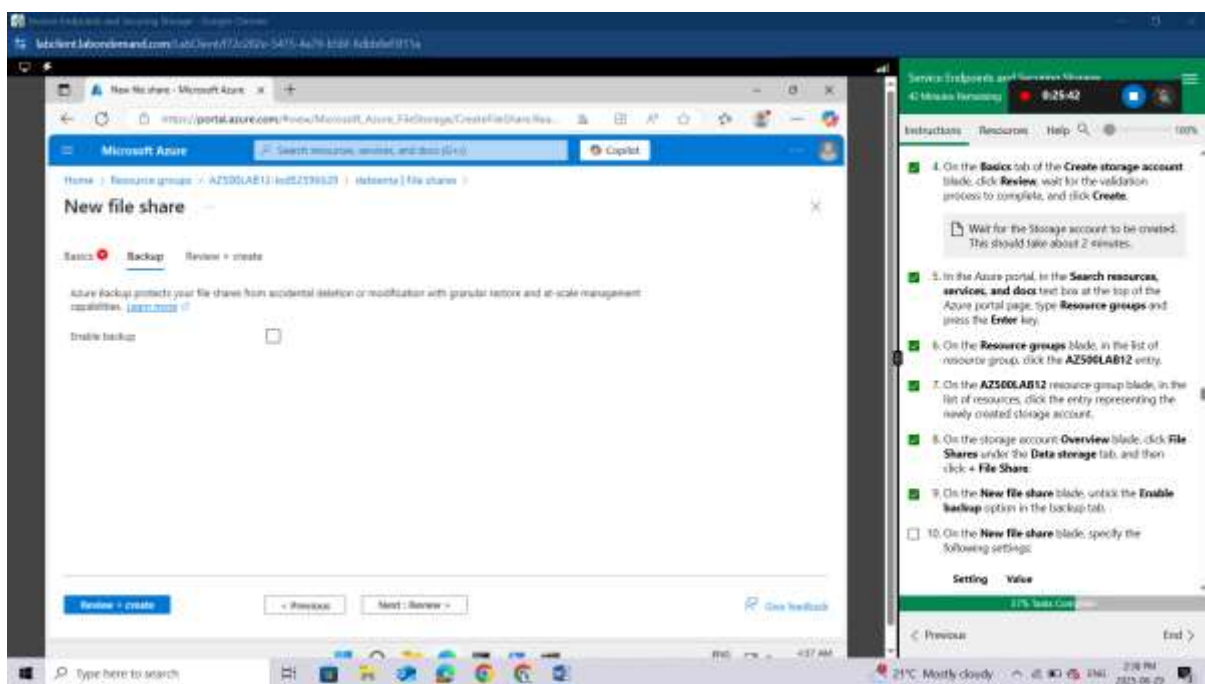
On the **AZ500LAB12** resource group blade, in the list of resources, click the entry representing the newly created storage account.



On the storage account **Overview** blade, click **File Shares** under the **Data storage** tab, and then click + **File Share**.

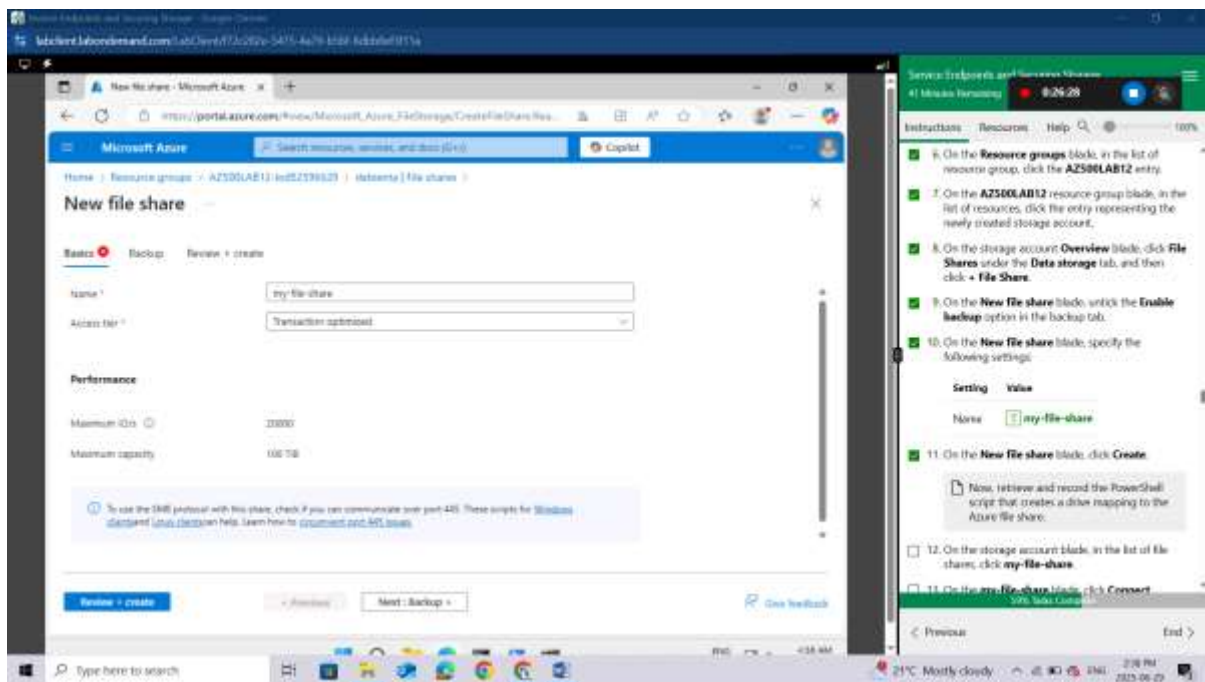


On the **New file share** blade, untick the **Enable backup** option in the backup tab.



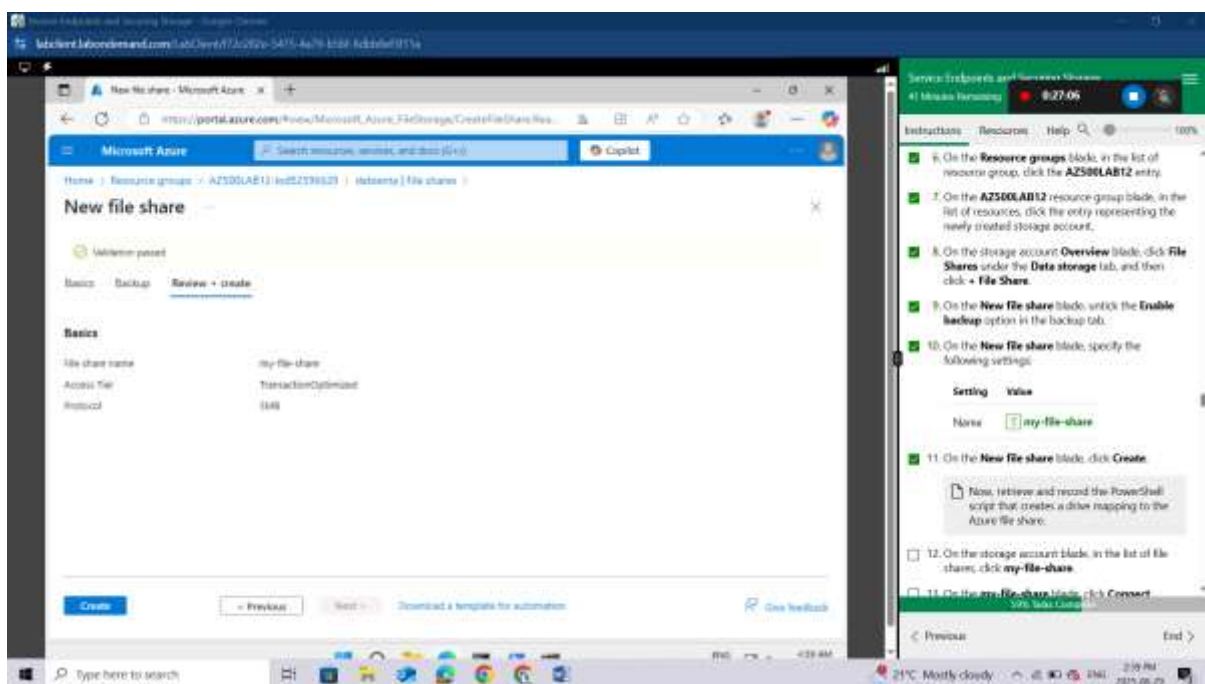
On the **New file share** blade, specify the following settings:

Setting	Value
Name	my-file-share

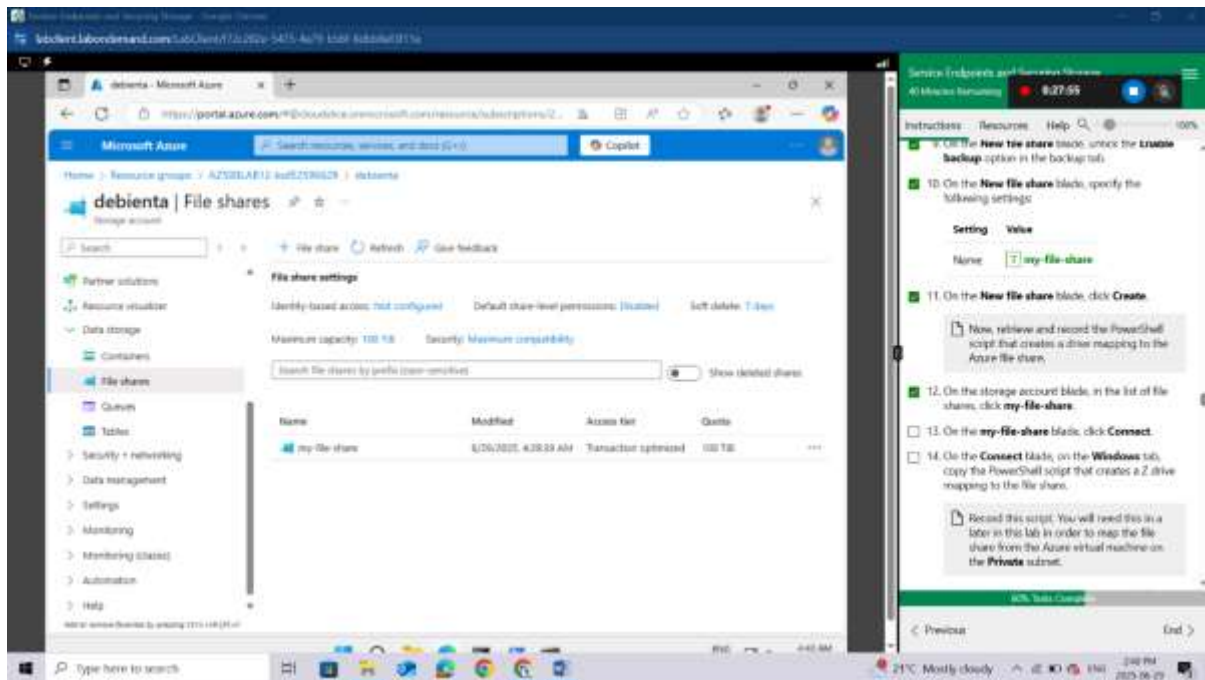


On the **New file share** blade, click **Create**.

Now, retrieve and record the PowerShell script that creates a drive mapping to the Azure file share.



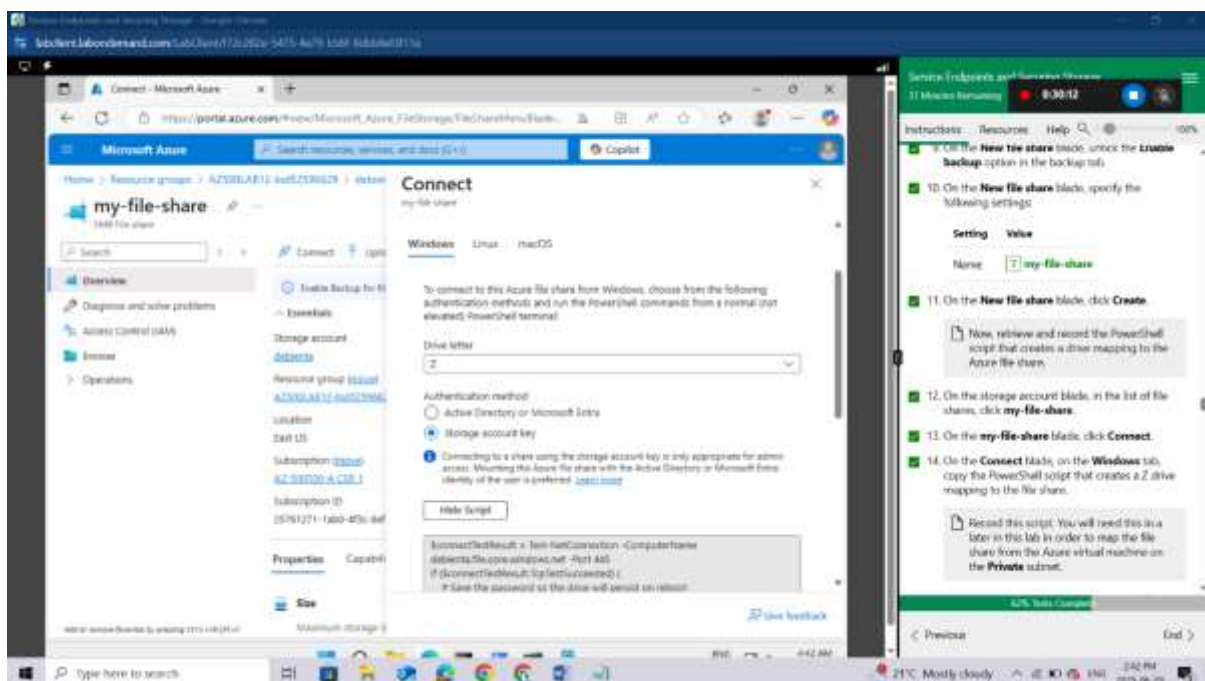
On the storage account blade, in the list of file shares, click **my-file-share**.



On the **my-file-share** blade, click **Connect**.

On the **Connect** blade, on the **Windows** tab, copy the PowerShell script that creates a Z drive mapping to the file share.

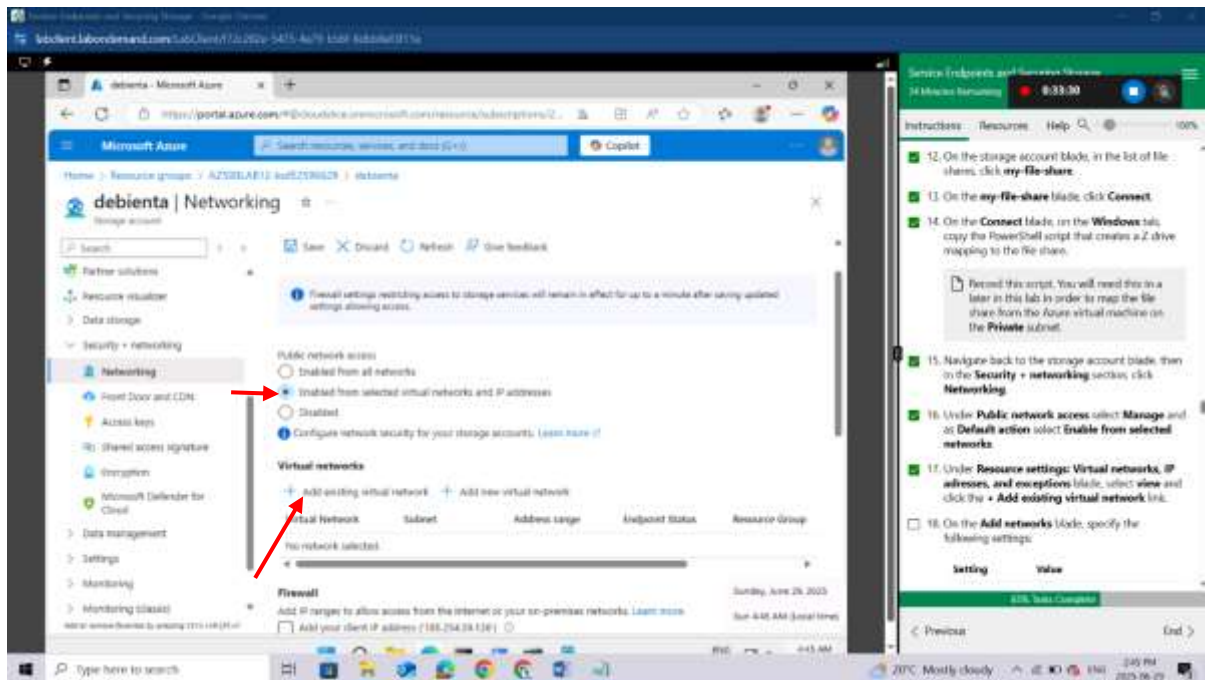
*Record this script. You will need this in a later in this lab in order to map the file share from the Azure virtual machine on the **Private** subnet.*



Navigate back to the storage account blade, then in the **Security + networking** section, click **Networking**.

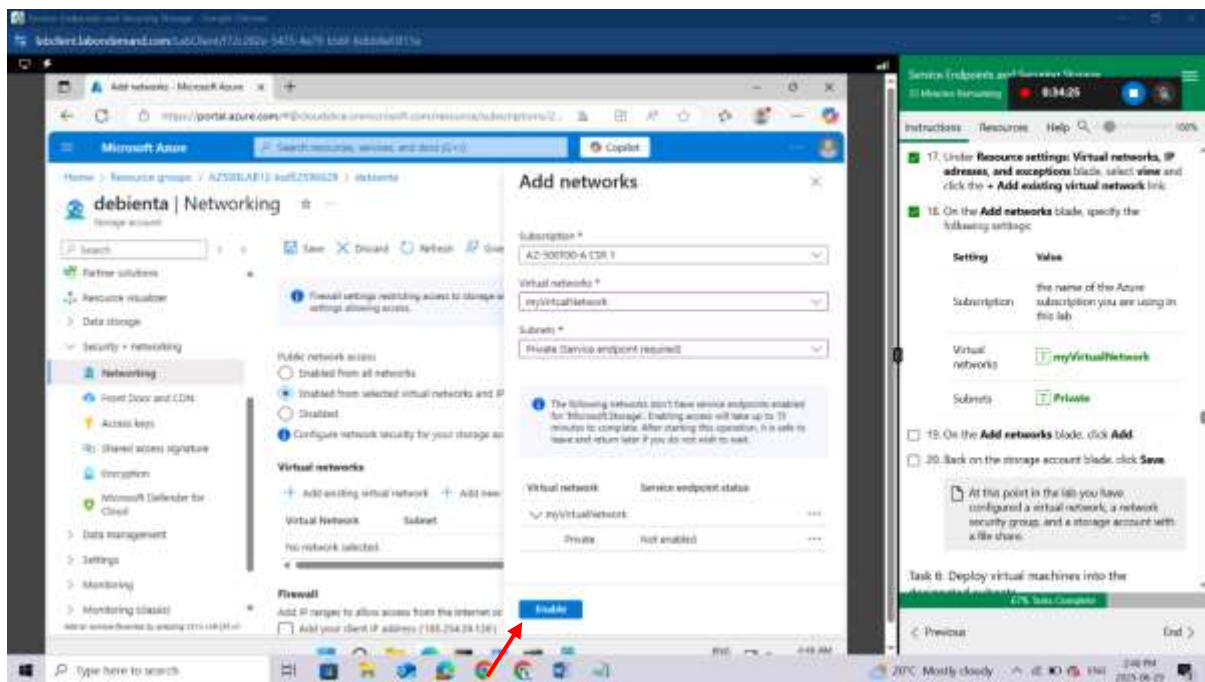
Under **Public network access** select **Manage** and as **Default action** select **Enable from selected networks**.

Under **Resource settings: Virtual networks, IP addresses, and exceptions** blade, select **view** and click the + **Add existing virtual network** link.

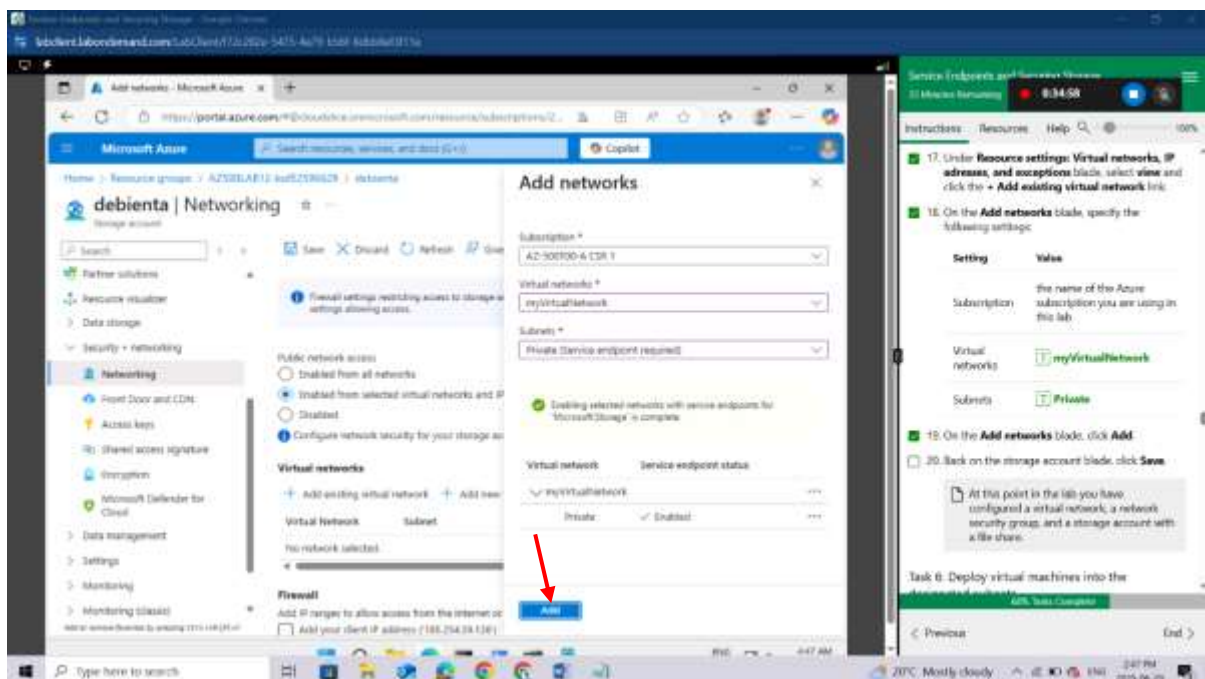


On the **Add networks** blade, specify the following settings:

Setting	Value
Subscription	the name of the Azure subscription you are using in this lab
Virtual networks	myVirtualNetwork
Subnets	Private

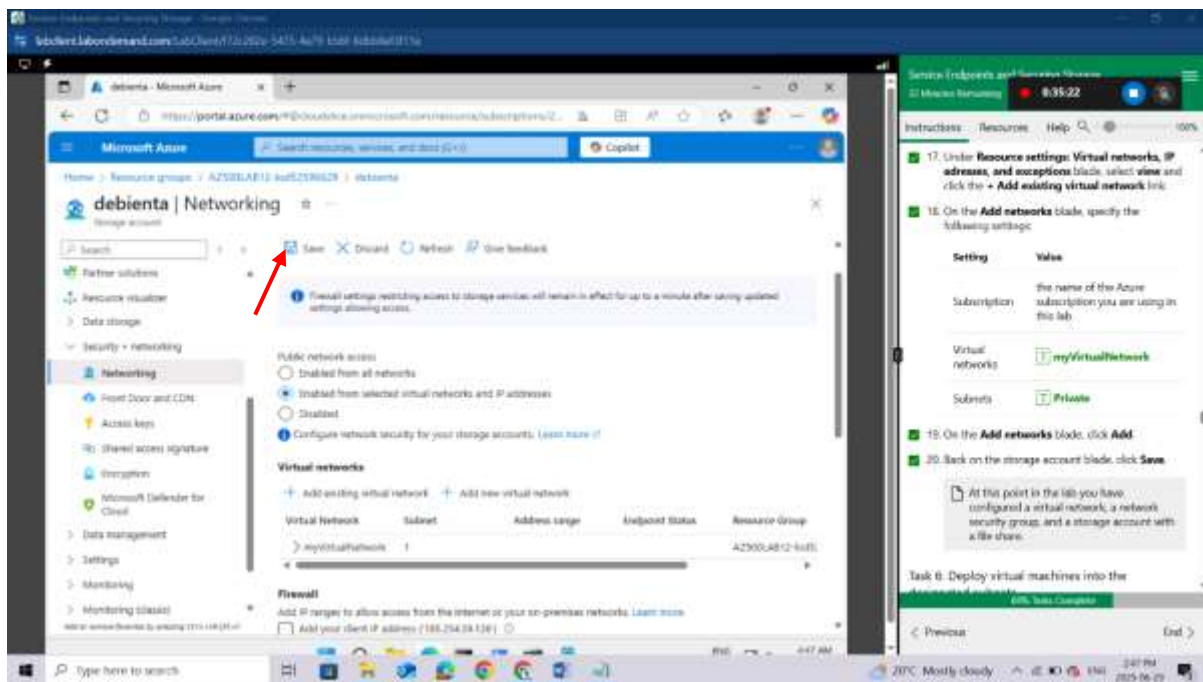


On the **Add networks** blade, click **Add**.



Back on the storage account blade, click **Save**.

At this point in the lab you have configured a virtual network, a network security group, and a storage account with a file share.

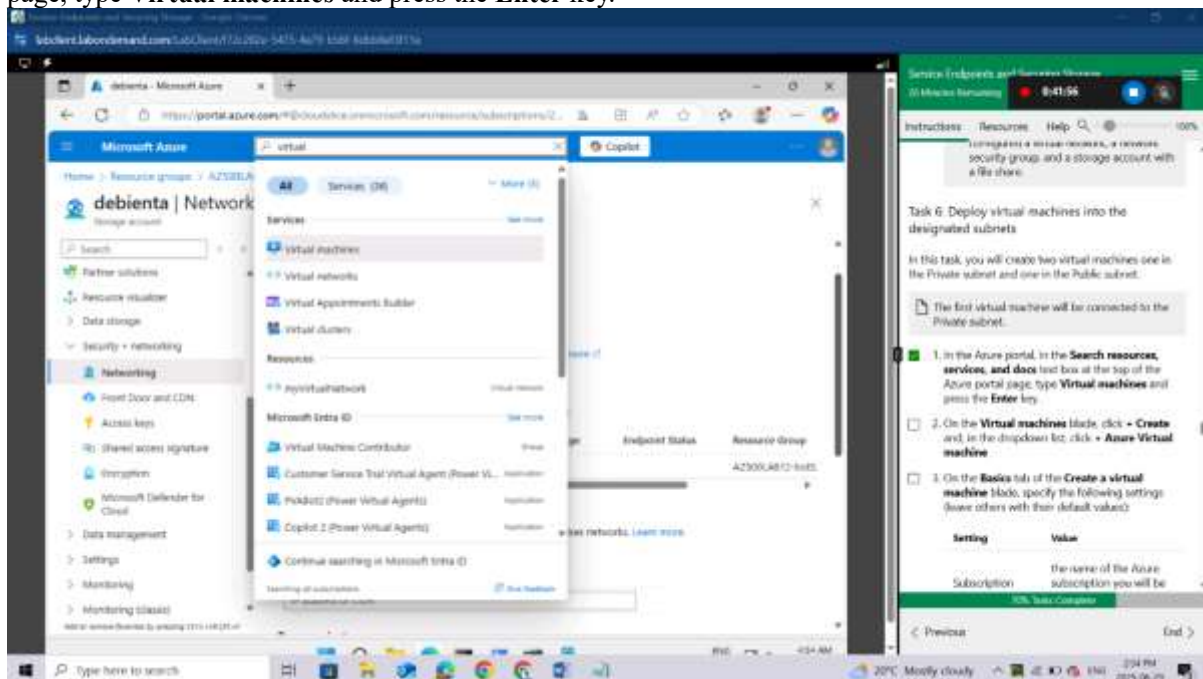


Task 6: Deploy virtual machines into the designated subnets

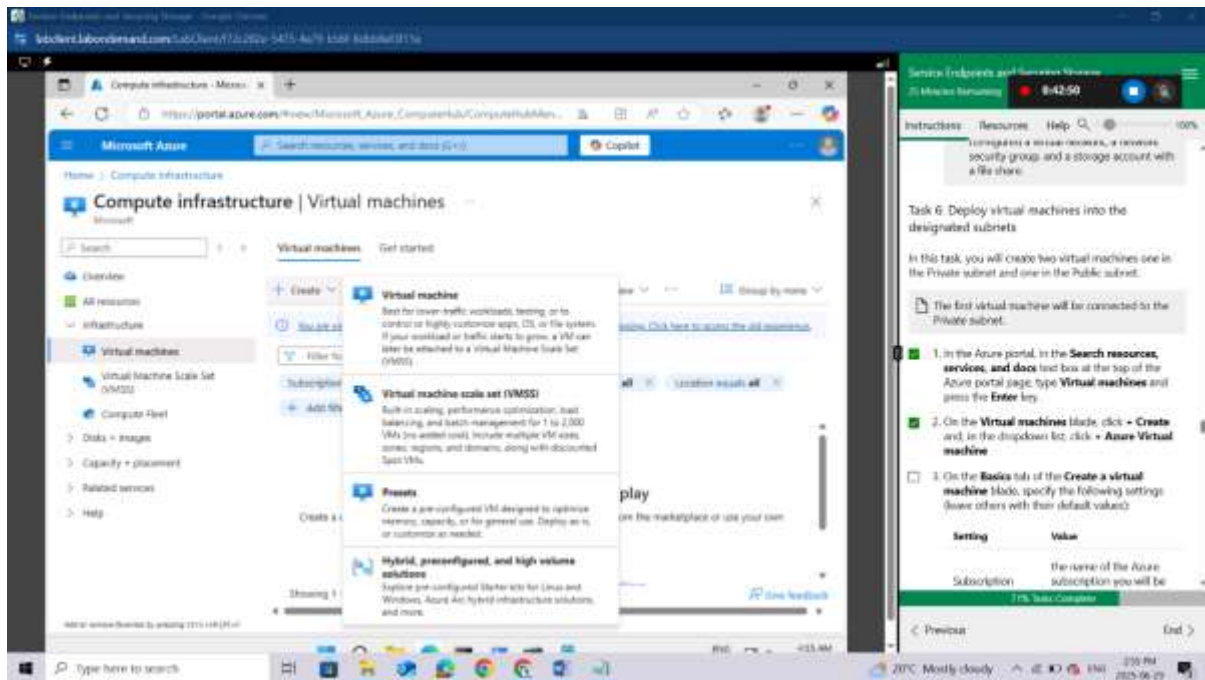
In this task, you will create two virtual machines one in the Private subnet and one in the Public subnet.

The first virtual machine will be connected to the Private subnet.

In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Virtual machines** and press the **Enter** key.

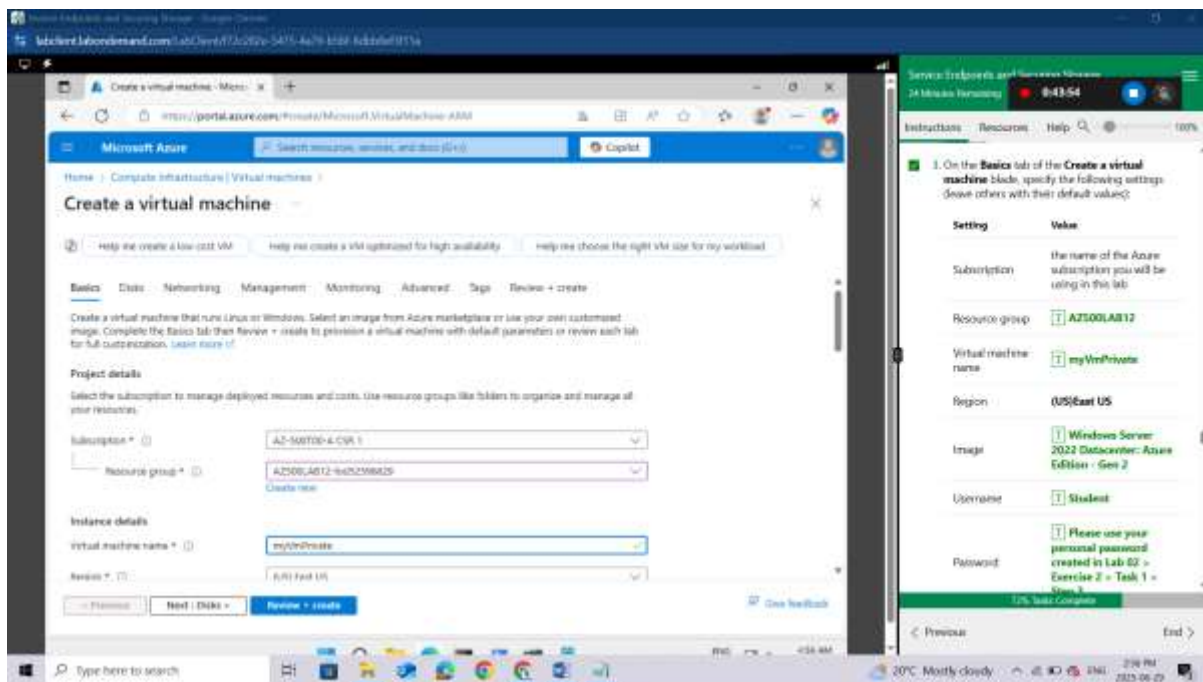


On the **Virtual machines** blade, click **+ Create** and, in the dropdown list, click **+ Azure Virtual machine**



On the **Basics** tab of the **Create a virtual machine** blade, specify the following settings (leave others with their default values):

Setting	Value
Subscription	the name of the Azure subscription you will be using in this lab
Resource group	AZ500LAB12
Virtual machine name	myVmPrivate
Region	(US)East US
Image	Windows Server 2022 Datacenter: Azure Edition - Gen 2
Username	Student
Password	Please use your personal password created in Lab 02 > Exercise 2 > Task 1 > Step 3.
Public inbound ports	None
Already have a Windows Server license	Not selected



Microsoft Azure | Search resources, services, and docs (Go)

Create a virtual machine

Help me create a low-cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Basics | Disks | Networking | Management | Monitoring | Advanced | Tags | Review + create

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription: AZ5000-4-CPU-1
Resource group: AZ5000LAB12-40206825 [Create new](#)

Instance details

Virtual machine name: myVmPrivate
Region: (US)East US
OS: Windows Server 2022 Datacenter: Azure Edition - Gen 2

Next | Disks | Review + create

Service Endpoints and Resource Monitor
24 Minutes Remaining | 8:43:54

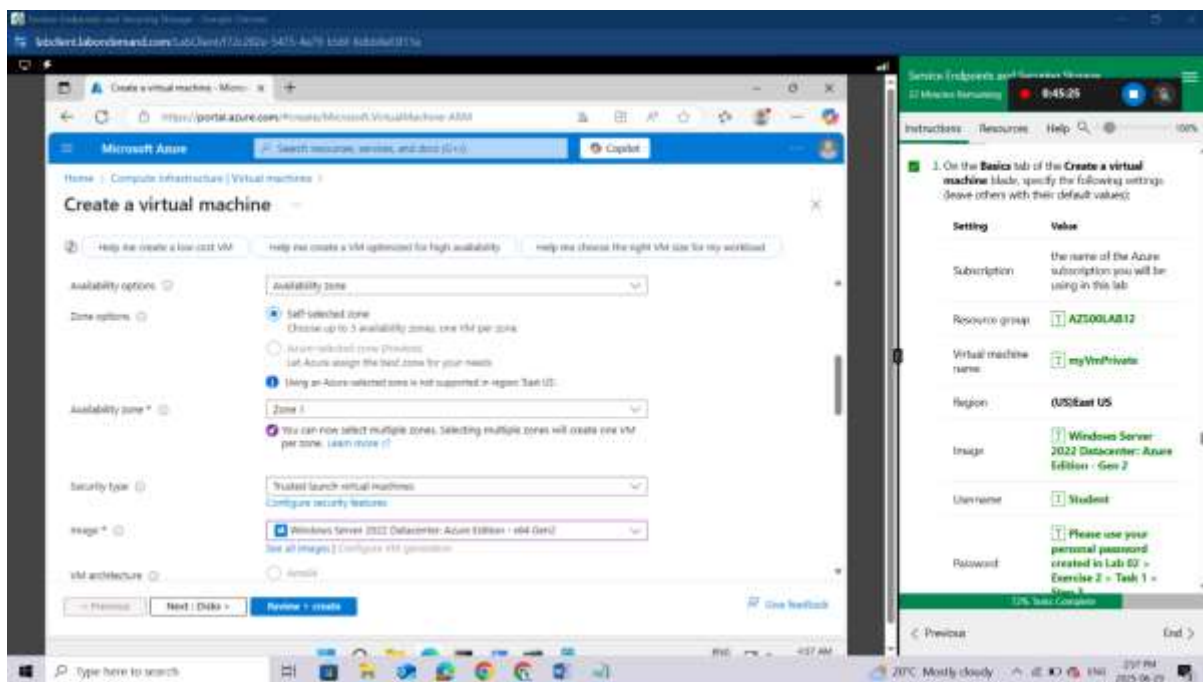
Instructions | Resources | Help | 100%

1. On the **Basics** tab of the **Create a virtual machine** blade, specify the following settings (leave others with their default values):

Setting	Value
Subscription	the name of the Azure subscription you will be using in this lab
Resource group	AZ5000LAB12
Virtual machine name	myVmPrivate
Region	(US)East US
Image	Windows Server 2022 Datacenter: Azure Edition - Gen 2
Username	Student
Password	Please use your personal password created in Lab 02 - Exercise 2 - Task 1 - Step 3.

10% Task Complete

Previous | End



Microsoft Azure | Search resources, services, and docs (Go)

Create a virtual machine

Help me create a low-cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Basics | Disks | Networking | Management | Monitoring | Advanced | Tags | Review + create

Availability options

Availability zone

Zone options

☒ Self-selected zone
Choose up to 3 availability zones, one VM per zone.

☐ Azure-selected zone (Preview)
Let Azure assign the best zone for your needs.

Using an Azure-selected zone is not supported in region East US.

Availability zone

Zone 1

You can now select multiple zones. Selecting multiple zones will create one VM per zone. [Learn more](#)

Security type

Trusted launch virtual machines
[Configure security features](#)

Image

Windows Server 2022 Datacenter: Azure Edition - v14 Gen2
[See all images](#) | [Configure VM generation](#)

VM architecture

☐ Arm64

Next | Disks | Review + create

Service Endpoints and Resource Monitor
24 Minutes Remaining | 8:45:25

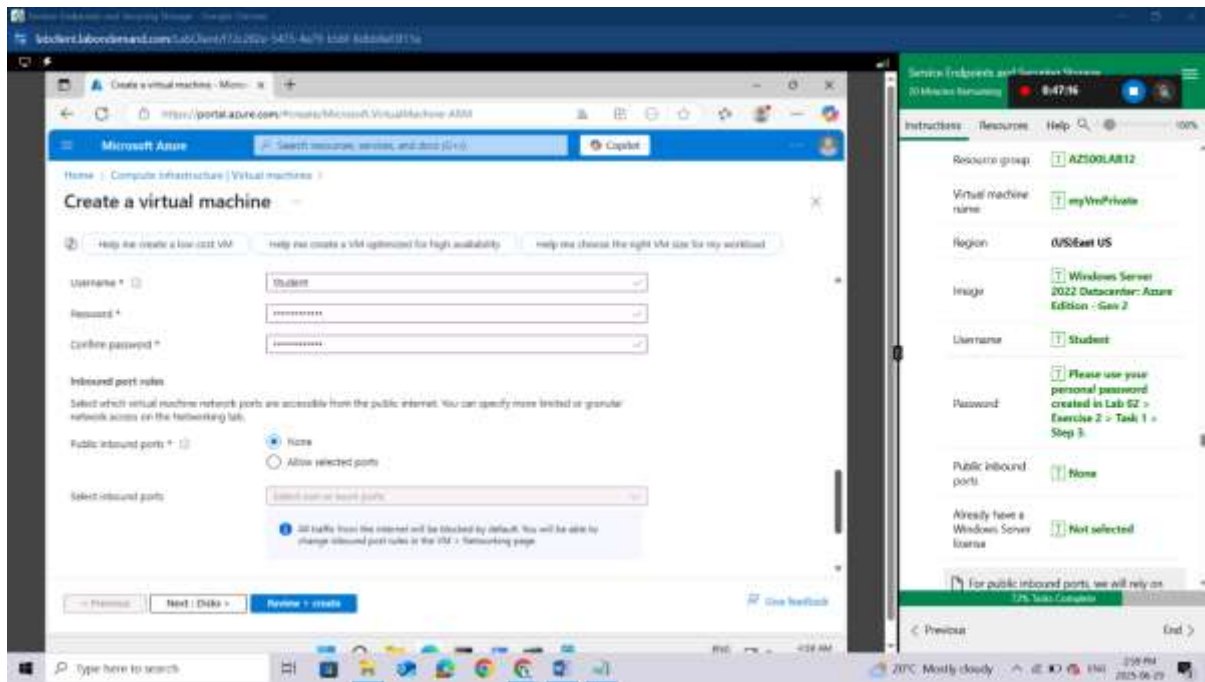
Instructions | Resources | Help | 100%

1. On the **Basics** tab of the **Create a virtual machine** blade, specify the following settings (leave others with their default values):

Setting	Value
Subscription	the name of the Azure subscription you will be using in this lab
Resource group	AZ5000LAB12
Virtual machine name	myVmPrivate
Region	(US)East US
Image	Windows Server 2022 Datacenter: Azure Edition - Gen 2
Username	Student
Password	Please use your personal password created in Lab 02 - Exercise 2 - Task 1 - Step 3.

10% Task Complete

Previous | End



Microsoft Azure | Search resources, services, and docs (0/10)

Home > Compute infrastructure > Virtual machines

Create a virtual machine

Help me create a low-cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Username: ✓

Resource: ✓

Cyber password: ✓

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the following tab.

Public inbound ports: ☒ None ☐ Allow selected ports

Select inbound ports: ✓

ⓘ All traffic from the internet will be blocked by default. You will be able to change inbound port rules in the VM > Networking page.

Previous | **Next: Disks** | Review + create

Give feedback

Service Endpoints and Resource Monitor: 10 Minutes Remaining 8:47:16

Instructions: Resources Help 100%

Resource group: AZ500LAB12

Virtual machine name: myVmPrivate

Region: USEast US

Image: Windows Server 2022 Datacenter: Azure Edition - Gen 2

Username: Student

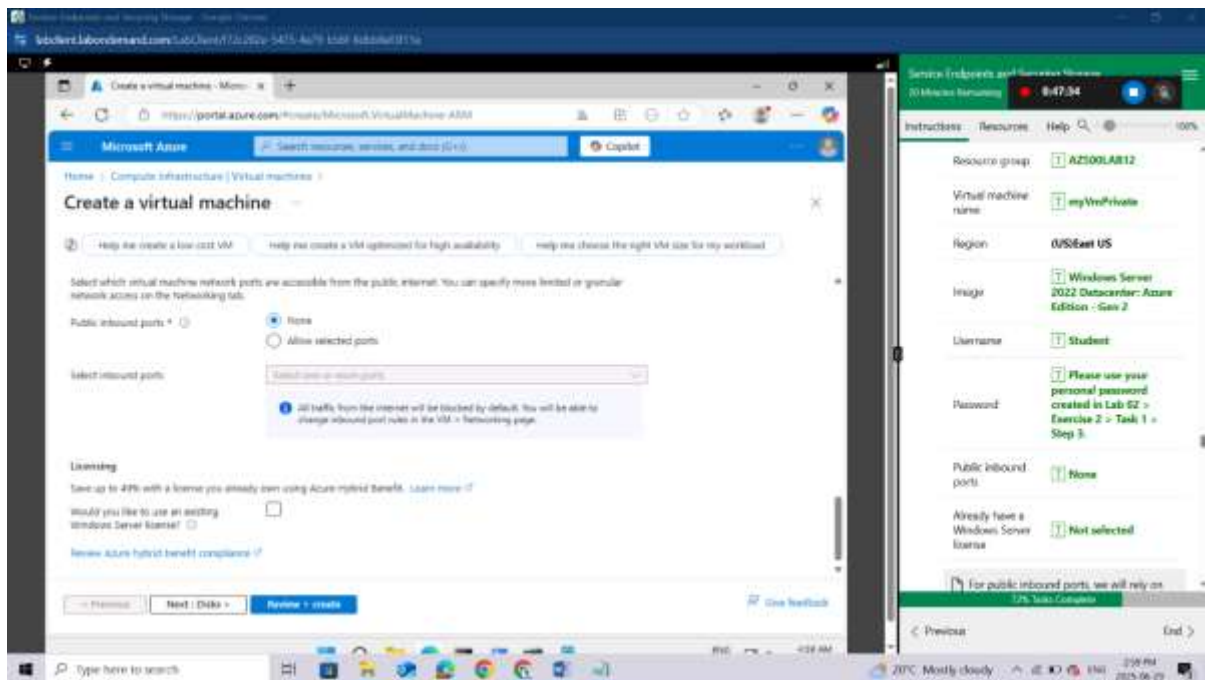
Password: Please use your personal password created in Lab 62 > Exercise 2 > Task 1 > Step 3.

Public inbound ports: None

Already have a Windows Server license: Not selected

For public inbound ports, we will rely on 12% Turbo Compute

Previous End



Microsoft Azure | Search resources, services, and docs (0/10)

Home > Compute infrastructure > Virtual machines

Create a virtual machine

Help me create a low-cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the following tab.

Public inbound ports: ☒ None ☐ Allow selected ports

Select inbound ports: ✓

ⓘ All traffic from the internet will be blocked by default. You will be able to change inbound port rules in the VM > Networking page.

Licensing

Save up to 40% with a license you already own using Azure Hybrid Benefit. Learn more >

Would you like to use an existing Windows Server license? ☐

Review Azure hybrid benefit compliance >

Previous | **Next: Disks** | Review + create

Give feedback

Service Endpoints and Resource Monitor: 10 Minutes Remaining 8:47:34

Instructions: Resources Help 100%

Resource group: AZ500LAB12

Virtual machine name: myVmPrivate

Region: USEast US

Image: Windows Server 2022 Datacenter: Azure Edition - Gen 2

Username: Student

Password: Please use your personal password created in Lab 62 > Exercise 2 > Task 1 > Step 3.

Public inbound ports: None

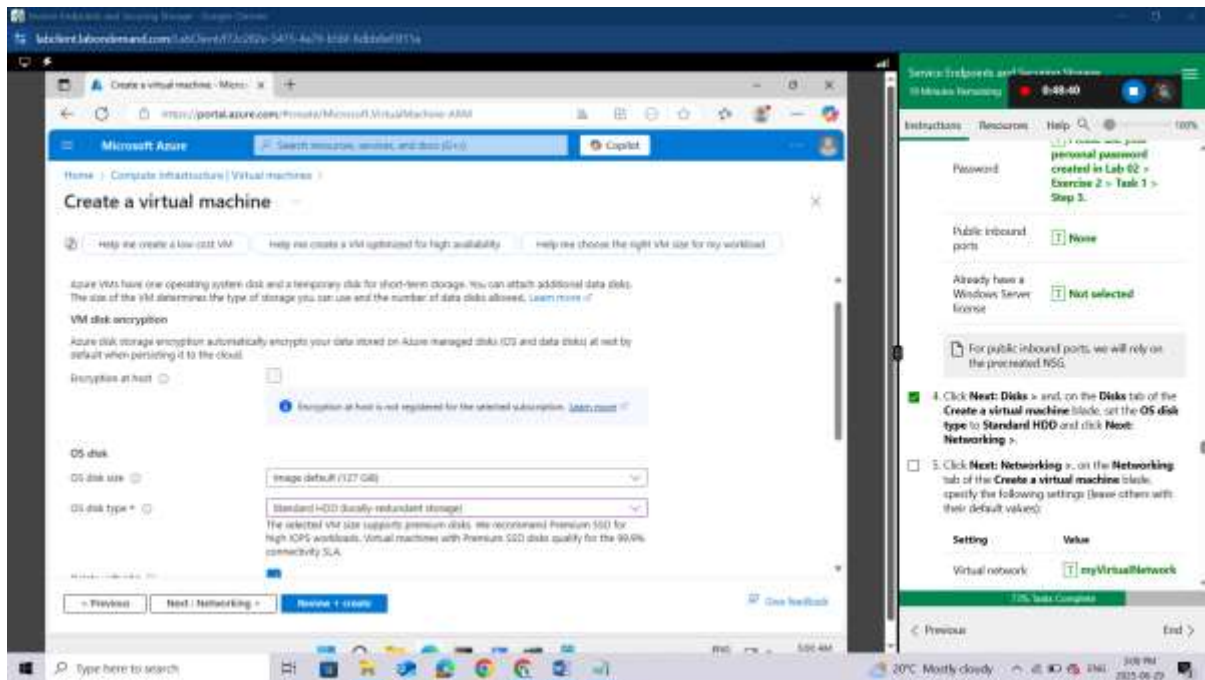
Already have a Windows Server license: Not selected

For public inbound ports, we will rely on 12% Turbo Compute

Previous End

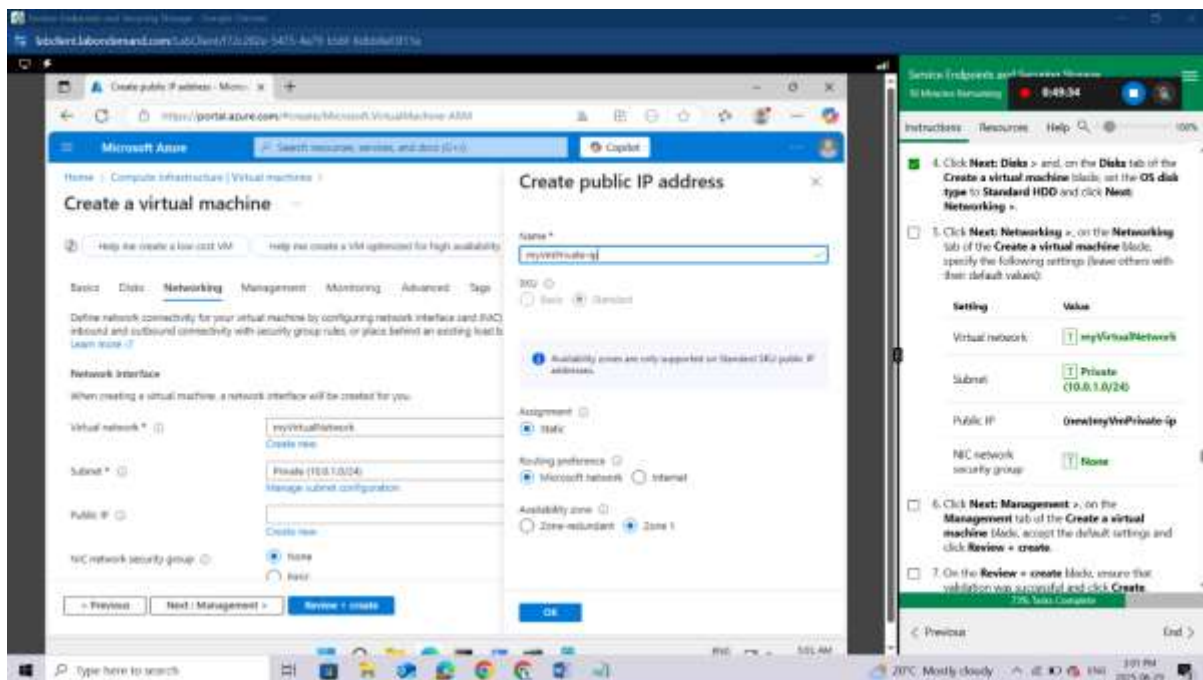
For public inbound ports, we will rely on the precreated NSG.

Click **Next: Disks** > and, on the **Disks** tab of the **Create a virtual machine** blade, set the OS disk type to **Standard HDD** and click **Next: Networking** >.

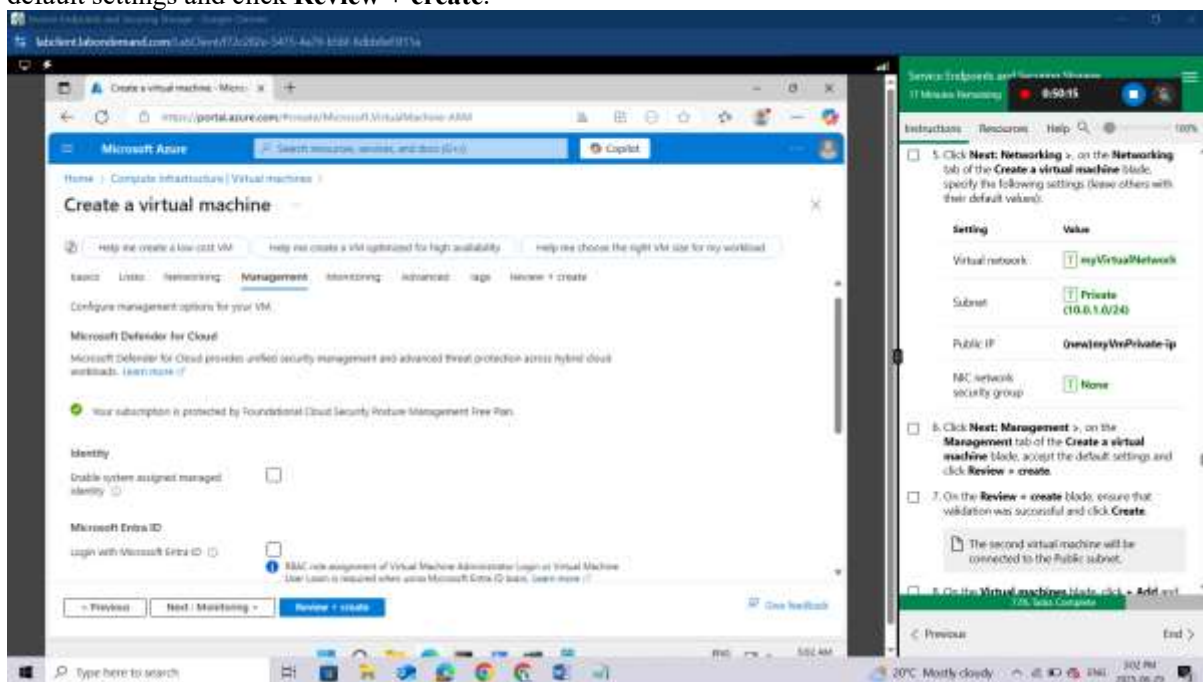


Click **Next: Networking >**, on the **Networking** tab of the **Create a virtual machine** blade, specify the following settings (leave others with their default values):

Setting	Value
Virtual network	myVirtualNetwork
Subnet	Private (10.0.1.0/24)
Public IP	(new)myVmPrivate-ip
NIC network security group	None

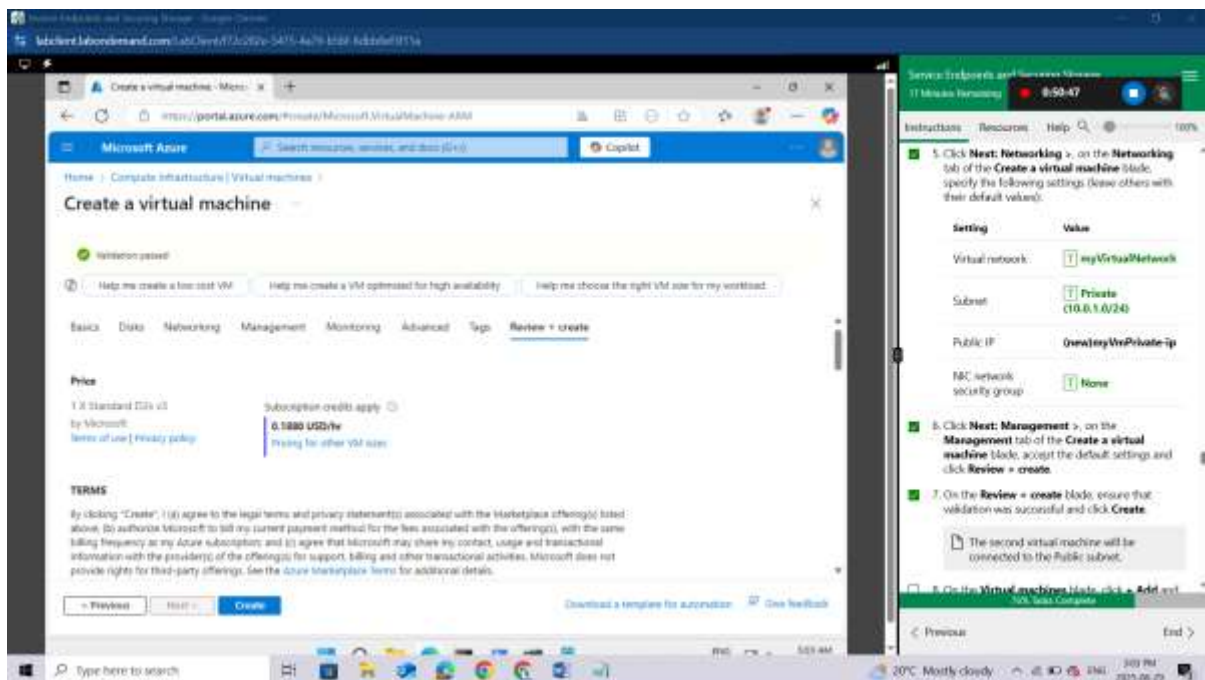


Click **Next: Management** >, on the **Management** tab of the **Create a virtual machine** blade, accept the default settings and click **Review + create**.

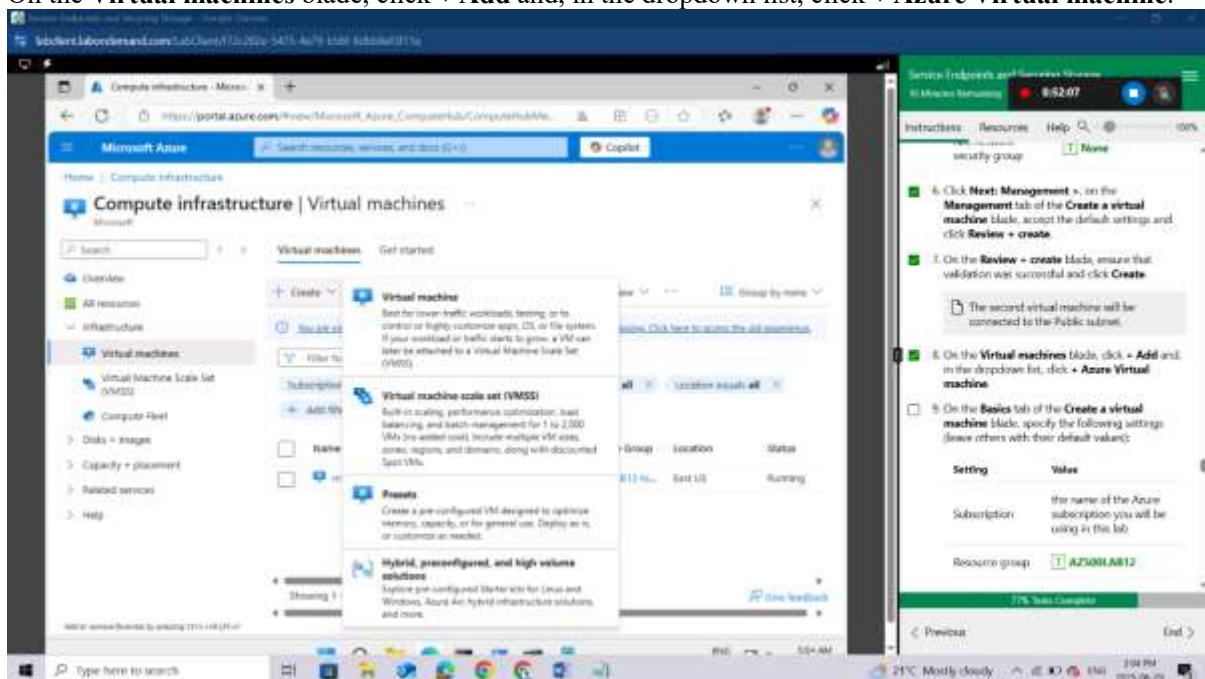


On the **Review + create** blade, ensure that validation was successful and click **Create**.

The second virtual machine will be connected to the Public subnet.



On the **Virtual machines** blade, click + **Add** and, in the dropdown list, click + **Azure Virtual machine**.

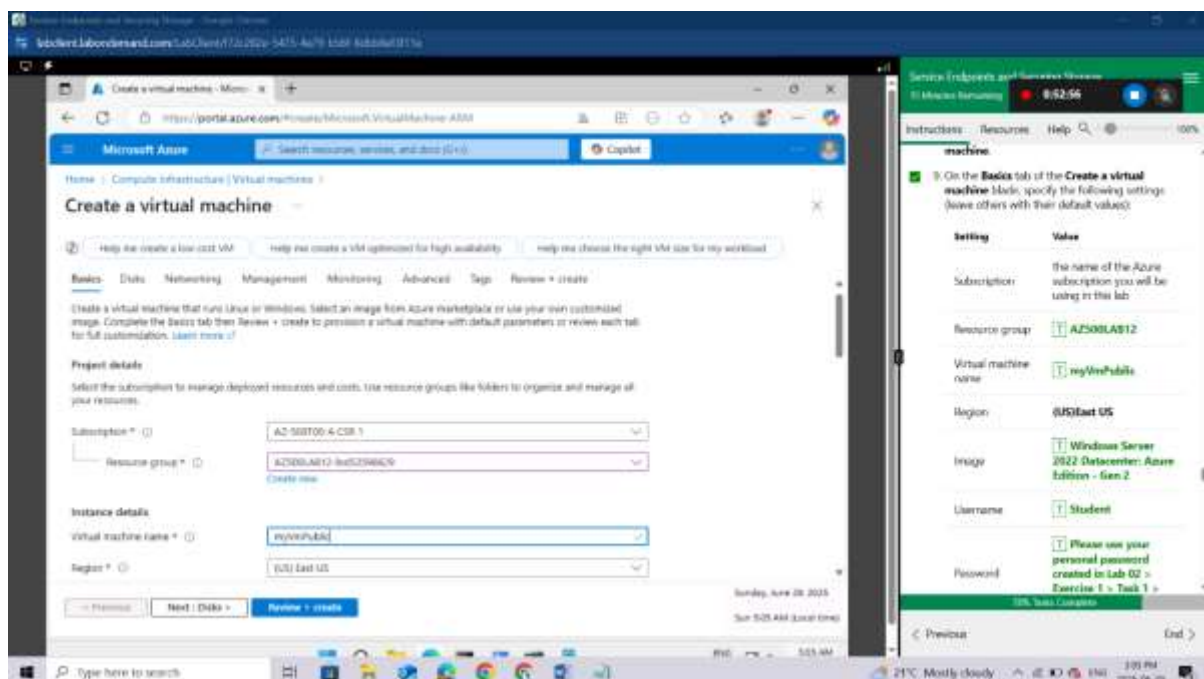


On the **Basics** tab of the **Create a virtual machine** blade, specify the following settings (leave others with their default values):

Setting	Value
Subscription	the name of the Azure subscription you will be using in this lab
Resource group	AZ500LAB12

Setting	Value
Virtual machine name	myVmPublic
Region	(US)East US
Image	Windows Server 2022 Datacenter: Azure Edition - Gen 2
Username	Student
Password	Please use your personal password created in Lab 02 > Exercise 1 > Task 1 > Step 9.
Public inbound ports	None
Already have a Windows Server license	Not selected

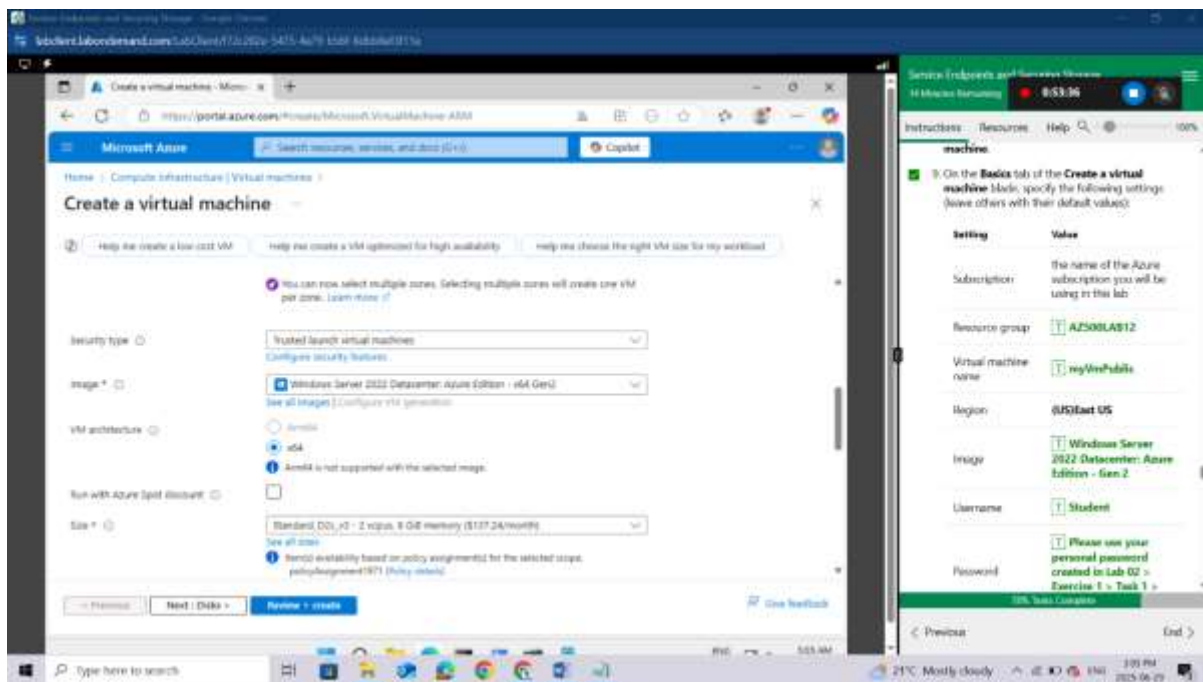
For public inbound ports, we will rely on the precreated NSG.



The screenshot displays the Microsoft Azure portal's 'Create a virtual machine' wizard. The 'Basics' tab is selected, showing the following configuration details:

- Subscription:** AZ-58BT00-4-C8R-1
- Resource group:** AZ58BT00-4-C8R-1
- Virtual machine name:** myVmPublic
- Region:** (US)East US
- Image:** Windows Server 2022 Datacenter: Azure Edition - Gen 2
- Username:** Student
- Password:** Please use your personal password created in Lab 02 > Exercise 1 > Task 1 > Step 9.

The sidebar on the right provides a summary of these settings in a table format, matching the data in the table above.



Create a virtual machine

Help me create a low-cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

You can now select multiple zones. Selecting multiple zones will create one VM per zone. [Learn more](#)

Security type: Trusted launch virtual machines (Configures security features)

Image: Windows Server 2022 Datacenter: Azure Edition - x64 Gen2 (See all images | Configure VM generation)

VM architecture: x64 (Note: ARM64 is not supported with the selected image.)

Run with Azure Spot Disks: ☐

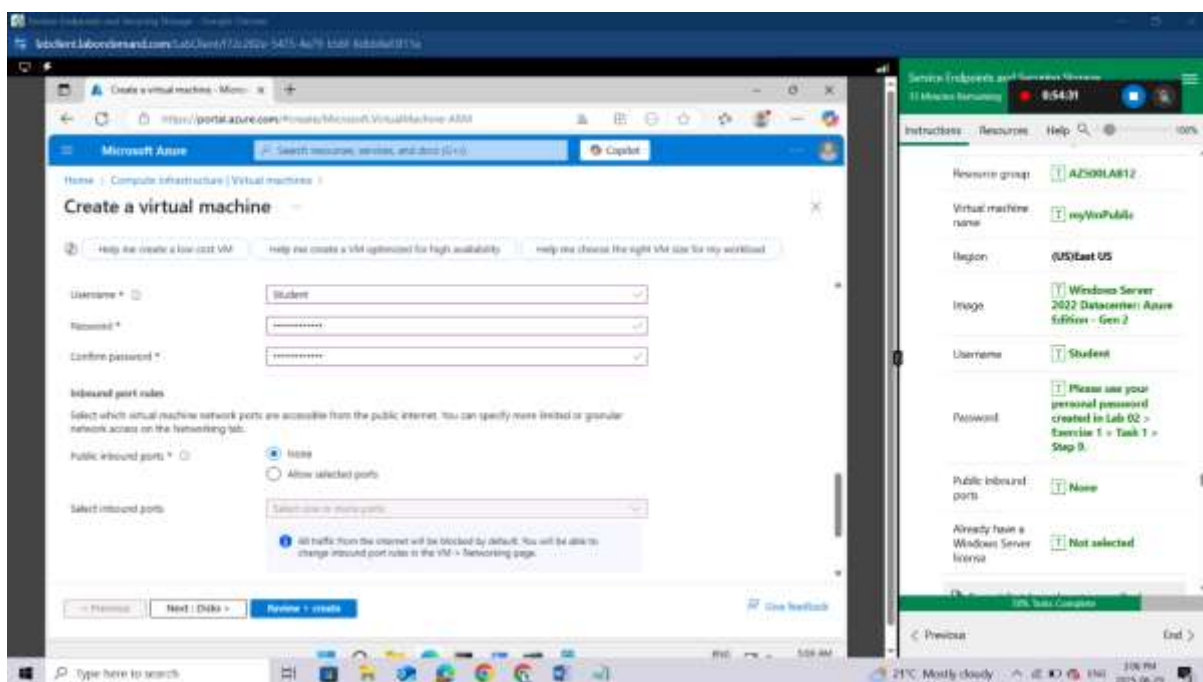
Size: Standard D2s v2 - 2 vCPUs, 8 GB memory (\$137.24/month) (See all sizes | Permitted availability based on policy assignment for the selected script: policyassignment/871 (Policy details))

Next (Disk) | **Review + create**

Summary

Setting	Value
Subscription	the name of the Azure subscription you will be using in this lab
Resource group	AZ508LAB12
Virtual machine name	myWinPublic
Region	(US)East US
Image	Windows Server 2022 Datacenter: Azure Edition - Gen 2
Username	Student
Password	Please use your personal password created in Lab 02 > Exercise 1 > Task 1

10% Task Complete



Create a virtual machine

Help me create a low-cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Username: Student

Password: [masked]

Confirm password: [masked]

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports: ☒ None (or ☐ Allow selected ports)

Select inbound ports: Select one or more ports

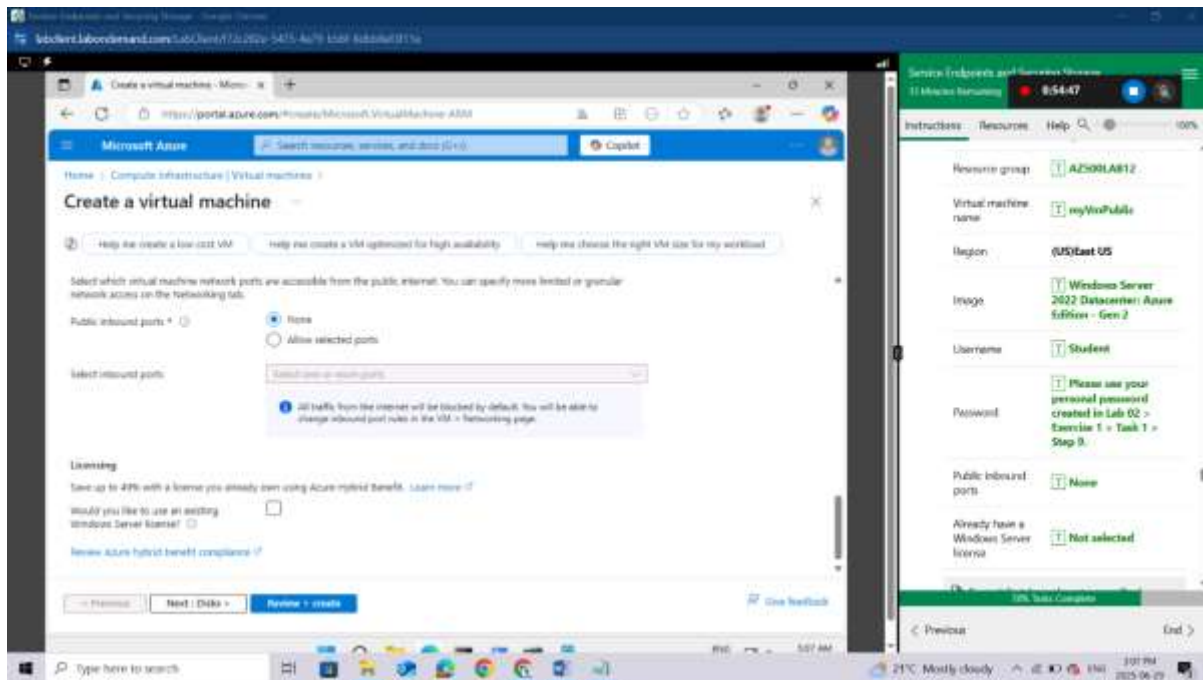
All traffic from the internet will be blocked by default. You will be able to change inbound port rules in the VM > Networking page.

Next (Disk) | **Review + create**

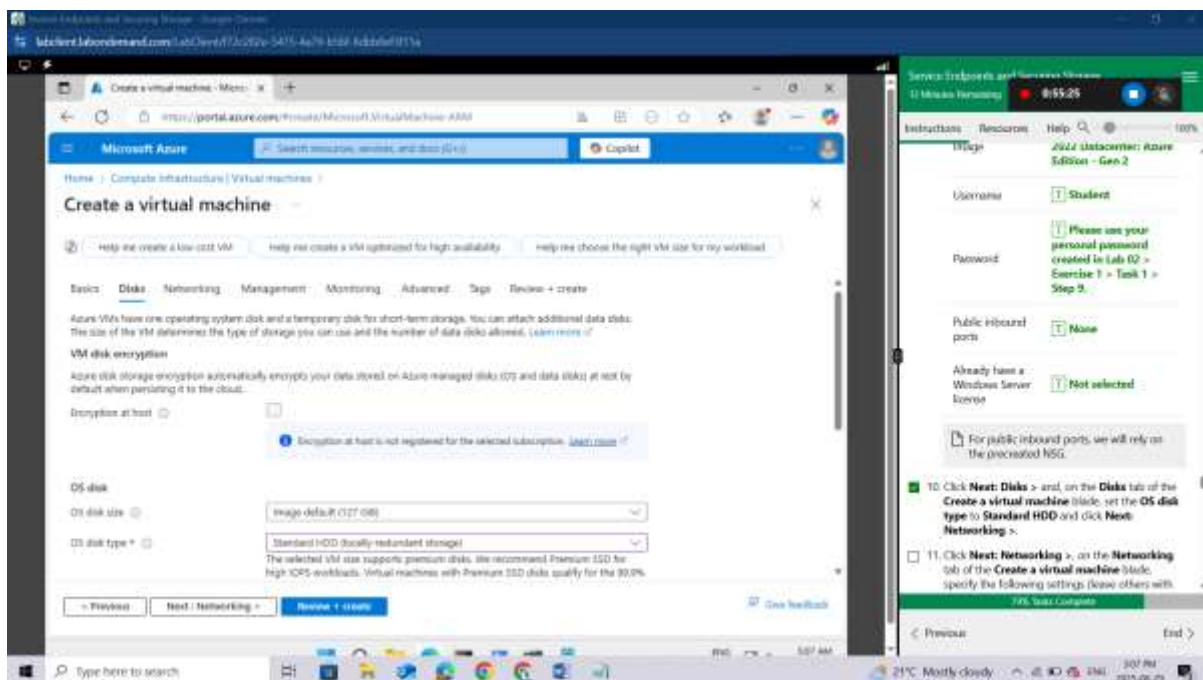
Summary

Resource group	AZ508LAB12
Virtual machine name	myWinPublic
Region	(US)East US
Image	Windows Server 2022 Datacenter: Azure Edition - Gen 2
Username	Student
Password	Please use your personal password created in Lab 02 > Exercise 1 > Task 1 > Step 9
Public inbound ports	None
Already have a Windows Server license	☐ Not selected

10% Task Complete



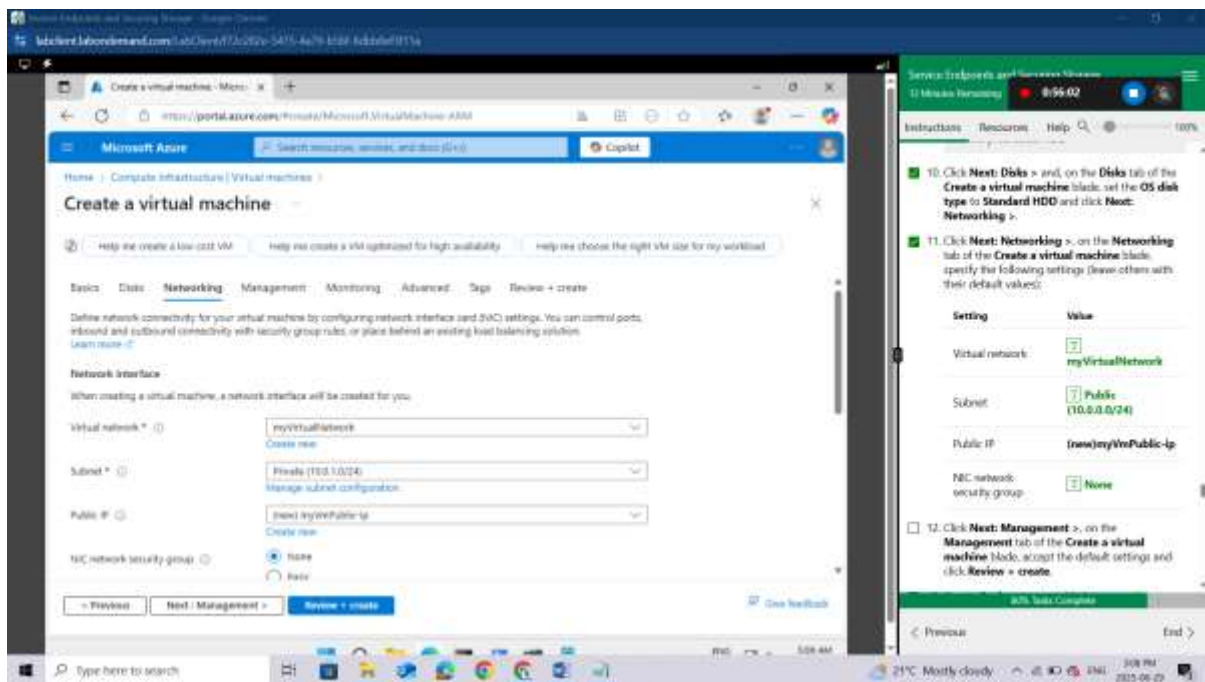
Click **Next: Disks** > and, on the **Disks** tab of the **Create a virtual machine** blade, set the OS disk type to **Standard HDD** and click **Next: Networking** >.



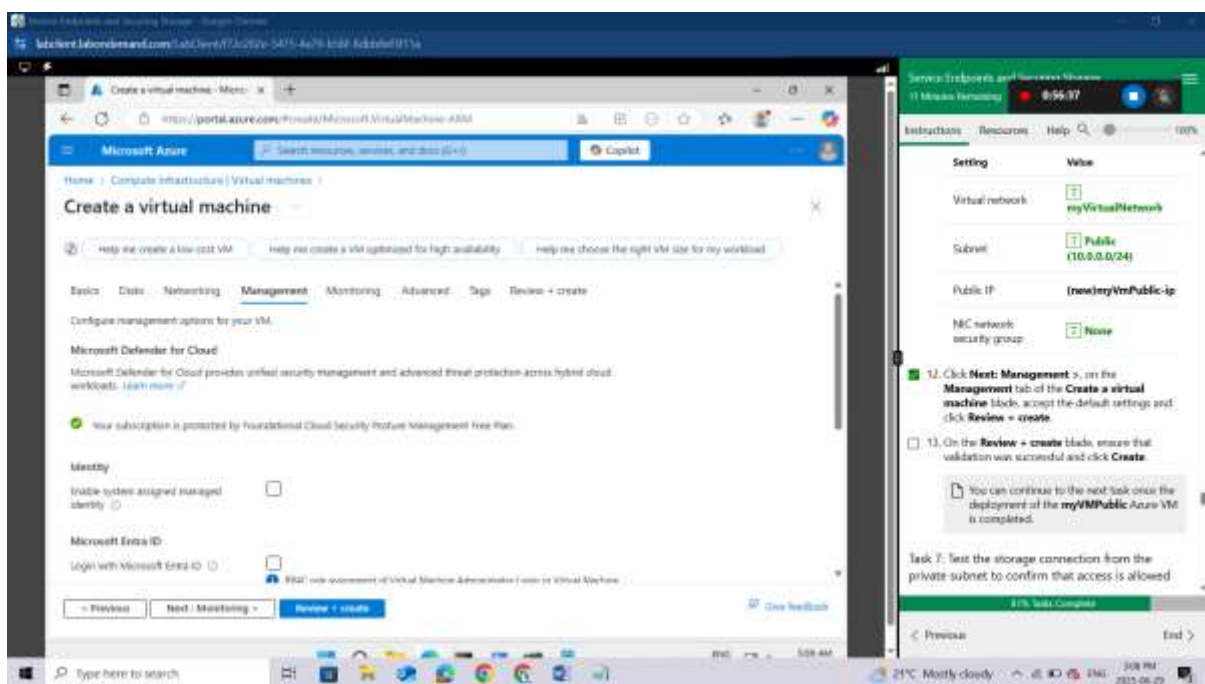
Click **Next: Networking** >, on the **Networking** tab of the **Create a virtual machine** blade, specify the following settings (leave others with their default values):

Setting	Value
Virtual network	myVirtualNetwork
Subnet	Public (10.0.0.0/24)

Setting	Value
Public IP	(new)myVmPublic-ip
NIC network security group	None

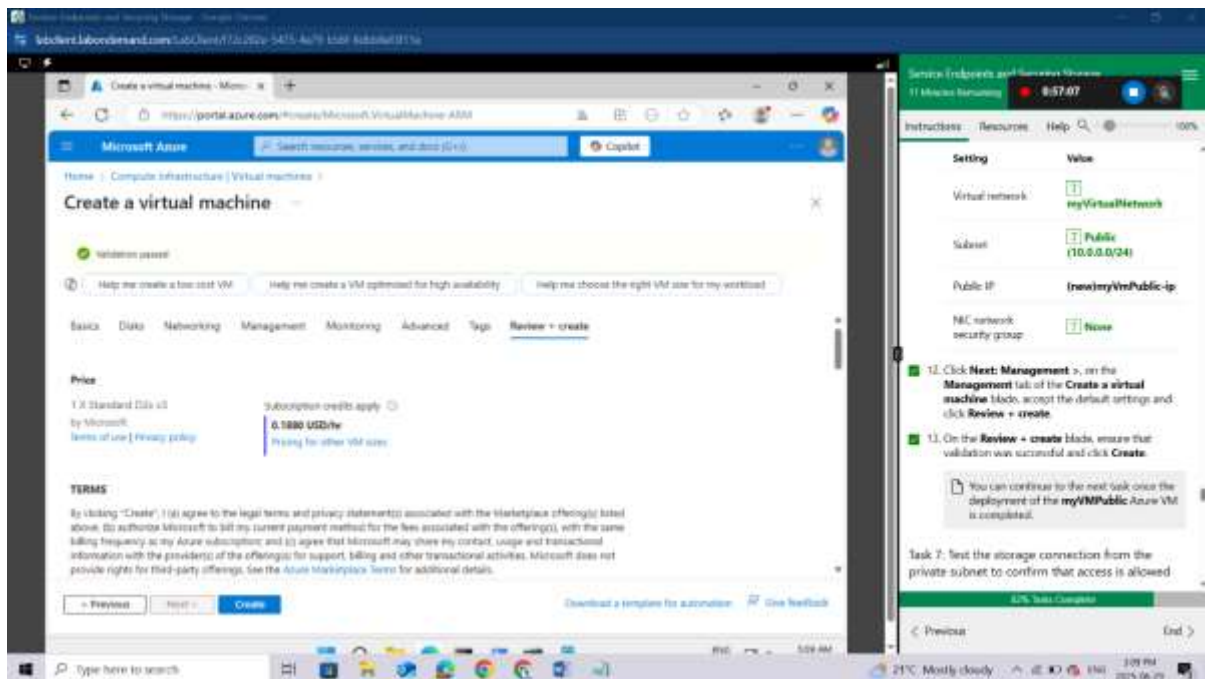


Click **Next: Management** >, on the **Management** tab of the **Create a virtual machine** blade, accept the default settings and click **Review + create**.



On the **Review + create** blade, ensure that validation was successful and click **Create**.

You can continue to the next task once the deployment of the **myVMPublic** Azure VM is completed.

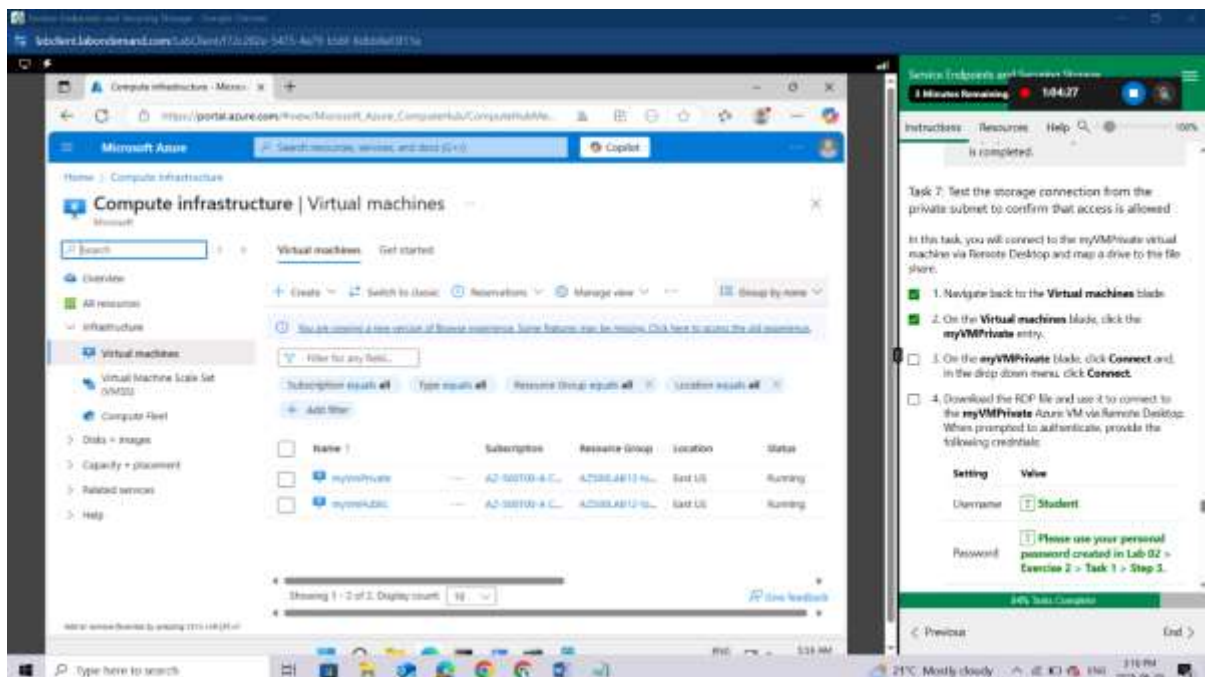


Task 7: Test the storage connection from the private subnet to confirm that access is allowed

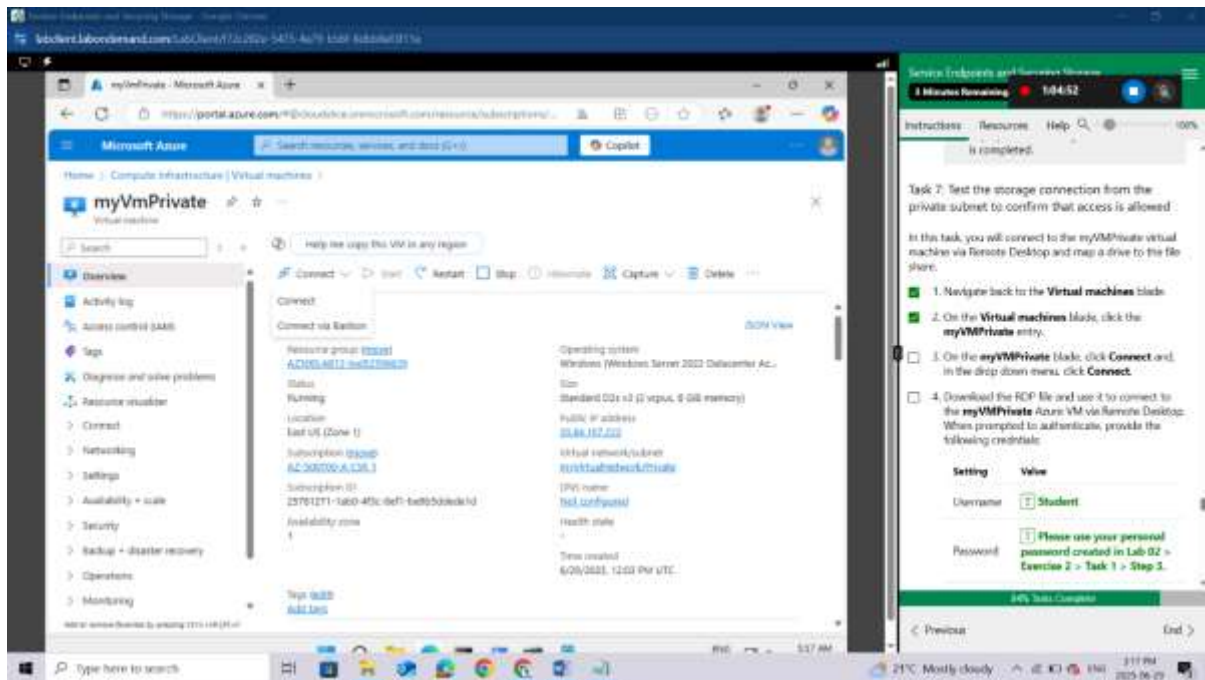
In this task, you will connect to the myVMPrivate virtual machine via Remote Desktop and map a drive to the file share.

Navigate back to the **Virtual machines** blade.

On the **Virtual machines** blade, click the **myVMPrivate** entry.



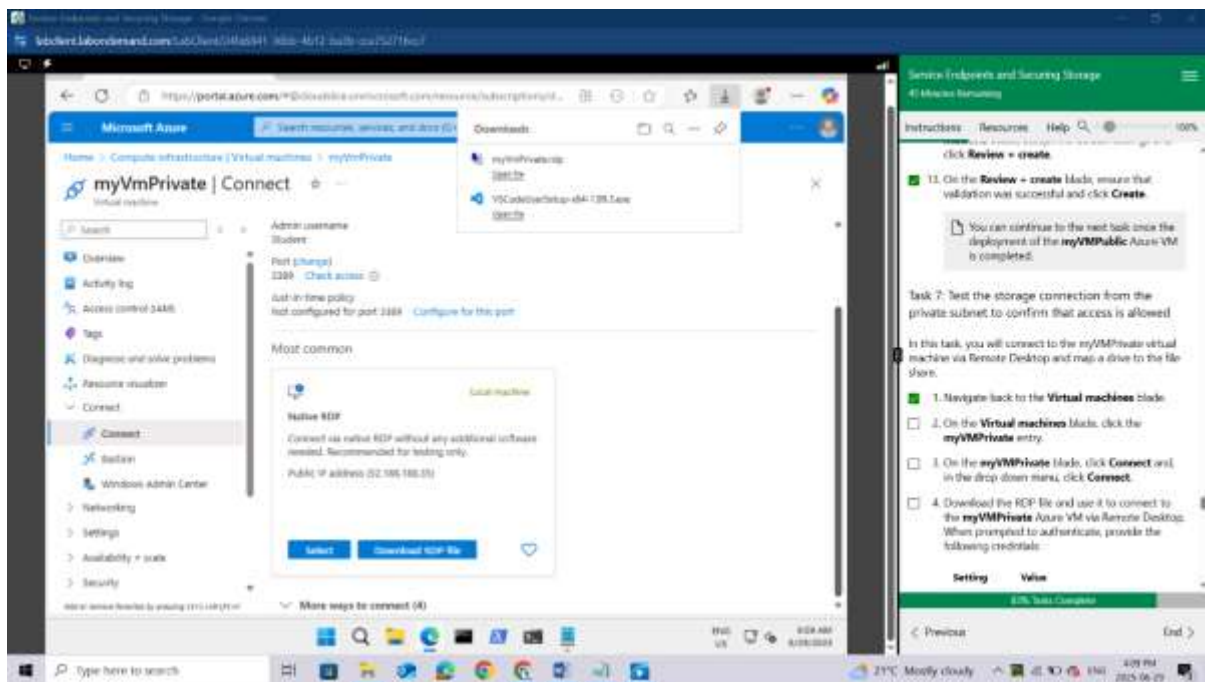
On the **myVMPrivate** blade, click **Connect** and, in the drop down menu, click **Connect**.



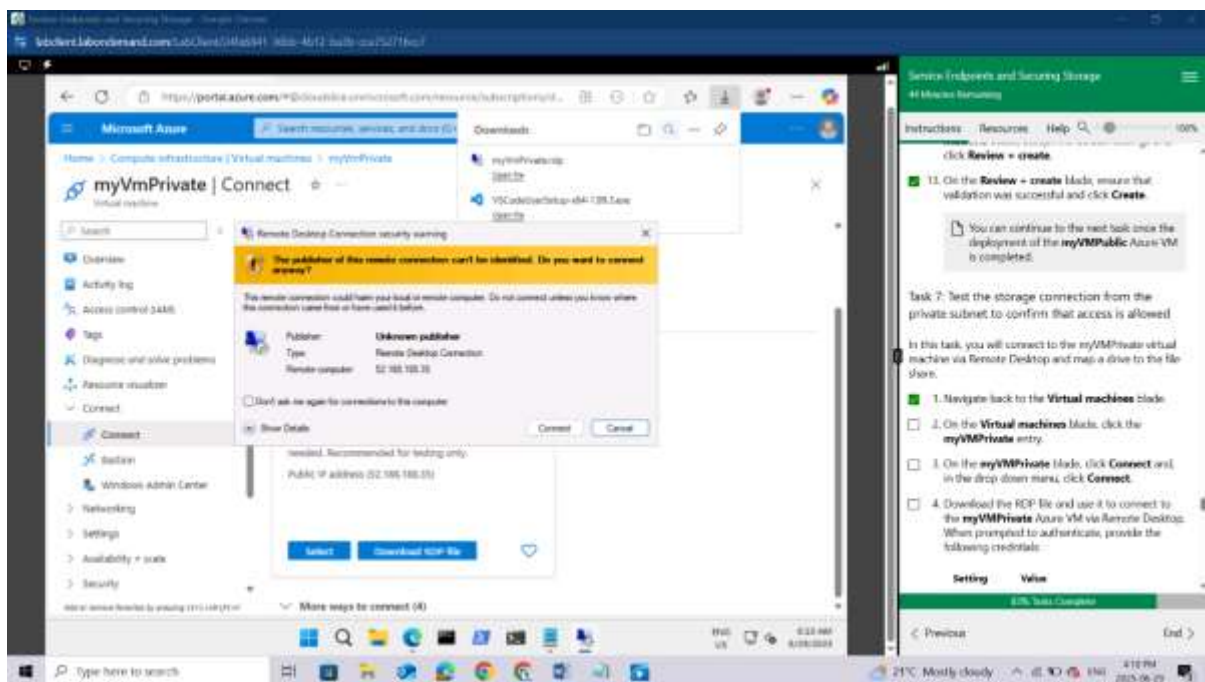
Download the RDP file and use it to connect to the **myVmPrivate** Azure VM via Remote Desktop. When prompted to authenticate, provide the following credentials:

Setting	Value
Username	Student
Password	Please use your personal password created in Lab 02 > Exercise 2 > Task 1 > Step 3.

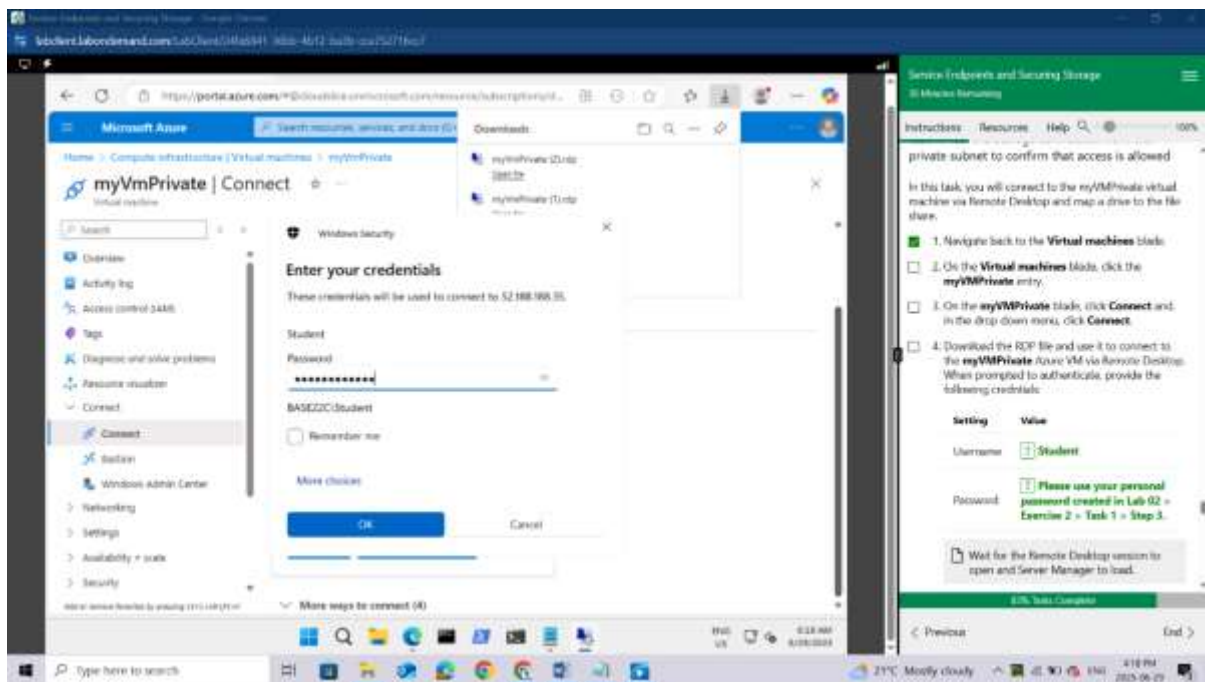
Wait for the Remote Desktop session to open and Server Manager to load.
 You will now map drive Z to an Azure File share within the Remote Desktop session to a Windows Server 2022 computer
 Click on the downloaded file to open



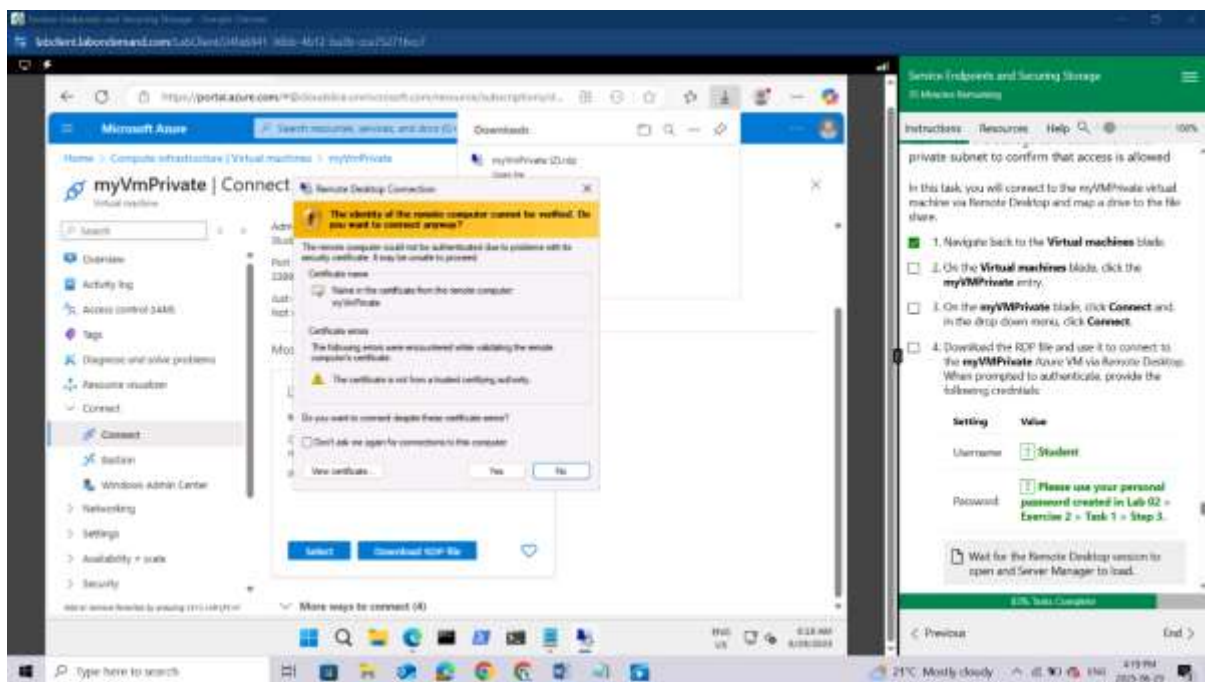
Click on connect



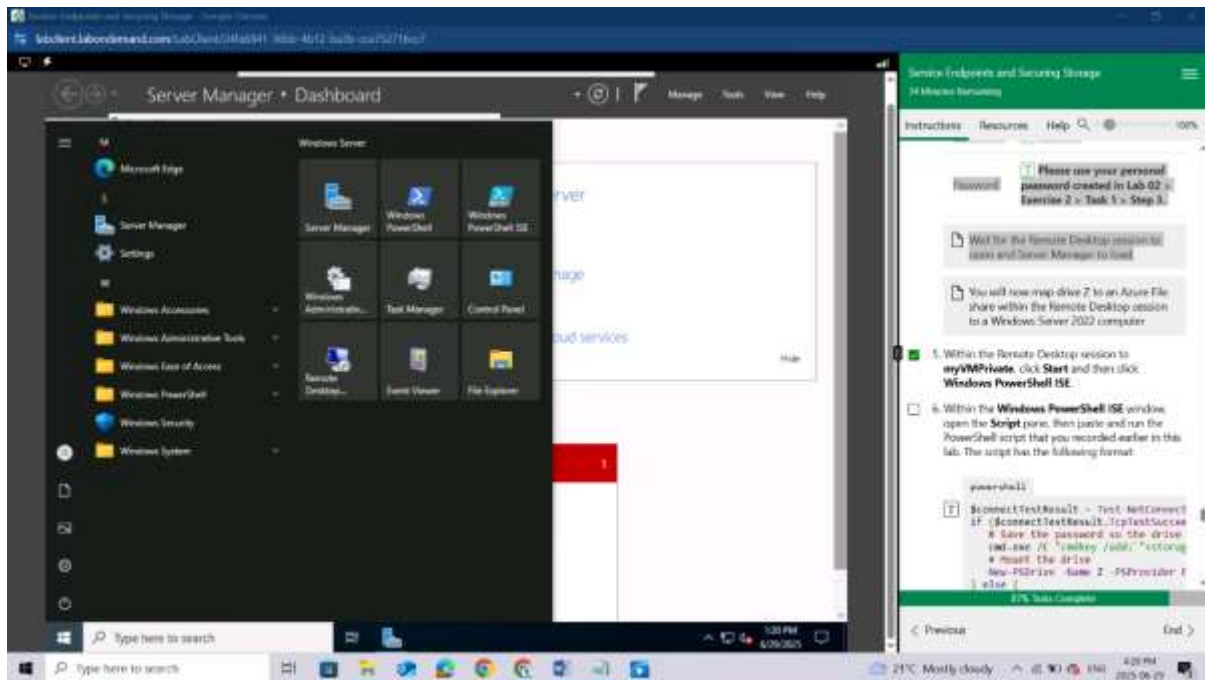
Login with your credentials:



When prompted with certification click yes



Within the Remote Desktop session to **myVMPrivate**, click **Start** and then click **Windows PowerShell ISE**.

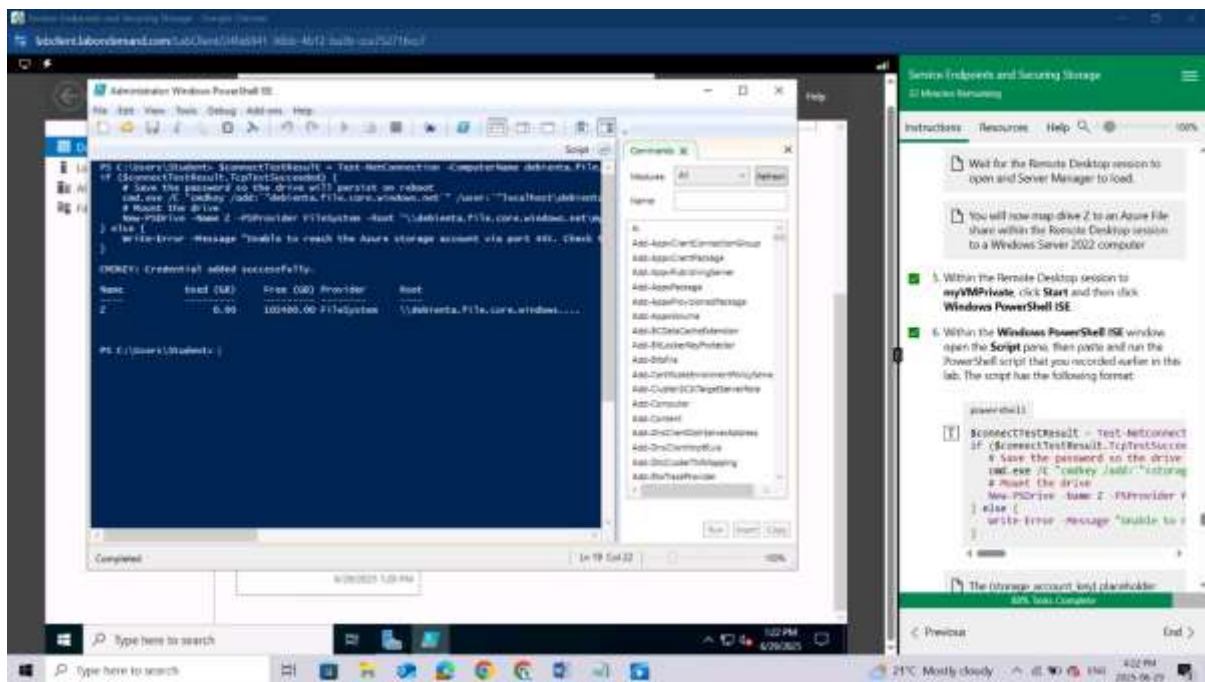


Within the **Windows PowerShell ISE** window, open the **Script** pane, then paste and run the PowerShell script that you recorded earlier in this lab. The script has the following format:

```
powershell
$connectTestResult = Test-NetConnection -ComputerName
<storage_account_name>.file.core.windows.net -Port 445
if ($connectTestResult.TcpTestSucceeded) {
    # Save the password so the drive will persist on reboot
    cmd.exe /C "cmdkey /add:"<storage_account_name>.file.core.windows.net`"
/user:"localhost\<storage_account_name>" /pass:"<storage_account_key>"`"
    # Mount the drive
    New-PSDrive -Name Z -PSProvider FileSystem -Root
"\\<storage_account_name>.file.core.windows.net\my-file-share" -Persist
} else {
    Write-Error -Message "Unable to reach the Azure storage account via port 445. Check to make sure your
organization or ISP is not blocking port 445, or use Azure P2S VPN, Azure S2S VPN, or Express Route to
tunnel SMB traffic over a different port."
}
```

The (storage_account_key) placeholder represents the name of the storage account hosting the file share and <storage_account_key> one its primary key

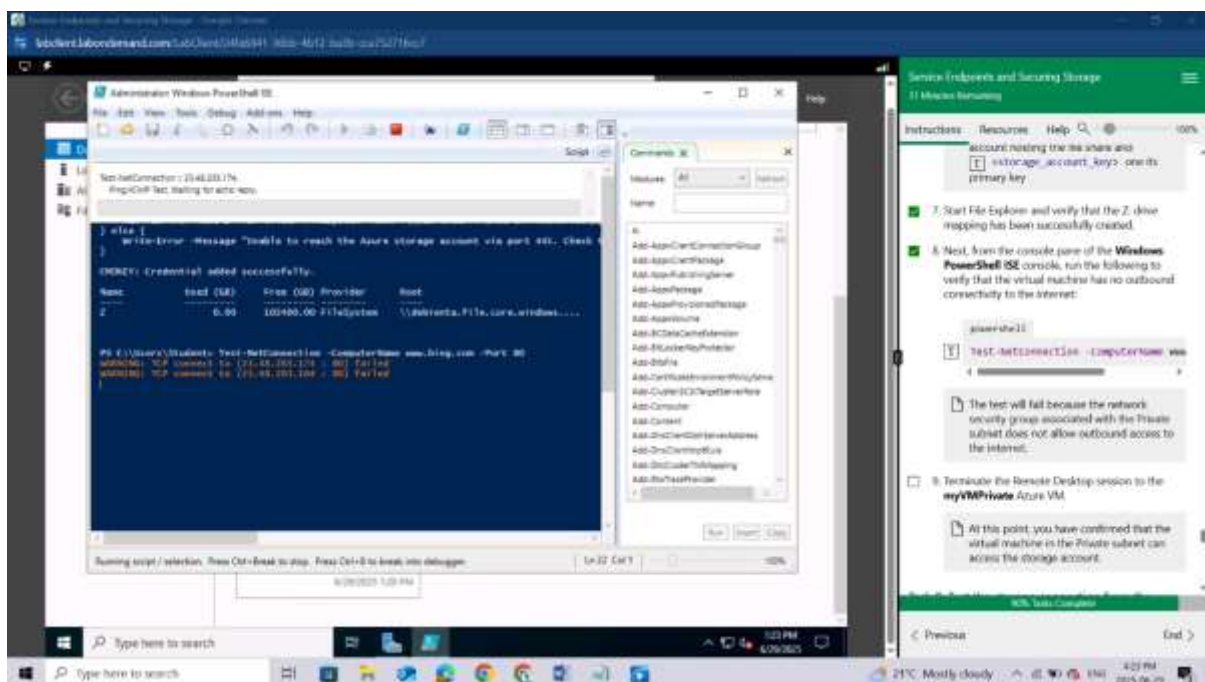
Start File Explorer and verify that the Z: drive mapping has been successfully created.



Next, from the console pane of the **Windows PowerShell ISE** console, run the following to verify that the virtual machine has no outbound connectivity to the internet:

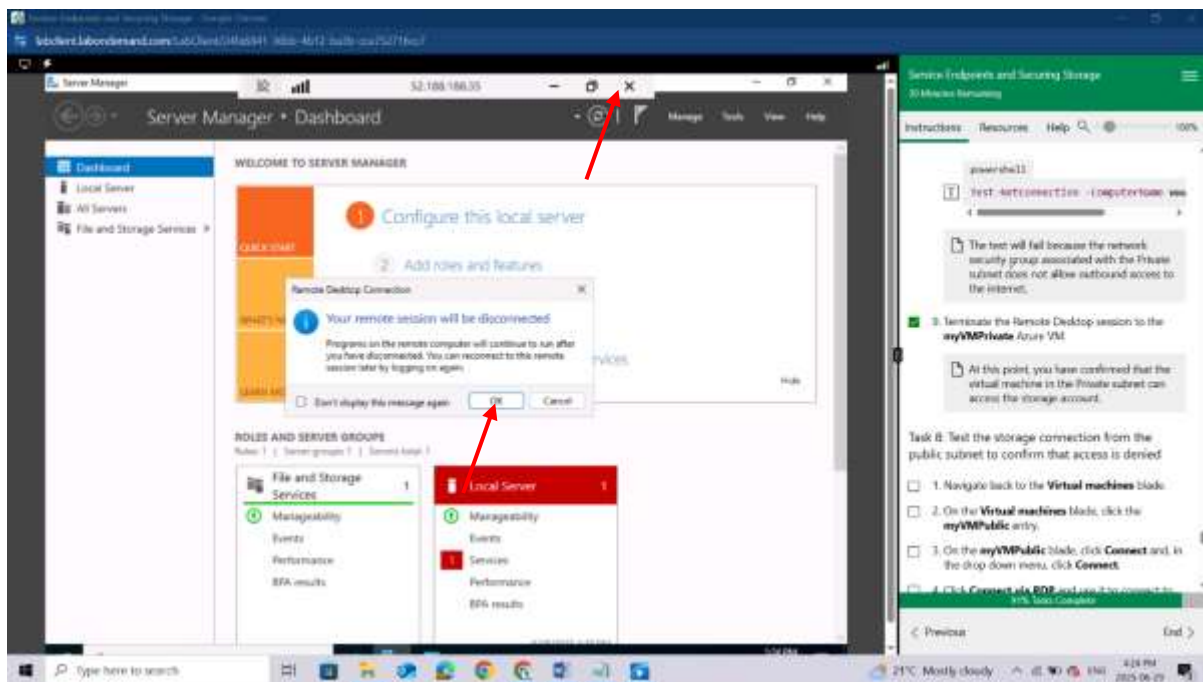
```
powershell
Test-NetConnection -ComputerName www.bing.com -Port 80
```

The test will fail because the network security group associated with the Private subnet does not allow outbound access to the internet.



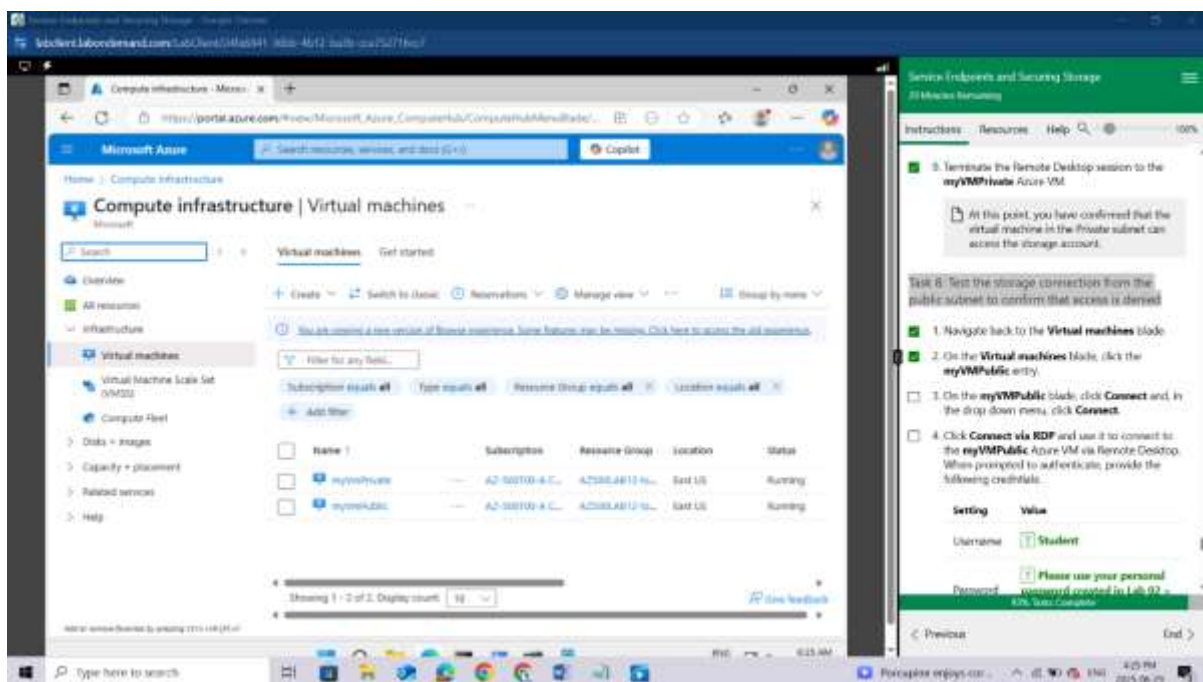
Terminate the Remote Desktop session to the **myVMPrivate** Azure VM.

At this point, you have confirmed that the virtual machine in the Private subnet can access the storage account.

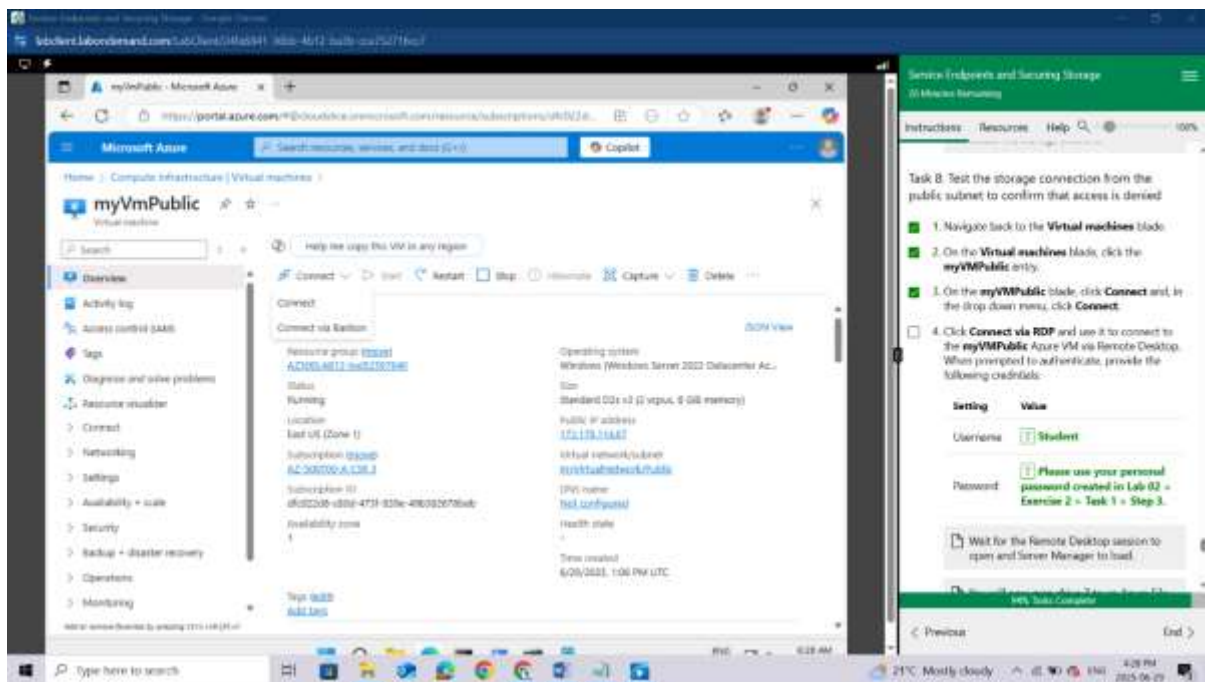


Task 8: Test the storage connection from the public subnet to confirm that access is denied
 Navigate back to the **Virtual machines** blade.

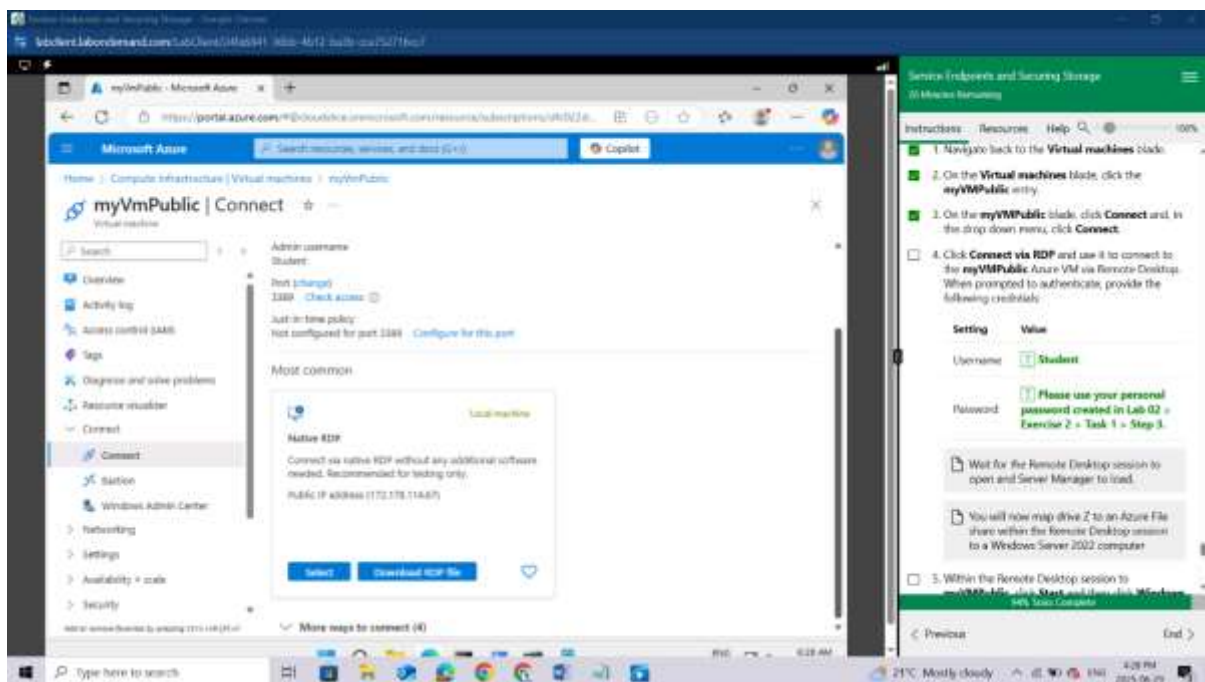
On the **Virtual machines** blade, click the **myVMPublic** entry.



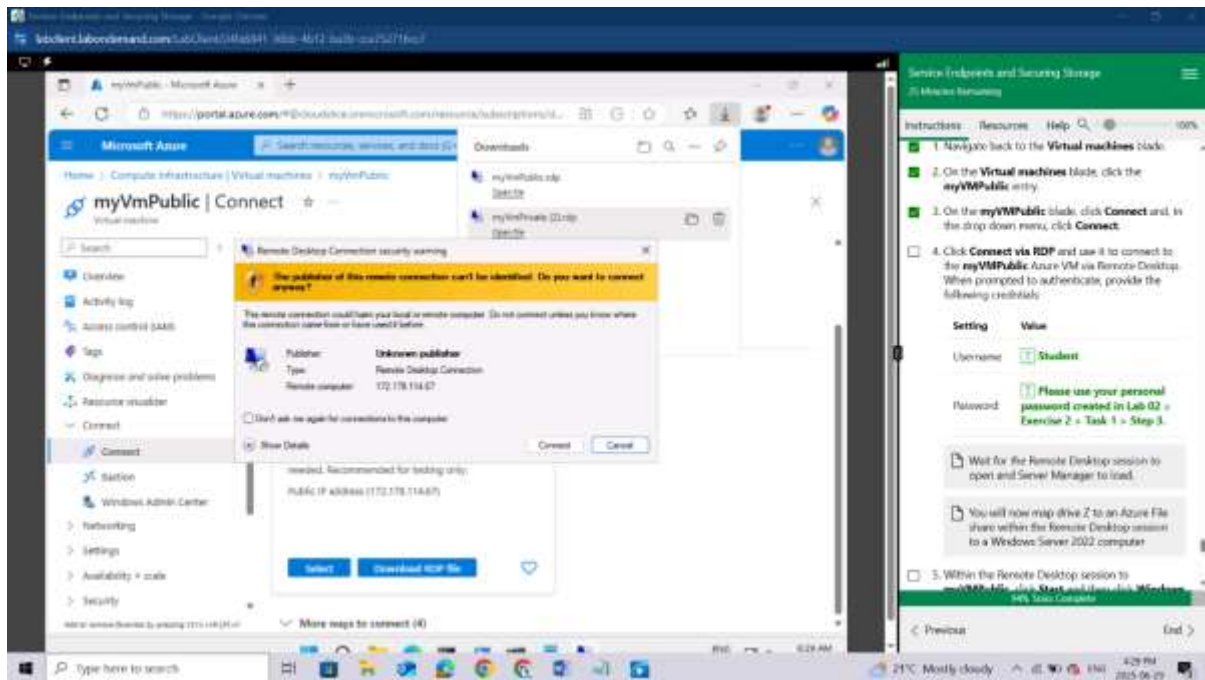
On the **myVMPublic** blade, click **Connect** and, in the drop down menu, click **Connect**.



Download the RDP file



After download is complete, click on the file then click connect

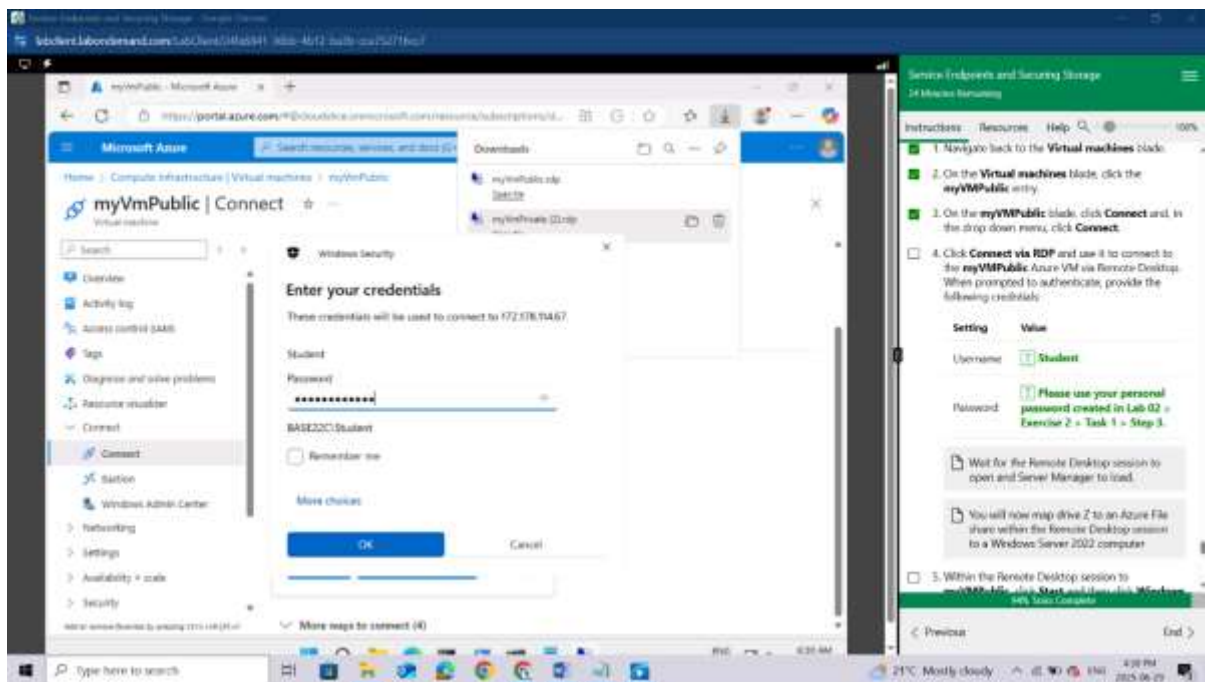


When prompted to authenticate, provide the following credentials:

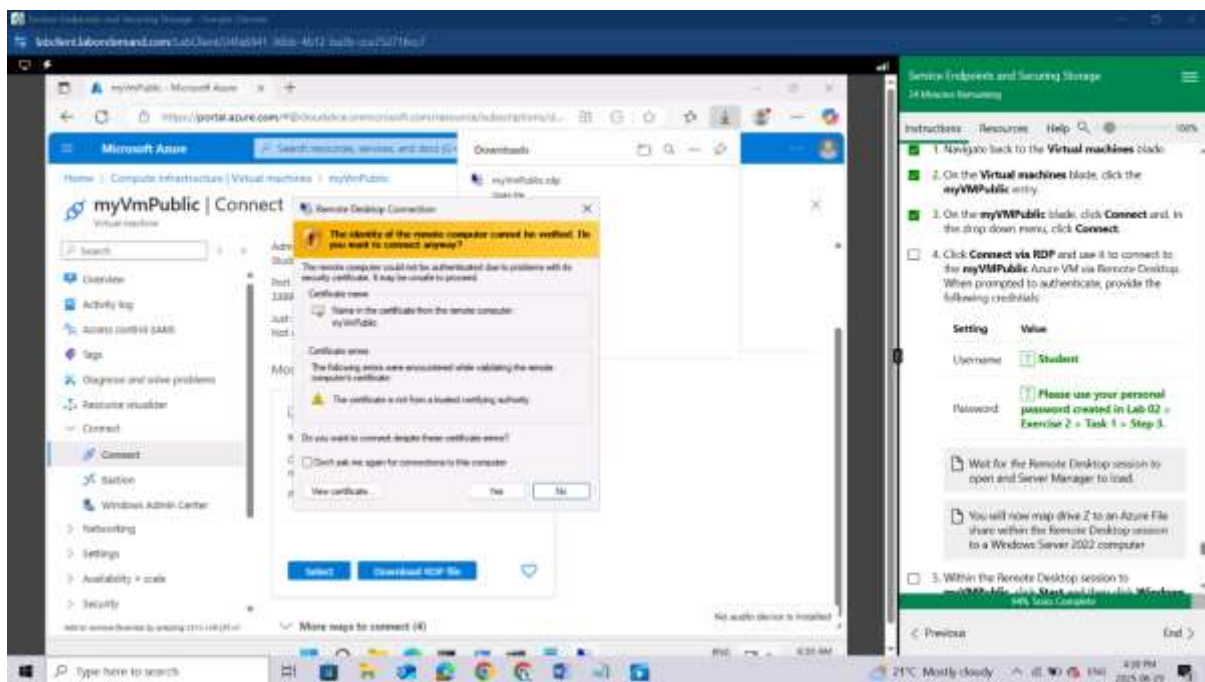
Setting	Value
Username	Student
Password	Please use your personal password created in Lab 02 > Exercise 2 > Task 1 > Step 3.

Wait for the Remote Desktop session to open and Server Manager to load.

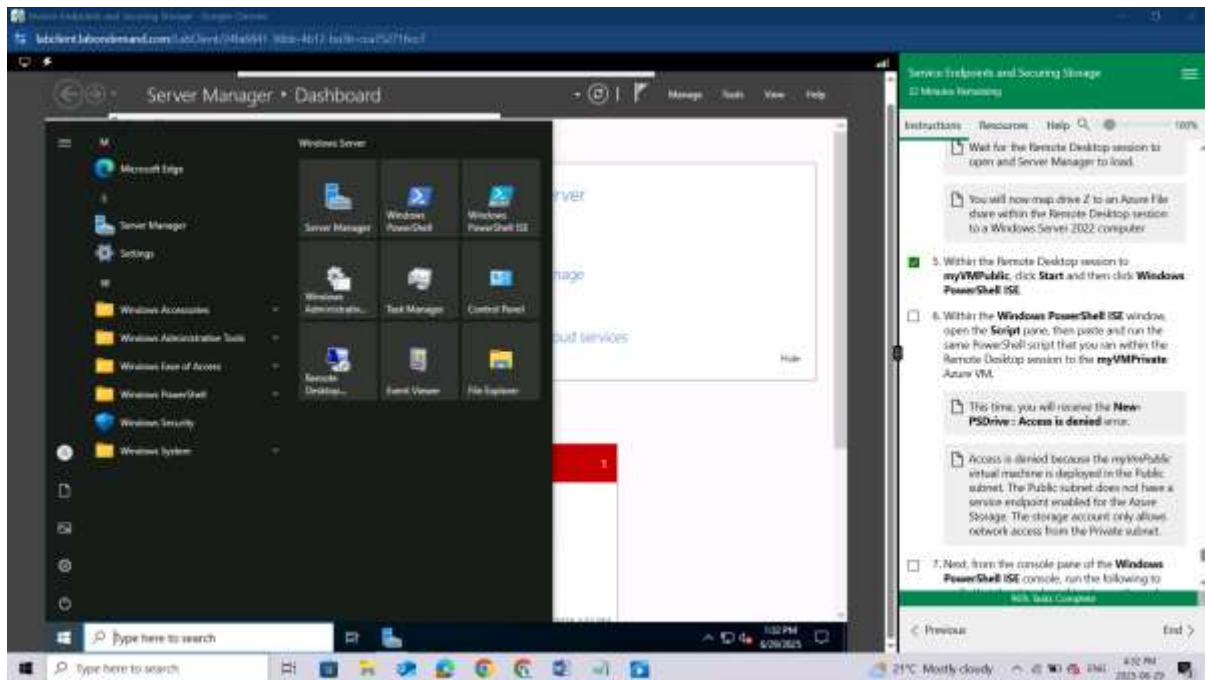
You will now map drive Z to an Azure File share within the Remote Desktop session to a Windows Server 2022 computer



When prompted on certificate click YES

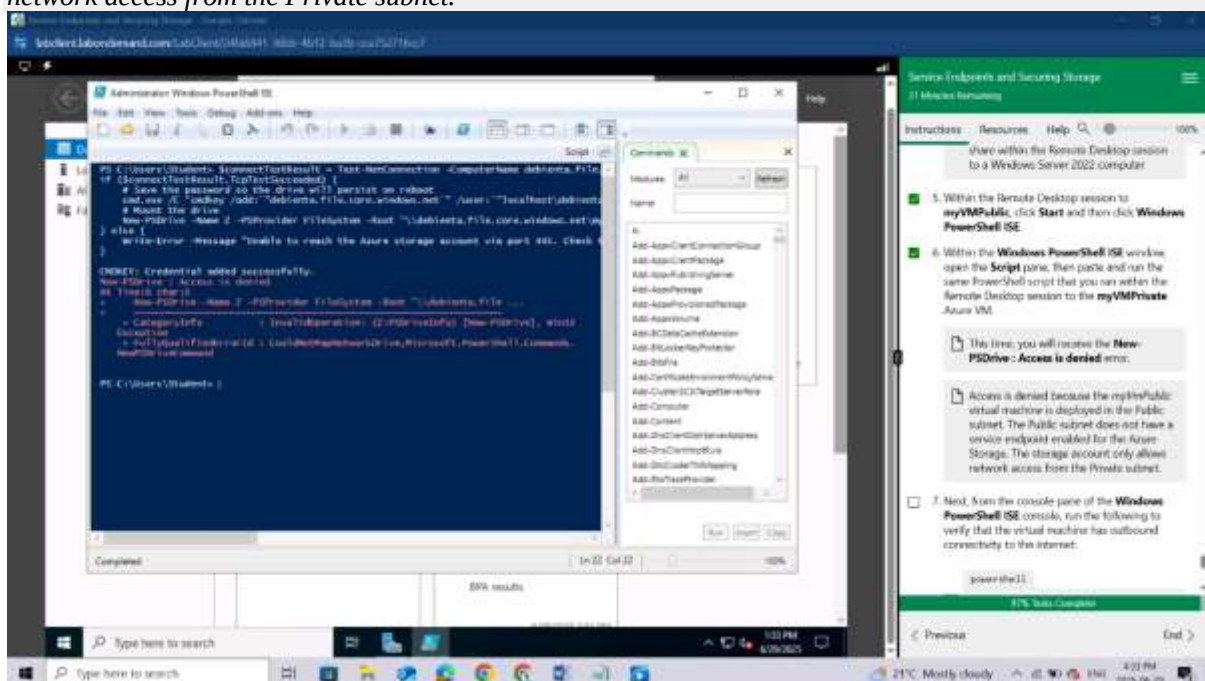


Within the Remote Desktop session to **myVMPublic**, click **Start** and then click **Windows PowerShell ISE**.



Within the **Windows PowerShell ISE** window, open the **Script** pane, then paste and run the same PowerShell script that you ran within the Remote Desktop session to the **myVMPrivate** Azure VM.

*This time, you will receive the **New-PSDrive : Access is denied** error. Access is denied because the myVmPublic virtual machine is deployed in the Public subnet. The Public subnet does not have a service endpoint enabled for the Azure Storage. The storage account only allows network access from the Private subnet.*

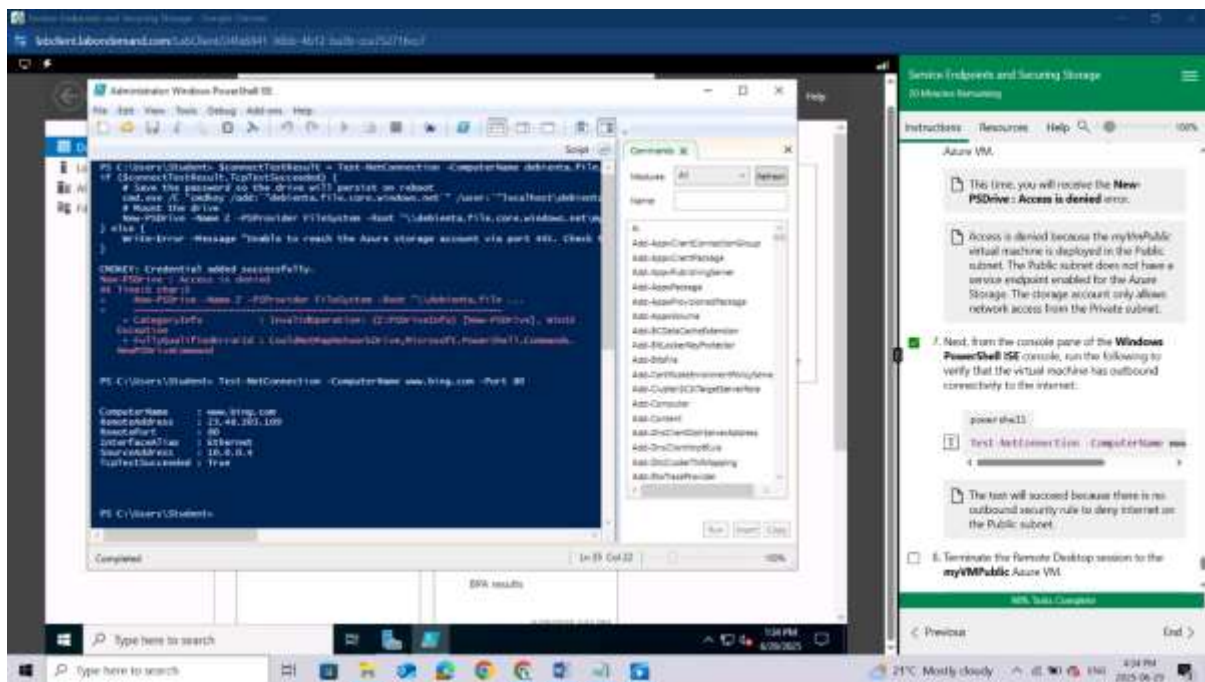


Next, from the console pane of the **Windows PowerShell ISE** console, run the following to verify that the virtual machine has outbound connectivity to the internet:

powershell

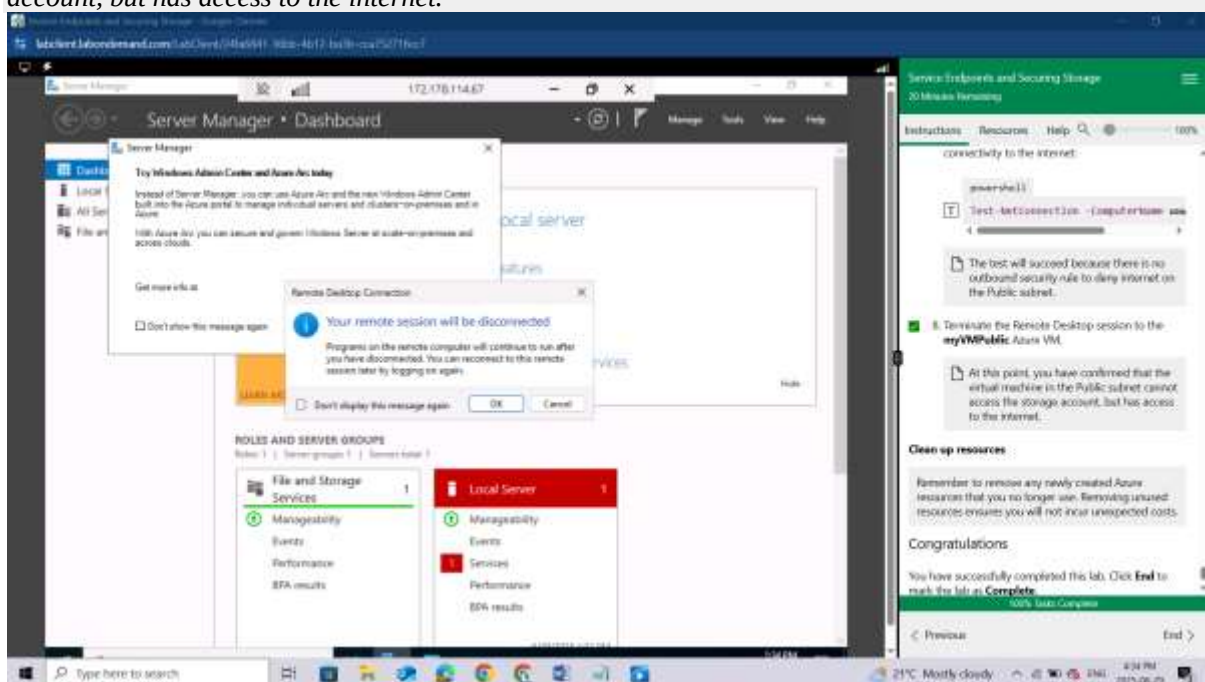
Test-NetConnection -ComputerName www.bing.com **-Port** 80

The test will succeed because there is no outbound security rule to deny internet on the Public subnet.



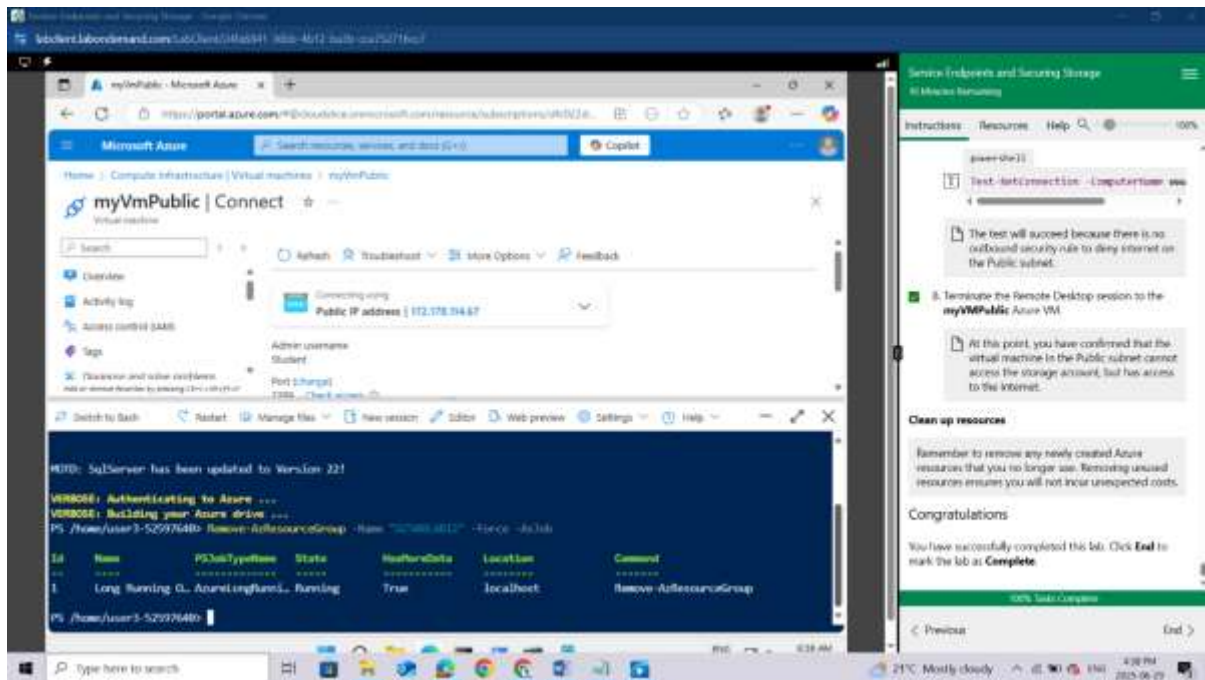
Terminate the Remote Desktop session to the **myVMPublic** Azure VM.

At this point, you have confirmed that the virtual machine in the Public subnet cannot access the storage account, but has access to the internet.



Clean up resources

Remember to remove any newly created Azure resources that you no longer use. Removing unused resources ensures you will not incur unexpected costs.



Congratulations! You have successfully completed this lab. Click End to mark the lab as Complete.

Conclusion

In this lab, we successfully implemented a **secure Azure networking environment** by using **service endpoints** and **network security groups (NSGs)** to control access to Azure Storage. We began by creating a virtual network and segmenting it into public and private subnets. Service endpoints were configured to allow secure connectivity from the private subnet to the storage account. To enforce access control, NSGs were applied to restrict and allow specific types of traffic, including enabling RDP only on the public subnet. A storage account with a file share was provisioned, and virtual machines were deployed into the designated subnets. Connectivity tests confirmed that access to the storage account was permitted from the private subnet and correctly denied from the public subnet. This hands-on exercise highlighted the importance of network isolation and secure resource access within Azure, providing practical experience in designing and implementing secure cloud architectures.